

REVIEW ARTICLE

Farmed cricket (*Acheta domesticus*, *Gryllus assimilis*, and *Gryllodes sigillatus*; Orthoptera) welfare considerations: recommendations for improving global practice

E. Rowe¹, K.Y. Robles López², K.M. Robinson², K.M. Baudier² and M. Barrett^{1,3*} 

¹Rethink Priorities, 530 Divisadero St, San Francisco, CA 94117, USA; ²School of Biological, Environmental, and Earth Sciences, The University of Southern Mississippi, Hattiesburg, MS 39406, USA; ³Department of Biology, Indiana University Indianapolis, Indianapolis, IN 46202, USA; *meghbarr@iu.edu

Received 4 December 2023 | Accepted 31 January 2024 | Published online 5 March 2024 |
Published in issue 29 July 2024

Abstract

Orthoptera, such as crickets, is currently the most reared group of hemimetabolous insects in the insects as food and feed industry, with over 370 billion individuals slaughtered and/or sold live annually. The most-farmed cricket species is *Acheta domesticus*, however there is growing interest in farming at least two additional species, *Gryllus assimilis* and *Gryllodes sigillatus*. Crickets are largely being explored for use as human protein, and exotic animal or pet feed – as well as, to a lesser extent, livestock and fish feed. Insect welfare is of growing interest to consumers who are considering incorporating insect protein into their diets, as well as to many producers. However, no studies have considered the welfare concerns of farmed crickets under current industry conditions. Using an established model for assessing farmed insect welfare, we assess potential welfare concerns for the three most-farmed cricket species, including: interspecific interactions (including parasites and pathogens), temperature and humidity, light cycles, electrical shocks, atmospheric gas levels, nutrition and hydration, environmental pollutants, injury and crowding, density, handling-associated stress, genetics and selection, enrichments, transport-related challenges, and stunning, anesthesia, and slaughter/depopulation methods. From our assessment of these factors, we make recommendations for improving cricket welfare now and as the industry continues to grow; in addition, we identify research directions that will improve our understanding of cricket welfare. We conclude by broadly discussing the importance of addressing the welfare challenges presented by the insects as food and feed industry for the animals and for the growth and health of the industry itself.

Keywords

insect welfare – rearing conditions – survival – slaughter methods – crickets

1 Introduction

The UN predicts there will be 9.7 billion people on the planet by 2050 (United Nations, 2022); as land, water, and resources for cultivating food are limited, new solutions are needed to sustainably and securely meet the

global protein demands of the growing human population (Hertel, 2015). Wild-caught Orthopterans such as crickets have long served as a popular source of food around the world, with dozens of species consumed (Magara *et al.*, 2021; Mott, 2016). The insects as food and feed (IAFF) industry promises to industrialize the rear-

ing of edible crickets at scale and may produce more sustainable protein than conventional animal agriculture (van Huis and Oonincx, 2017; although this is dependent on using sustainable feed substrates; Lundy and Parrella, 2015).

Popular farmed cricket species are 60–70% protein by dry weight (da Rosa Machado and Thys, 2019; Finke, 2002; Zielińska *et al.*, 2015). Powdered cricket meal is already incorporated into some food products, though it may also pose some safety concerns (e.g. bioaccumulation of heavy metals or toxins, allergic reactions, or bacterial contamination; Magara *et al.*, 2021, though see Pener, 2016; Fernandez-Cassi *et al.*, 2018). More commonly in the West, cricket protein has been used in pet/exotic animal feed, and, to a lesser extent, livestock or fish feed (Cortes Ortiz *et al.*, 2016; van Huis, 2020).

An estimated 370–430 billion crickets (or grasshoppers) were slaughtered or sold live by the IAFF industry each year, as of 2020 (Rowe, 2020), with a particular focus on *Acheta domesticus*, *Gryllus assimilis*, and *Gryllobates sigillatus* as edible insects (Rumbos and Athanassiou, 2021). The IAFF industry is expected to grow substantially in the coming decades (de Jong and Nikolic, 2021), and thus represents one of the largest ever undertakings of industrial livestock rearing in terms of the number of individual animals. Despite some uncertainty about insect sentience (Adamo, 2019, 2016), recent literature suggests insect sentience is plausible (Gibbons *et al.*, 2022), and therefore many academics, consumers, producers, and professional societies have expressed an interest in the welfare of insects (Barrett *et al.*, 2022, 2023; Bear, 2021, 2019; de Goede *et al.*, 2013; Delvendahl *et al.*, 2022; Erens *et al.*, 2012; Gjerris *et al.*, 2016; IPIFF, 2019; Klobučar and Fisher, 2023; Voulgari-Kokota *et al.*, 2023).

Several publications have explored the welfare concerns of popular farmed insects (e.g. *Hermetia illucens*: Barrett *et al.*, 2022; *H. illucens* and *Musca domestica*: Kortsmit *et al.*, 2023; *Tenebrio molitor*: Barrett *et al.*, 2023); however, each species has unique welfare concerns and may be reared using different practices. In addition, cricket species are hemimetabolous (i.e. incomplete metamorphosis), while all insects investigated to date are holometabolous (i.e. complete metamorphosis); these different developmental strategies may bear significantly on the question of animal welfare at the juvenile life stage due to critical differences in the developmental timing of discrete, sensory integrative brain regions and, possibly, connections between them (Barrett and Fischer, 2023; Gibbons *et al.*, 2022; Fischer and Sandall, 2023).

Here, we apply a model for assessing farmed insect welfare to the three most popular farmed species of crickets (*A. domesticus*, *G. assimilis*, and *G. sigillatus*), generating specific recommendations for improving welfare in industry conditions.

2 Welfare framework

Although there are many definitions of animal welfare, a fundamental characteristic in this study is the subjective experience (feelings) of the individual animal (Mellor, 2016). For example, behavioral expression is a component of animal welfare, but it is the subjective experience associated with behavior that determines welfare state (e.g. *pleasure* from engaging in reward-motivated behaviors such as exploration) – rather than the “naturalness” of the behavior per se (Mellor, 2015; Browning, 2020). Thus, the general agreement is that only animals capable of subjective experience – i.e. sentience – have welfare states (Bentham, 1789; Broom, 2019; Dawkins, 1990; Singer, 2002; Webster, 1994).

The empirical evidence does not provide a clear answer to the question of insect sentience (Adamo, 2016; Birch, 2022; Klein and Barron, 2016; Lambert *et al.*, 2021), although a recent review found strong evidence for pain in several insect taxa despite current research gaps (Gibbons *et al.*, 2022). One approach, including for insect mini-livestock (Röcklinsberg *et al.*, 2017; van Huis, 2021), to managing uncertainty around animal sentience is the “precautionary principle”, which suggests sensible precautionary measures should be taken to protect the welfare of animals when there is evidence that they may be sentient (Birch, 2017; Knutsson and Munthe, 2017). Alongside the moral importance of safeguarding animal welfare (DeGrazia, 1996; Fischer, 2021; Thompson, 2020), improving animal welfare can also improve the economic productivity of the farming industry and provide other social benefits (Dawkins, 2017), including ensuring the social acceptability of the use of insects for human benefit (Barrett and Adcock, 2023).

The ‘Five Domains’ welfare framework (Mellor and Reid, 1994) provides four physical/functional domains (nutrition, environment, physical health, and behavior) that influence, and therefore can serve as proxies, for a fifth domain of mental state, i.e. an animal’s valenced subjective experiences, which ultimately determine welfare status (see Figure 1). Using the model’s four physical/functional domains, we can assess welfare state by assessing an animal’s physical state (e.g. presence of

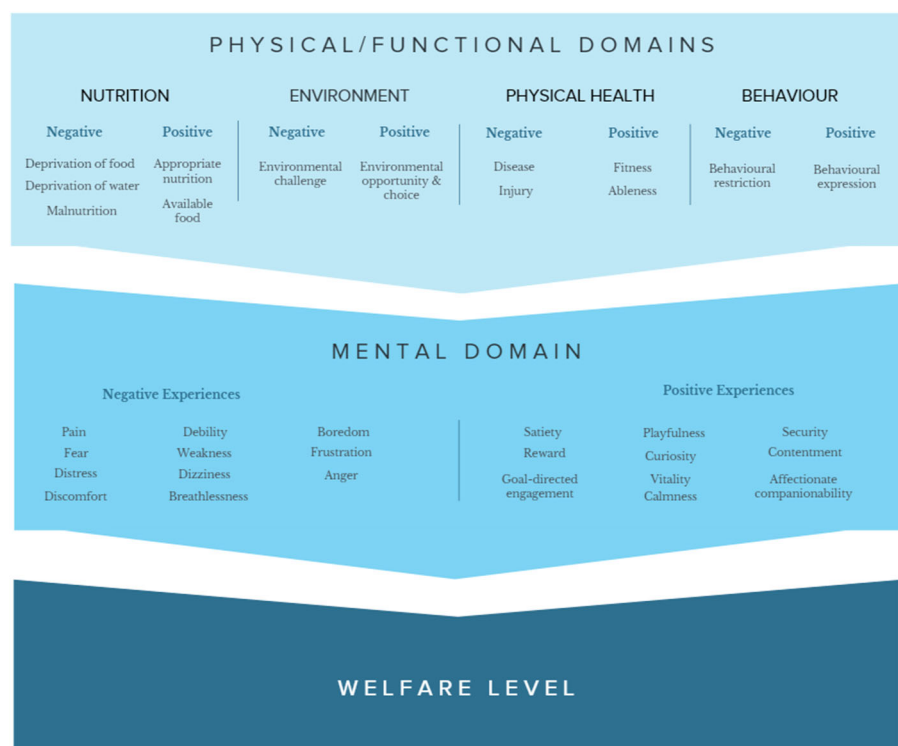


FIGURE 1 The Five Domains framework of animal welfare Adapted from Mellor *et al.* (2015); from Barrett and Fischer (2023).

disease), and the inputs affecting both their physical state (e.g. provision of food/water) and their external circumstances (e.g. provision of environmental enrichment). Moreover, as the Five Domains model is inclusive of physical/functional states associated with positive subjective states as well as negative ones, welfare guidelines can promote at least a life worth living (Farm Animal Welfare Council, 2009) for insect mini-livestock.

Therefore, to assess cricket welfare in this paper, we will use the 'Five Domains' model to determine the resources needed to: (1) ensure appropriate nutrition and avoid malnutrition (domain 1); (2) increase environmental opportunity and choice and reduce environmental challenge (domain 2); (3) increase fitness and ableness and reduce disease and injury (domain 3); and (4) increase behavioral expression (especially of putative rewarding behaviors) and reduce behavioral restriction and behavioral indicators of negative states (domain 4); all of these domains are expected to influence the mental state of the animal (domain 5) which is thus considered implicitly included in all recommendations and challenges.

3 Cricket biology

Understanding the basic biology of each cricket species is an important step in considering welfare concerns in

the farmed cricket industry. All three species belong to the subfamily Gryllinae (field crickets) and share similarities in habitat, nocturnal activity, and diet (Table 1).

Acheta domesticus (Linneo 1758; synonym: *Gryllus domesticus*), the European house cricket, probably originates from southwest Asia or northern Africa (Ghouri, 1961). *Gryllus assimilis* (Fabricus 1775; subject to many taxonomic reclassifications: Rehn and Hebard, 1915; Weissman *et al.*, 2009), the Jamaican field cricket, is neotropical in origin (Alexander, 1968; Alexander and Walker, 1962). Accurate identification requires sonograms and/or DNA barcoding (Masson *et al.*, 2020; Weissman and Gray, 2019); as a result, the farmed cricket industry has also been known to mistakenly identify other species (such as *Gryllus bimaculatus*) they are farming as *G. assimilis* (Weissman *et al.*, 2012), which could make the application of information reviewed here problematic if species are misidentified. *Grylloides sigillatus* (Walker & F. 1869), the tropical house cricket, is originally from the Asian tropics and subtropics (Broder *et al.*, 2023; Ghouri and McFarlane, 1958a; Ivy and Sakaluk, 2005: most relevant taxonomic synonym: *Grylloides supplicans*; Otte, 2007).

The natural diet of wild ground-dwelling crickets is not well understood (Fontanetti and Zefa, 2000), but these crickets are considered non-specialized omnivores with a preference for plant material, fungi, and scavenging on dead insects or even cannibalism (Collavo

TABLE 1 Summary of biological information of *A. domesticus*, *G. assimilis*, and *G. sigillatus*

Life history traits	<i>A. domesticus</i>	<i>G. assimilis</i>	<i>G. sigillatus</i> / <i>G. sup- plicans</i>	Sources
Range and habitat				
Natural origin point	Northern Africa/Southwest Asia	Neotropical	Asian tropics	<i>A. domesticus</i> (Ghour, 1961; Weissman and Rentz, 1977) <i>G.</i> <i>assimilis</i> (Alexander, 1968) <i>G. sigillatus</i> (Broder <i>et al.</i> , 2023; Ghour and McFarlane, 1958b; Ivy and Sakaluk, 2005)
Dispersal ability, frequency of dispersing	Higher ratio of macropterous individuals in population	No wing polymorphism; all individuals are macropterous	Typically, all micropterous	<i>A. domesticus</i> (Alexander, 1961) <i>G.</i> <i>assimilis</i> (Walker and Sivinski, 1986; Zera <i>et al.</i> , 1998) <i>G. sigillatus</i> (Ghour and McFarlane, 1958a)
Geographic range	Cosmopolitan worldwide Farmed worldwide	Eastern Mexico and Central America, Southern Texas; introduced in Florida, Caribbean islands, Brazil, and possibly elsewhere in South America* Farmed across the Americas, reported to be farmed in Europe, although unlikely to be the correct species	Found around the world near human habitations, otherwise found in tropical regions Farmed worldwide	<i>A. domesticus</i> (Weissman and Rentz, 1977) <i>G.</i> <i>assimilis</i> (Alexander and Walker, 1962; Weissman <i>et al.</i> , 2009, 2012, 2019; Weissman and Gray, 2019; Zefa <i>et al.</i> , 2022) <i>G. sigillatus</i> (Alexander, 1968; Ghour and McFarlane, 1958a)
Habitat type	Woodlands, caves, high grasses, below logs; Associates with humans	Grasslands (pastures, fields, yards, roadsides); Associates with humans	Roadsides, hills, also secondary vegetation in forest; Associates with humans	<i>A. domesticus</i> (Weissman and Rentz, 1977) <i>G.</i> <i>assimilis</i> (Weissman <i>et al.</i> , 2009, 1980; Weissman and Gray, 2019) <i>G. sigillatus</i> (Cadena-Castañeda, 2011; Otte, 2007; Weissman <i>et al.</i> , 2012)

TABLE 1 (Continued)

Life history traits	<i>A. domesticus</i>	<i>G. assimilis</i>	<i>G. sigillatus</i> / <i>G. sup- plicans</i>	Sources
Feeding Habits	Non-specialized omnivorous	Non-specialized omnivorous	Non-specialized omnivorous	<i>A. domesticus</i> (Gutiérrez <i>et al.</i> , 2020; Hanboonsong <i>et al.</i> , 2013) <i>G. assimilis</i> (Fontanetti and Zefa, 2000; Gangwere, 1961) <i>G. sigillatus</i> (Gangwere, 1961)
Development**				
Average life cycle duration (days from hatching until death)	100-115	82-103 (potentially up to 138.34*)	92-111	<i>A. domesticus</i> (Clifford and Woodring, 1990; Lyn <i>et al.</i> , 2011) <i>G. assimilis</i> (Jobin, 1961; Limberger <i>et al.</i> , 2021; Loranger and Bertram, 2016a; Masson <i>et al.</i> , 2020) <i>G. sigillatus</i> (Archer <i>et al.</i> , 2012; Ghouri and McFarlane, 1958b; Houslay <i>et al.</i> , 2015; Mallard and Barnard, 2004; Sakaluk <i>et al.</i> , 2002)
Egg Incubation period (days)	10-14	Up to 19-21 days, (averaging potentially as low as 11.87*)	13-14	<i>A. domesticus</i> (Fernandez-Cassi <i>et al.</i> , 2019; Nix and Bass, 1973) <i>G. assimilis</i> (Masson <i>et al.</i> , 2020; Walker, 1980) <i>G. sigillatus</i> (Subramaniam and Subramoniam, 1990; Vandeweyer <i>et al.</i> , 2018)
Juvenile stage (days)	42-49	44-64*	34-60	<i>A. domesticus</i> (Fernandez-Cassi <i>et al.</i> , 2019; Lyn <i>et al.</i> , 2011; Nix and Bass, 1973) <i>G. assimilis</i> (Jobin, 1961; Masson <i>et al.</i> , 2020) <i>G. sigillatus</i> (Mallard and Barnard, 2004; Sakaluk <i>et al.</i> , 2002)

TABLE 1 (Continued)

Life history traits	<i>A. domesticus</i>	<i>G. assimilis</i>	<i>G. sigillatus</i> / <i>G. sup- plicans</i>	Sources
Number of instars	6-9, documented up to 14	5-10*	8	<i>A. domesticus</i> (Clifford and Woodring, 1990; Ghouri and McFarlane, 1958b; Masaki and Walker, 1987; Nix and Bass, 1973) <i>G. assimilis</i> (Jobin, 1961; Masson <i>et al.</i> , 2020) <i>G. sigillatus</i> (Ghouri and McFarlane, 1958b)
Adult stage lifespan (days)	30-64	28-56	43-74	<i>A. domesticus</i> (Ghouri and McFarlane, 1958b; Nowosielski and Patton, 1965; Patton, 1978) <i>G. assimilis</i> (Limberger <i>et al.</i> , 2021; Loranger and Bertram, 2016a) <i>G. sigillatus</i> (Archer <i>et al.</i> , 2012; Ghouri and McFarlane, 1958b; Houslay <i>et al.</i> , 2015)
Generations per year	Multivoltine; 2 generations per year with overlapping continuous generation; 6-7 in captivity	Multivoltine; 2-3 generations per year with overlapping continuous generations; up to 6 generations per year in captivity	Multivoltine; unknown number of continuous, overlapping generations per year	<i>A. domesticus</i> (Alexander, 1968; Ismail, 1978; Makkar <i>et al.</i> , 2014) <i>G. assimilis</i> (Alexander, 1968; Weissman <i>et al.</i> , 2012; Weissman and Gray, 2019) <i>G. sigillatus</i> (Alexander, 1968)
Reproduction**				
Lifetime number of eggs	1200-2994 ± 245	1250-3100	713.4 ± 73.3–2328 ± 248	<i>A. domesticus</i> (Clifford and Woodring, 1990; Patton, 1978) <i>G. assimilis</i> (Limberger <i>et al.</i> , 2021) <i>G. sigillatus</i> (Arnqvist and Nilsson, 2000; Sakaluk, 1987)

TABLE 1 (Continued)

Life history traits	<i>A. domesticus</i>	<i>G. assimilis</i>	<i>G. sigillatus</i> / <i>G. sup- plicans</i>	Sources
Eggs per day	95 at the start of the egg laying period, decreasing to 55 until death	28.07 ± 3.05 to 69 ± 8.16, the majority are laid in the first days after mating	27-75	<i>A. domesticus</i> (Clifford and Woodring, 1990; Patton, 1978) <i>G. assimilis</i> (Limberger <i>et al.</i> , 2021) <i>G. sigillatus</i> (Archer <i>et al.</i> , 2012; Sakaluk, 1987)
Egg viability (proportion)	0.75 ± 0.11	0.57 ± 0.03	0.66-0.87, depending on quality of adult diet	<i>A. domesticus</i> (Murtaugh and Denlinger, 1985) <i>G. assimilis</i> (Loranger and Bertram, 2016a) <i>G. sigillatus</i> (Mallard and Barnard, 2004; Sakaluk, 1987)
Number of mating partners	Multiple	Multiple*	Multiple	<i>A. domesticus</i> (Gray, 1997; Wilson and Walker, 2019) <i>G. assimilis</i> (Masson <i>et al.</i> , 2020) <i>G. sigillatus</i> (Sakaluk <i>et al.</i> , 2002; Vasanth, 1988; Wilson and Walker, 2019)
Special behaviors associated with mating	Male calls/chirps are distinct from other species. Aggressive chirps, calling song to attract females, and courtship songs are distinct from one another. Males engage in mate guarding behavior. Non-preferred males may harass females.	Male calls/chirps are distinct from other species. Aggressive chirps, calling song to attract females, and courtship songs are distinct from one another. Males, as well as females, exhibit mate choice preferences.	Male calls/chirps are distinct from other species. Aggressive chirps are hard to distinguish from calling song. Males may provide a spermatophylax nuptial gift to females.	<i>A. domesticus</i> (Alexander, 1961; Gray, 1997; Wilson and Walker, 2019) <i>G. assimilis</i> (Alexander, 1961; Bertram <i>et al.</i> , 2017) <i>G. sigillatus</i> (Alexander, 1961; Sakaluk <i>et al.</i> , 2019; Vasanth, 1988)

TABLE 1 (Continued)

Life history traits	<i>A. domesticus</i>	<i>G. assimilis</i>	<i>G. sigillatus</i> / <i>G. sup-</i> <i>plicans</i>	Sources
Sex ratios in natural populations (M:F)	Density-dependent; often 1:1.25 under farmed conditions	Variable; potentially 4:6*	1:1.36-1:0.83, dependent on quality of diet	<i>A. domesticus</i> (Crocker and Hunter, 2018; Ismail, 1978) <i>G. assimilis</i> (Villarreal et al., 2018) <i>G. sigillatus</i> (Houslay et al., 2015)

* This information is sourced from publications that provide specific biological data using a species identification listed as *G. assimilis*, but the identification cannot be verified either because the publication is too old to verify with modern information or because the species identification has not yet been verified for that specific study.

** The information on development is highly variable across temperatures, humidities, and diet quality. Most of these studies did not use a standard set of conditions, therefore, we present a range of average ($\pm$ SE, where found) values for each trait found across studies.

et al., 2005; Gangwere, 1961; Gutiérrez et al., 2020; Hanboonsong et al., 2013). All three species are active year-round (Alexander, 1968; Weissman et al., 2009, 2012).

Cricket species can vary in the prevalence of short-winged (micropterous) and long-winged (macropterous) morphs. Macropterous individuals can disperse longer distances upon adulthood and may be obligate or facultative. *Acheta domesticus* typically has a higher proportion of macropterous individuals relative to micropterous individuals (Alexander, 1961) while *G. sigillatus* typically has a higher proportion of micropterous individuals relative to macropterous individuals (Ghouri and McFarlane, 1958a). All *G. assimilis* are macropterous, suggesting a life history of dispersal following eclosion (Walker and Sivinski, 1986; Zera et al., 1998).

If dispersal occurs, mating will begin shortly thereafter. Female crickets can mate several times before ovipositing (Alexander, 1961) and can continue to mate throughout their lifetime with different males, storing the sperm internally. Males of all three species produce songs by rubbing their forewings together (Alexander, 1961; Walker, 1962; Weissman et al., 2009): there are long-distance calling songs (for females to locate males), short-distance courtship songs, and aggression sounds (Alexander, 1962).

Male crickets may engage in aggressive intrasexual interactions, with larger males tending to be more aggressive and preferred by females (Bertram et al., 2016; Bertram and Rook, 2012; Brown et al., 2006; Lorange and Bertram, 2016b). Females oviposit individual eggs into a damp substrate (Bertram and Rook, 2012; Clifford and Woodring, 1990; Reverberi, 2020). Dampness appears to be necessary and preferred for oviposition (Alexander, 1961; Murtaugh and Denlinger, 1985;

Smith and Thomas, 1988) and possibly for egg development as well (Bertram et al., 2016; Masaki and Walker, 1987). Females may oviposit for several weeks (three, in *A. domesticus*), breeding and producing young multiple times during the year (Ismail, 1978; Weissman et al., 2012; Weissman and Gray, 2019). Crickets are hemimetabolous, so newly-hatched juveniles (nymphs) are morphologically similar to adults, albeit smaller in size and without wings or developed genitalia.

4 Industry history and practices

Industrial cricket farming was introduced in Thailand in 1997, which is now considered the leading country in cricket farming with over 20,000 farms (Halloran et al., 2016) producing 3,000-7,000 tons/yr (Halloran et al., 2018). These farms are mostly small-scale, independent, and rural (Hanboonsong et al., 2013; Magara et al., 2021); small and medium-sized enterprises are also the most common facility type in Africa (Tanga et al., 2021). While small-scale cricket farming has occurred since the mid-20th century in the West (e.g. Armstrong Crickets Georgia, n.d.), industrial insect farming has only occurred in earnest in the past decade (Magara et al., 2021; Reverberi, 2020). Eight farms in the USA rear over 1 billion crickets/year each (Reverberi, 2020), and the world's largest cricket farm in Canada is expected to house four billion crickets (Zandbergen, 2022).

Cricket production takes place in either closed or semi-open facilities. Facilities that are completely closed to the outdoor environment (i.e. a building with windows, doors and other openings kept closed) represent the majority of Western cricket production systems (Eilenberg et al., 2018). Semi-open facilities con-

tain large openings to the outdoor environment, and are common in Thai cricket operations (Eilenberg and Jensen, 2018).

Incubation and hatching

Eggs are laid into a moist substrate (e.g. coconut husk or moss) where they incubate. Depending on the scale of the operation, incubation can occur in simple manual systems (i.e. a box that allows for air exchange) or large, fully climate-automated chambers (EntoCube, n.d.; Kyllönen and Manzanares, 2022). Collection of newly hatched nymphs (known as pinheads) is usually achieved passively, as they are very fragile and any handling can cause mortality (Mott, 2016). Nymphs can 1) crawl out of the egg substrate into (Kyllönen and Manzanares, 2022), or 2) be shaken or blown into (Mott, 2016), a collection container, measured to a certain volume, and transferred to the rearing units at a standard density.

Juvenile rearing

Rearing of pinheads to adults is the main operation of a cricket farm (Kyllönen and Manzanares, 2022). The one step approach involves placing pinhead crickets into one rearing unit and harvesting them at the desired age. The two-step approach involves rearing pinheads until they reach instar four or five, at which point the population is divided into further rearing units.

In mass-production operations, crickets are usually reared in stackable containers with an internal volume over 500 liters (Cortes Ortiz *et al.*, 2016; Kyllönen and Manzanares, 2022). In smaller operations, plywood boxes or concrete pens may be used (Fernandez-Cassi *et al.*, 2019; Reverberi, 2020). If units are open, crickets will be exposed to the climate of the external rearing room; closed containers allow for more precise, automated climate control. Entire rooms may serve as rearing containers in 'free-range' production, reducing labor/automation costs, but increasing the risk of pathogen spread (Entomofarms, 2016; Kyllönen and Manzanares, 2022). A single rearing unit may have several thousands to millions of crickets in medium to large scale operations (Kyllönen and Manzanares, 2022), with larger units reducing the ease of inspecting the health or welfare status of the colony (Mott, 2016).

Harvesting

Harvesting involves removing the crickets from the rearing matrices in order to be slaughtered, usually just before or just after ecdysis from last instar nymph to adult (Kyllönen and Manzanares, 2022). As not all

adults emerge simultaneously, harvesting usually occurs when at least 85% of the population has reached adulthood, in order to maximize output (Mott, 2016). Crickets are 40-60 days old at harvest depending on species and conditions (Fernandez-Cassi *et al.*, 2019; Halloran *et al.*, 2017; Reverberi, 2020; Vandeweyer *et al.*, 2018), although one producer reports harvesting *A. domesticus* after only 28-33 days (Kyllönen and Manzanares, 2022).

In smaller scale production, rearing matrices are shaken by hand into a collection container. Live crickets then separated from dead crickets and frass by placing vertical material ('sorters' e.g. egg trays, cardboard tubes; Mott, 2016) in the container to allow the live crickets to climb, which are then shaken into a separate container. Alternatively, the first container may be exposed to light with no hiding substrate causing crickets to attempt to hide by moving out of the exposed container and into an adjacent collection container (Figure 2A; Kyllönen and Manzanares, 2022). In larger scale production, rearing matrices may be fed manually into a machine as a continuous "sheet" and crickets are collected via suction (Figure 2B; Kyllönen and Manzanares, 2022).

Adult rearing, mating, and oviposition

To obtain eggs for the next cycle of rearing, a population of adults is either separated from the main population for reproduction (forming a propagation colony, or broodstock) or the propagation colony is established and reared separately but simultaneously to the rearing colony. After the terminal molt, provision of a moist and porous substrate in the rearing matrices stimulates oviposition (Cortes Ortiz *et al.*, 2016; Kyllönen and Manzanares, 2022; Reverberi, 2020; Vandeweyer *et al.*, 2018). This substrate may be left with the crickets for 1-3 days. Alternatively, it may be provided and removed every other day (Clifford, 1985).

Although an adult female will continue to lay over her lifetime, which can be over a month (Behrens *et al.*, 1983; Murtaugh and Denlinger, 1985), egg numbers start to decrease after one to two weeks (Parajulee *et al.*, 1993). Therefore, breeding adults tend to be harvested and slaughtered after two weeks (Kyllönen and Manzanares, 2022).

Processing for slaughter and transport

Crickets may be transported live to markets in small mesh pens, such as in Thailand, for home consumption (Halloran *et al.*, 2016). Crickets may also be transported live for exotic pet feed – in the USA alone, an estimated 2.6 billion live crickets were shipped for this purpose in 2012 (Weissman *et al.*, 2012). Some, though not all, pro-

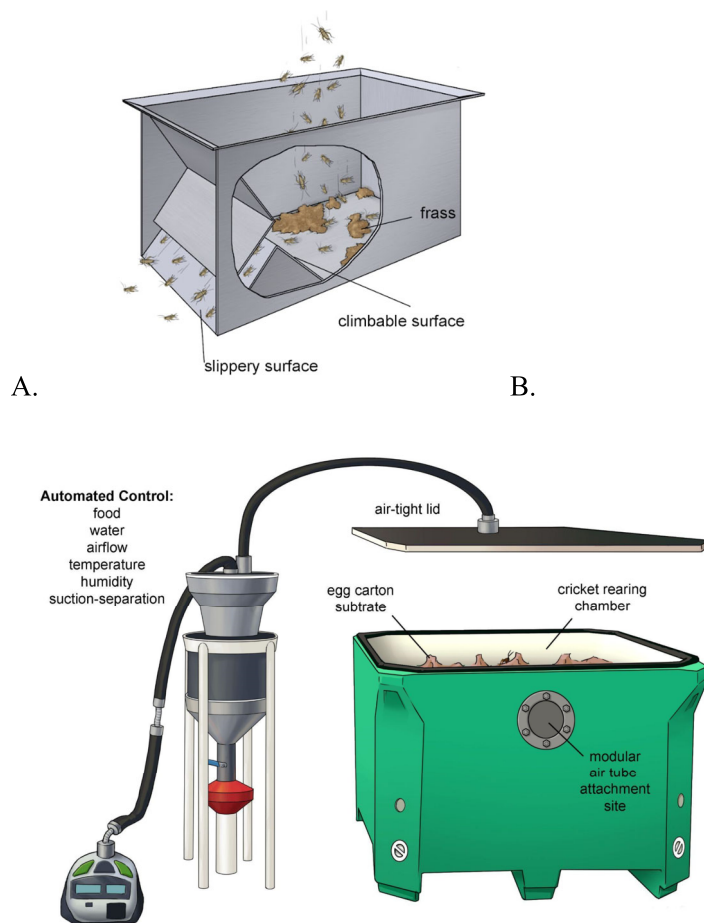


FIGURE 2 Two cricket harvesting devices used in industry. (A) collection using stress response to light to cause crickets to manually separate themselves from frass and other materials; (B) collection by suction.

ducers may provide feed and moisture sources during transit (Cricket King, n.d.; Josh's Frogs, n.d.; Suckling *et al.*, 2020).

Crickets used for processed food/feed applications are slaughtered before transport (Vandeweyer *et al.*, 2018). Crickets may be starved for 24–48 hours prior to slaughter to clear their digestive systems (Bear, 2019), improve taste, and decrease microbial load, although the efficacy of this in crickets has not been established (Fernandez-Cassi *et al.*, 2019; and see Caparros Medigo *et al.*, 2017). Preventing oviposition (through absence of substrate) for 12–24 hours before slaughter is also expected to improve taste (Hanboonsong *et al.*, 2013).

Reported slaughter methods for crickets include boiling (described as the most common method in Thailand; Reverberi, 2020), immersion in pressurized steam (Tatarova, 2017), immersion in hot, non-boiling water (e.g. 60 °C; Vandeweyer *et al.*, 2018), drowning in non-boiling water (Fernandez-Cassi *et al.*, 2019; Miech, 2018), freezing in air (e.g. 24 hours at –18 °C, Bear, 2019; Fernandez-Cassi *et al.*, 2019), heating (presumably in dry air; Fernandez-Cassi *et al.*, 2019), shredding (Bear,

2019), and asphyxiation (Singh *et al.*, 2020). Freezing in air is reported to be the main slaughter method for mass-produced crickets (van Huis *et al.*, 2013). Anesthetic use is not widely reported before slaughter; however, at least one cricket producer has self-reported using carbon dioxide gas as an anesthetic before freezing (Bear, 2019).

5 Current welfare concerns for crickets and recommendations

We reviewed the scientific literature for information on crickets and interspecific interactions (including parasites and pathogens), abiotic conditions, nutrition and hydration, environmental pollutants, injury and crowding, density, handling-associated stress, genetics and selection, enrichments, transport-related challenges, and stunning, anesthesia, and slaughter/depopulation methods. Not all information reviewed was relevant to the welfare concerns discussed further in this paper at this time, however the information may still be relevant

to producers or researchers looking to improve production goals; therefore, we have included it, organized by variable, as Supplemental File S1. This information may also serve as the basis of future investigations into the species' biology and welfare recommended in section 7.

A few caveats inform all the following recommendations:

1. Most studies we reference have been conducted at laboratory scales, not mass-rearing scales; however, scale can influence many aspects of insect biology (Sørensen *et al.*, 2012). Producers should consider the effect of their facility's scale and set up when implementing any of the recommendations provided.
2. Population can affect many welfare-relevant characteristics due to genetic variability (as a result of selective breeding, genetic drift, etc.); producers should assess their population's performance under these recommendations and monitor their animals, making adjustments to continue to improve their specific population's welfare.
3. We often rely on improved production outcomes (e.g. increased viable reproduction or growth) as possible indicators of high-welfare conditions; however, improved production outcomes do not always mean better welfare. In some cases, they can be negatively correlated (e.g. fast-growing chicken breeds increase productivity but reduce welfare; Abdourhamane and Petek, 2022). We can be most confident welfare is improving when other measures, such as survival, health, or expression of rewarding behaviours, improve alongside production-relevant outcomes like growth.
4. Many welfare concerns interact; therefore, it is important to implement welfare recommendations across all domains. Further, addressing one welfare concern could have unintended impacts on another aspect of cricket production/welfare.
5. Species identification of crickets in studies is not always reliable when conducted morphologically, particularly for *G. sigillatus* (as mentioned in the biology section); therefore, caution should be employed when adopting results from the literature for this species.

For each identified concern, we review the relevant biological information that undergirds it before making recommendations to improve current practice.

Concern 1. Intentional or accidental dietary restriction (quantity) leading to hunger

Appropriate nutrition is integral to good animal welfare (Mellor, 2015). Inadequate nutrition is likely to have knock-on effects on other welfare domains such as health (domain 3) and behavior (domain 4) which may affect growth and reproduction (e.g. in crickets: Lundy and Parrella, 2015).

Providing appropriate nutrition for animals has three considerations: (1) quantity of provisions able to be consumed (e.g. made available in a form accessible to the animal); (2) composition of the diet provided (see concern 2); and (3) how, where, and when the diet is provided (e.g. to ensure accessibility and promote natural foraging behaviours). Data are currently only available to make recommendations re: 1 and 2 for farmed crickets.

Recommendation 1a (domain 1: nutrition). Provide adequate quantities of nutrition at all times during juvenile rearing. Avoid fasting whenever possible. If needed for product safety reasons, fasting should be limited to 24 hours

The economic incentive to avoid feed waste may cause dietary restrictions for crickets in industry settings, where the strongest financial limitation is the cost of cricket feed – accounting for up to half of the farming expenses (Durst and Hanboonsong, 2015). Food may also be removed for 24–48 hours (Bear, 2019), or only vegetables provided (Hanboonsong *et al.*, 2013; Miech, 2018; Vandeweyer *et al.*, 2018), prior to slaughter. Adult dietary restriction significantly reduces the body condition of adult males (Whattam and Bertram, 2011) and decreases the energy allocated to ovaries and reproductive development in females (Zera and Harshman, 2001). Across cricket species, dietary restriction (from quantity or composition) during the juvenile period increased mortality, slowed growth, worsened adult body condition (body mass: body size ratio), reduced adult fitness, and increased cannibalistic behavior (Collavo *et al.*, 2005; Lyn *et al.*, 2011; Patton, 1967; Whattam and Bertram, 2011).

Recommendation 1b (domain 1: nutrition). To avoid accidental dietary restriction caused by inaccessible feed, diets should be ground to a 20 mesh particle size for juveniles

Food particle size can also lead to juvenile dietary restriction, with too-large particle sizes being more difficult to eat and potentially leading to cannibalistic behavior (Patton, 1967); diets should thus be ground to a

20 mesh particle size to enable juvenile crickets to easily eat (Nakagaki and Defoliart, 1991). Juvenile males on restricted diets produced lower-quality mating songs as adults, a trait that is likely directly related to body condition (larger body sizes correlate with more attractive mating songs; Whattam and Bertram, 2011) alongside the other fitness consequences for juveniles listed in recommendation 1a.

Concern 2. Inappropriate macronutrient or micronutrient provisioning, leading to malnutrition

Beyond dietary accessibility and digestibility, dietary composition is important to animal welfare. Not much information exists on the natural diet of crickets (Fontanetti and Zefa, 2000) aside from being generalized omnivores with preferences for plant material, fungi, and scavenging dead insects (Gangwere, 1961; Gutiérrez *et al.*, 2020; Hanboonsong *et al.*, 2013). As a result of a desire to use cost-effective, sustainable feeds where possible, cricket diets in the industry may be highly variable and lack essential macronutrients or micronutrients necessary to support cricket health and welfare (Patton, 1967; Cortes Ortiz *et al.*, 2016; Sorjonen *et al.*, 2019).

More commercial cricket feeds are being developed around the world as a response to the rising popularity of cricket farming (Halloran *et al.*, 2016). Fluker's®, Mazuri®, Timberline, and Repashy provide cricket diets with ingredients like wheat middlings, corn meal, soybean meal, oats, brewer's yeast, alfalfa meal, molasses, beet pulp, plant oil, whey, and various salts, vitamins, and minerals. Occasionally, animal products are added, such as fish meal, fish oil, porcine meal, bone meal, and animal fat (however, the use of animal products in farmed cricket feed may be limited by legislation in some countries; Veldkamp *et al.*, 2021).

Although correct macronutrient levels are more important for direct impacts on survival, balancing micronutrients can also be important to animal health and welfare (Harrison *et al.*, 2014; Loaiza *et al.*, 2008; Poissonnier *et al.*, 2014; Ritchot and McFarlane, 1961). Little information is available on micronutrient provisioning during cricket development in industry conditions.

Recommendation 2a (domain 1: nutrition): provide minimally 20%, preferably closer to 30-35% protein, in cricket diets

Improper diets may result in negative welfare impacts due to inappropriate macronutrient provisioning (Lyn *et al.*, 2011). Farmed crickets may be fed commercial live-

stock feed, such as chicken feed, which ranges from 15.2-22% protein (Fernandez-Cassi *et al.*, 2019; Lundy and Parrella, 2015; Nakagaki and Defoliart, 1991; Sorjonen *et al.*, 2019), and most commercial cricket diets report protein levels of less than 20% (Cortes Ortiz *et al.*, 2016). This is below the recommended range of protein levels for maximizing survival and growth and below the preferences of *A. domesticus* (Patton, 1967; Sorjonen *et al.*, 2019).

Acheta domesticus diets that maximize both juvenile survival and growth are at least 20-30% protein, 32-47% carbohydrate, and 3.2-5.2% fat (Patton, 1967; and see Oonincx *et al.*, 2015). When *A. domesticus* juveniles are allowed to self-select the ingredients that make up their diets, the average percentage of macronutrients is 30.7% protein, 58.7% carbohydrate, and 10.6% lipids (Morales-Ramos *et al.*, 2020). Barley mash and turnip rape diets with 22.5-30% protein best supported growth and development for *A. domesticus* (as well as *Gryllus bimaculatus*; Sorjonen *et al.*, 2019).

High protein diets lead to faster development, larger body sizes, and early reproductive investment for male crickets while high carbohydrate diets lead to longer-term reproductive investment and increased longevity (Hunt *et al.*, 2004; Reifer *et al.*, 2018). Female *A. domesticus* fed high-protein diets (3P:1C) lay more eggs over their lifetimes but also die younger than those fed on lower-protein diets (1P:1C) – likely due to the high protein demands of egg production (Gutiérrez *et al.*, 2020). The effects of macronutrient levels on reproduction can vary by sex: *G. sigilatus* male signaling effort was maximized on 1P:7C ratio diet, but female egg production was maximized on a 1P:1C diet (Rapkin *et al.*, 2018).

Overall, these results suggest that current feeds do not adequately meet cricket protein needs; diets should provide protein levels closer to 30-35% protein, with a minimum of 20% protein.

Recommendation 2b (domain 1: nutrition; domain 4: behaviour): supplement cricket diets with heterogeneous materials that increase opportunities for nutrient self-selection and improve welfare via enrichment

Homogenous diets are suggested to be detrimental to insect welfare by reducing the ability to self-select for optimal nutrition (Barrett *et al.*, 2022) or fulfil dietary preferences. Crickets do have dietary preferences. In order of preference, *A. domesticus* juveniles ate yeast, soybean meal, animal liver powder, and wheat middlings (Patton, 1967); *A. domesticus* adults were found to prefer rice bran, corn, buckwheat, and dry cabbage out of a variety of options given (Morales-Ramos *et al.*,

2020). The reasons for these preferences (texture/particle size, nutrient composition, etc.) and their relationship to welfare have not been explored. However, preference data in itself is a valuable welfare indicator, and good welfare can be ensured by catering to animal preferences in order to give animals the things they want, positive physical and mental experiences, choice, and agency (Dawkins, 2008; Špinka and Wemelsfelder, 2018). There are, however, limitations to preference tests (e.g. see Kirkden and Pajor, 2006).

Occasionally, cricket diets are supplemented with various fruits, vegetables, fish meal, grains, and agricultural by-products (Durst and Hanboonsong, 2015; Megido *et al.*, 2016; Miech, 2018; Oloo *et al.*, 2020; Reverberi, 2020; van Huis and Tomberlin, 2016). These supplements may improve health and longevity, especially by giving crickets the opportunity to self-select optimal macro- and micronutrients for their sex and life stage (Halloran *et al.*, 2016). Dietary variety may also confer positive welfare in animals through pleasure from different tastes/smells/textures (Mellor, 2017).

Recommendation 2c (domain 1: nutrition). For crickets fed plant-based diets, supplement vitamin B-12 using brewer's yeast or pumpkin in feed. Consider micronutrient supplementation using phosphorus, vitamin C, sterol, and manganese, for all crickets

Micronutrient supplementation can improve animal health and welfare, especially when diets are lacking in appropriate levels of essential nutrients. For example, vitamin B-12 can be challenging for insects to obtain from plants; this may explain why crickets fed diets with animal proteins exhibited faster development (Morales-Ramos *et al.*, 2020; Patton, 1967) and increased oviposition (Ismail, 1978). One non-animal source that can provide B-12 is Brewer's yeast. Further, feeding pumpkin or squash can serve as a source of B vitamins as well as increasing cricket body mass (compared to 16% protein chicken feed; Bawa *et al.*, 2020). Crickets showed preferential consumption of both fresh and dried pumpkin over chicken feed (with the effects of novelty ruled out; Bawa *et al.*, 2020), suggesting supplemental pumpkin may support high welfare.

Other micronutrients, such as phosphorus, vitamin C, sterol, and manganese, may improve cricket welfare in all cases. When reared on a diet containing 1% phosphorus compared to a diet of 0.2% phosphorus, *A. domesticus* had better body condition and increased mass (Visanuvimol and Bertram, 2011), adult males signaled more often (Bertram *et al.*, 2009), and females laid more eggs (Visanuvimol and Bertram, 2010), suggest-

ing potential welfare benefits through increased health and behavioral expression. The phosphorus content of some commercially available cricket feeds is 0.72-0.84% (Finke *et al.*, 2005, 2004), which is approaching the 1% level recommended for *A. domesticus*. The benefits of phosphorus supplementation vary across cricket species (Harrison *et al.*, 2014), suggesting further research may be needed to optimize dietary phosphorus in cricket diets.

Phosphorus can be added to a cricket's diet in the form of calcium phosphate (Bertram *et al.*, 2009). Natural sources that are high in phosphate include fish meal (which varies on type of fish) and dairy products; plant sources include corn gluten meal and peanut meal (Bertram *et al.*, 2009; Riche and Brown, 1996).

Finally, Vitamin C, sterol, and manganese had a positive relationship with increasing biomass in *A. domesticus* whereas vitamins B-1 and B-5 had a negative relationship with biomass. The ideal proportions of these micronutrients for optimising biomass at harvesting were 310.2 (sterol), 49.9 (vitamin C), 5.8 (manganese), 1.2 (vitamin B-1), and 2.2 mg (vitamin B-5) per 100 g; more data on individual growth, and not total biomass, would be beneficial in interpreting the optimal micronutrient levels from a welfare perspective. Foods high in vitamin C, sterol, or manganese include rice bran, wheat bran, cabbage, brewer's yeast, and spirulina, many of which were preferred by crickets in a self-selection trial (Morales-Ramos *et al.*, 2020).

Concern 3. Too high stocking densities

Our focal species are gregarious, where individuals may aggregate even when given more space (Kieruzel, 1976). Population densities can be so high in nature that they "literally cover the ground", though the prevalence and persistence of different densities are not well documented (Alexander, 1961; Ismail, 1978). However, high density conditions cause an increase in macropterous individuals in wing-polymorphic species, enabling individuals to seek resources elsewhere (Alexander, 1961; see concern 4). This suggests that, although high densities may be found in nature, they are not locally persistent across many generations and are self-limiting in density. Therefore, while aggregation is a natural behavior for the focal species, and can even benefit growth in small groups (e.g. "group effect"; Kieruzel, 1976), there is a threshold above which population density is expected to negatively affect growth, behavior, survival and welfare.

High density wild populations may also undergo "self-thinning" due to competition-driven mortality, often

increasing the size of remaining individuals (Fréchette *et al.*, 1995; Jonsson, 2017). Alongside mortality, density may negatively affect growth, which may affect the ability to cope with environmental variation such as intermittent food supply due to competition, thus reducing welfare. Starvation resistance was reduced in *A. domesticus* reared under high densities (Mahavidanage *et al.*, 2023), likely due to reduced body mass and reduced energy reserves. Individual weight and/or growth rate has been found to decrease with increasing rearing density in *A. domesticus* (Kieruzel, 1976; Mahavidanage *et al.*, 2023; Tennis *et al.*, 1977) and *G. sigillatus* (Mazurkiewicz *et al.*, 2013). Increased mortality alongside rearing density occurs in *A. domesticus*, even when food and water were provided *ad libitum* (Gutiérrez *et al.*, 2020; Tennis *et al.*, 1977), suggests that overall resource limitation is unlikely to be the cause of the reduced survival. However, overcrowding could lead to restricted access to even *ad libitum* water, resulting in dehydration as well as aggression, cannibalism, and fewer instars during development (Clifford and Woodring, 1990).

High densities can also affect behavior, increasing aggression and injury that may result in mortality (Mahavidanage *et al.*, 2023). To date, there are no specific reports on the effects of crowding or aggression on injury rates in *A. domesticus*, *G. assimilis*, or *G. sigillatus*. Injuries to the antennae, cerci, and legs resulting from fighting have been reported anecdotally for some cricket species (Alexander, 1961; Sandford, 1987; Simmons, 1986), and incidence of bodily injury (especially damage to antennae) has been shown to increase with rearing density for the cricket *Velarifictorus micado* in laboratory colonies, most likely through fighting (although this species is particularly aggressive and may not be representative of the focal species' behaviors; Wu *et al.*, 2014). Crowding has also been reported to result in considerable damage to wings in *Gryllus bimaculatus* in laboratory colonies (Montealegre-Z *et al.*, 2011), potentially affecting mating behaviors and productivity.

Furthermore, too high a population density may negatively affect adult reproductive output alongside welfare. *Grylloides sigillatus* female spermatophore retention time has been found to decrease significantly if exposed to other crickets, in part due to a decline in mate-guarding as males pursue other females (Bateman and MacFadyen, 1999). Furthermore, crowded environments can cause wing-polymorphic species to retain wings for dispersal. Flight-fecundity trade-offs are common in crickets (Roff, 1984), and long-winged *G. sigillatus* display decreased insemination success due to

smaller spermatophores and spermatophylaxes, which were more quickly discarded by females than those of short-winged (flightless) males (Sakaluk, 1997). It is also worth noting that these long-winged morphs may not have the environmental choice (i.e. opportunity to disperse) that would optimise their welfare (domain 2 and 5).

Reported rearing densities vary greatly. In the USA, some reports have placed cricket nymph densities at 0.04–0.07 crickets/cm² using stacked egg cartons (Cortes Ortiz *et al.*, 2016); rearing densities can be read as number of crickets per amount of surface area provided for crickets to rest via egg cartons and flooring, and are thus not volumetric. Other US consultancies report densities of 0.4–2.6 crickets/cm² (Big Cricket Solutions, 2019b), or recommend 1 cricket/20 cm³ or 1/2.5 cm² (Orinda *et al.*, 2021). Big Cricket Farms (now Big Cricket Solutions) have self-reported using troughs with 3,000 crickets each (a population density of 1.35 crickets/cm², according to their unit dimensions and yields). They report that this high stocking density can generate poor welfare conditions for the crickets: “They’ll find a way to escape, they’ll bite each other, they’ll eat each other” (Wiedemann, 2014). Medium-sized *A. domesticus* farms have an average density of approximately 6,000–12,000 crickets/m² (0.6–1.2 crickets/cm²; Hanboonsong *et al.*, 2013). Given farmers’ reports of cricket attempted escape, aggression and cannibalism, it seems that whatever stocking densities actually are, they may frequently exceed the limit for high-welfare conditions.

Disturbance from conspecifics may not impact successful transfer of sperm, as touch-evoked escape responses are suppressed during copulation in *A. domesticus* (Killian *et al.*, 2006); however, disturbance before copulation e.g. during courtship, when escape responses are still active, could affect mating success. Furthermore, too high a population density may negatively affect both welfare and reproductive output. *Grylloides sigillatus* female spermatophore retention time has been found to decrease significantly if exposed to other crickets, in part due to a decline in mate-guarding as males pursue other females (Bateman and MacFadyen, 1999). Furthermore, crowded environments can cause wing-polymorphic species (including *A. domesticus* and *G. sigillatus*) to retain wings for dispersal, imposing a physiological and potentially behavioral cost on reproduction.

Recommendation 3a (domain 2: environment). *Rear A. domesticus at densities near 0.09 crickets/cm²; do not use densities above 0.4 crickets/cm². G. assimilis can be reared at densities up to 0.37 ml crickets/L space, and possibly at slightly higher densities than this*

Despite all the possible welfare effects, no consensus exists on optimal stocking densities for farmed crickets (Mahavidanage *et al.*, 2023). Beyond the densities reported above by producers, many lab studies have looked into the effects of rearing density on survival to generate recommended rearing conditions. Patton (1978) recommends a maximum density of 1 cricket/2.5 cm² (0.4 crickets/cm²) for *A. domesticus* with mortality increasing from 19.2% at densities of 1 cricket/2.7 cm² (0.37 crickets/cm²) to 33% at 1 cricket/0.67 cm² (1.49 crickets/cm²). Mortality increased with *A. domesticus* rearing density across a range of 5–400 individuals/660 cm² (or 76–6,061 individuals/m²; Tennis *et al.*, 1977). Housing nymphs together at a density of 800 individuals/m² decreased survival compared to those housed alone (133 individuals/m²; Gutiérrez *et al.*, 2020). Parajulee *et al.* (1993) reports a density of 1 cricket/7.5 cm³ (however, few rearing matrices were provided to allow utilization of all the provided space; 2D space provided was only 1 cricket/0.37 cm²). They note that it was necessary to seed the rearing unit with enough eggs to allow for mortality (but the percentage mortality was not reported). Ultimately, increased mortality is an indicator of poor welfare under high stocking densities.

Mahavidanage *et al.* (2023) found significantly increased mortality at all densities above 0.09 cricket/cm². They recommend an optimal stocking density is <0.93 cricket/cm² for economic productivity; despite significantly increased mortality at high densities, the total number of crickets that could be produced was greatest at high densities. The authors therefore advocate for the higher densities despite increased mortality, representing a potential trade-off between economic productivity and animal welfare similar to the trade-off seen in many vertebrate mass production systems.

Importantly, in *G. assimilis*, increased stocking density did not affect survival (at least, across a range of 7.5 ml to 30 ml of crickets per 81 liters; Mazurkiewicz *et al.*, 2013), suggesting that each species of cricket should be studied independently to determine optimal stocking densities. There was not enough data to generate a recommendation for *G. sigillatus* at this time.

Concern 4. Macropterous morphs in closed facilities

High densities and temperature can affect cricket morphology. A higher incidence of wing shedding in *A. domesticus* (which peaks at around 20 days after eclosion) was observed to occur at low population densities (Chen *et al.*, 2019), consistent with the hypothesis that crickets retain wings to disperse from stressful crowded environments. In favorable conditions, crickets may shed wings in order to allocate resources into other fitness-relevant behaviors/tissues (e.g. wing shedding and subsequent histolysis of queen ants' flight muscles provides protein to support reproduction; Hölldobler and Wilson, 1990; Wheeler and Buck, 1996; Wheeler and Martinez, 1995).

Grylodes sigillatus also displays wing dimorphism, with short-winged flightless morph being the norm but long-winged morphs occurring in the field and laboratory settings (Sakaluk, 1997); however *G. assimilis* are reported to be wing monomorphic (long-winged only) in the field (and, presumably, on the lab/farm as well; Walker and Sivinski, 1986 and see 'Cricket Biology' section). The inability for macropterous individuals to disperse (e.g. behavioural restriction), as presented by the farmed context, may result in significant welfare challenge (in addition to the stressors they may be attempting to escape).

Recommendation 4a (domain 2: environment).

Monitor density and temperature to avoid producing macropterous individuals in wing-polymorphic species (*A. domesticus* and *G. sigillatus*)

Although there are not specific reports of the densities that tend to cause the development of macropterous individuals in *A. domesticus* and *G. sigillatus* at this time, producers should monitor the densities they use in their facilities and reduce densities if many adults are consistently macropterous. Further, in *G. sigillatus*, rearing juveniles at 35 °C is also more likely to cause growth of large wings as compared to 28 °C (Mathad and McFarlane, 1967). Producers should use lower rearing temperatures in this species to discourage macropterous morph generation.

Recommendation 4b (domain 4: behaviour). Provide opportunities for dispersal-like behaviours in *G. assimilis*

All *G. assimilis* are macropterous. Providing opportunities for dispersal-like behaviors (such as flight) could help mitigate the impact of a closed facility design or rearing pen on behavioral restriction. This may include

options like flight cages or increased open vertical space in adult rearing pens.

Concern 5. Aggression, injury, and cannibalism

Aggression can negatively impact welfare by causing injury and negative mental states. Aggressive behaviors documented in *A. domesticus*, *G. assimilis*, *G. sigillatus* and *G. bimaculatus* include: antennal fencing (opponents rapidly antennating their antennae), body rocking, leg kicks towards rival, threat posture (raising body onto forelegs), mandible flaring (hyperextension of mandibles), lunging, chasing, biting, and grappling (butting heads and/or interlocking mandibles and pushing), some of which may cause injuries (Adamo and Hoy, 1995; Albers and Reichert, 2022; Baringer and Hamilton, 2020; Bertram *et al.*, 2011). In the wild, *A. domesticus* males are known to fight over resources such as shelter (burrows), mating partners, and food (Nosil, 2002), and are observed to assume and defend territories in the form of hiding spaces (tubes) in laboratory settings (Kieruzel, 1976). Male *G. sigillatus* also aggressively defend territories and increase aggressiveness in the presence of females (Baringer and Hamilton, 2020). Fighting has also been observed to lead to higher levels of longer lasting aggression between winners of contests in *G. bimaculatus*, mediated by an increase in octopamine (Rillich and Stevenson, 2011).

Fighting is energetically costly for crickets. Agonistic behaviors in *A. domesticus* raise oxygen consumption six to eight times above resting levels, making them more energetically costly than walking, grooming, calling for mates, and courtship (Hack, 1997a). A single fight in *A. domesticus* male crickets uses on average less than 1% of the average daily energy budget (Hack, 1997a), but it is likely that farmed crickets will fight several times a day over resources such as females (Brown *et al.*, 2007; Montroy *et al.*, 2016), and territories (Rillich *et al.*, 2010), or establishing consistent dominance relationships (Hack, 1997b). Frequent intense competition of 52 fights/hour have been observed in *A. domesticus* under laboratory conditions (Alexander, 1961). This suggests a significant impact on energy budget over the cricket's lifespan.

In small enough populations, crickets may be able to recognize other individuals and establish stable dominance hierarchies, reducing aggression and energetically costly behaviours (Alexander 1961; Stevenson and Rillich, 2013). Crickets grouped together in individual mesh cages that allowed for visual, olfactory, mechanical and antennal contact but prevented fighting, were as aggressive as crickets initially housed solitarily, until

a dominance hierarchy was established and aggression declined within 10 minutes.

Alongside aggression, cannibalism is reported as a frequent behavior in many cricket species (Nakajima and Ogura, 2022), though both cannibalism and aggression are reported as rare in *G. sigillatus*, specifically, on farms (Mazurkiewicz *et al.*, 2013). Cricket producers (species not specified) in the UK have reported cannibalism as an issue on their farms (Bear, 2019), and some have observed larger *A. domesticus* cannibalize smaller conspecifics (Crocker and Hunter, 2018). However, it is presently unclear whether cannibalism is a cause of mortality, or simply a consequence of mortality caused by other factors, in *A. domesticus* and *G. assimilis*.

Recommendation 5a (domain 1: nutrition; domain 2: environment). Reduce fasting period length to a maximum of 24 hours (see concerns 1 & 2) and provide adequate food and water during rearing

Cricket producers in the UK report that during periods of starvation prior to harvesting, crickets “start eating each other” if the period is too great (Bear, 2019). A maximum fasting length of 24 hours is thus recommended to reduce cannibalism (and see concern 1).

Both juvenile and adult *G. assimilis* housed in laboratory settings engaged in cannibalism, even when food and water were provided (but only of dead or dying individuals; Ackert and Wadley, 1921; Masson *et al.*, 2020). The rate of cannibalism in *A. domesticus* was only marginally affected by dietary protein (Gutiérrez *et al.*, 2020), suggesting that protein may not be the main driver of cannibalistic behavior, though other nutrient deficiencies and protein do deserve further testing (see also Vaga *et al.*, 2020). Still, providing adequate food and water could contribute minorly to reducing cannibalism. For this reason, some commercial cricket rearing systems provide automated food and water dispensers, and minimize fecal contamination of food through separating housing/feeding/feces collection areas, in order to minimize cannibalism (Breedinginsects.com, n.d.)

Recommendation 5b (domain 2: environment; domain 3: health). Maintain stocking densities below the species-specific recommendations designed to avoid self-thinning behaviour (see concern 3) and aggression. Monitor levels of aggression and cannibalism

Aggression levels may increase when crickets are housed at very high densities. Olzer *et al.* (2019) suggest this could be due to increased population densities leading to increased perceived (but not actual) competition for

resources (such as burrows), and/or an increased ability to monopolize an abundant population of females. Chemical cues from females increase aggression in male *A. domesticus* (Buena and Walker, 2008; Otte and Cade, 1976), with variation in male aggression associated with dominance and mating status (Stevenson and Rillich, 2013). Dominance may also be increasingly difficult for males to establish or maintain in very large populations, increasing aggressive interactions. Therefore, increased competition for abundant females or other resources at high densities could lead to increased aggression, injury, and/or death.

In *A. domesticus*, males produce distinctive “aggressive songs” consisting of a brief sharp signal that can elicit fighting, although sometimes rival males retreat without fighting (Alexander, 1962). Aggressive songs have been found to signal aggressiveness and male ability to win in *A. domesticus* and *G. assimilis* (Bertram and Rook, 2012; Brown *et al.*, 2006), which may enable crickets to avoid energetically costly fighting by assessing the likelihood of defeating an opponent; with appropriate validation, these songs could also be used to monitor levels of aggression on farms.

Crowding increases the risk of cannibalism in some insect species (Cottrell and Yeargan, 1998; Modanu *et al.*, 2014; Van Buskirk, 1989). However, there is presently no scientific information available on the prevalence of cannibalism on cricket farms, or the impact of crowding on this behavior. Anecdotal reports of cannibalism in crickets (probably, *A. domesticus*) used as pet feed abound (e.g. Arachnoboard, 2021; Dizor, 2022), with some claims that cannibalism may relate to overcrowding (Anonymous, 2022). One cricket breeding company reports that “cannibalism is a significant problem when crickets are placed in high densities”. Producers should monitor cannibalistic behaviour and reduce their rearing densities as needed.

Recommendation 5c (domain 3: health). Monitor populations for live individuals being cannibalized (or otherwise grossly injured), which should be removed and humanely euthanize

It is unclear whether insects feel pain in response to mechanical injuries, such as cannibalism (Barrett, 2024; Eisemann *et al.*, 1984; Gibbons *et al.*, 2022). In the absence of conclusive data, producers are advised to act cautiously to preserve the welfare of their animals, and humanely euthanize individuals observed with severe injuries or being eaten alive. The removal of dead and dying conspecifics has other welfare benefits in reducing diseases spread via cannibalism (concern 6) and

reducing cricket stress responses to necromones (concern 14). We recognize that it may not be possible for producers to monitor each individual’s injury status across a farm; therefore, producers may want to monitor the overall cannibalism levels in bins – and, if cannibalism is found to be high, address the situation by euthanizing all animals in the bin or euthanizing only those being eaten/injured (if possible).

Gutiérrez *et al.* (2020) speculated that cannibalism may happen during the molting period when nymphs are soft-bodied and unable to defend themselves; it may therefore be prudent for producers to be especially vigilant during peak molting periods for their population.

Concern 6. Mortality or reductions in physical health due to disease or parasitism

Crickets’ interactions with microorganisms can often be detrimental; though not always lethal, many sublethal symptoms may still represent a threat to both the individual’s welfare and economic productivity.

Viruses have already posed a major welfare and economic productivity concern for farmed crickets, and therefore deserve special attention in this review. There are several viruses that have been reported, mostly in *A. domesticus* (Table 2). Many have been identified from production facilities around the globe or in laboratory screening trials. However, their isolation is infrequent and welfare-relevant information (prevalence, mortality, symptoms, etc.) are often not reported. The most studied are *A. domesticus* densovirus (AdDNV), cricket paralysis virus (CrPV), cricket iridovirus (CrIV), and *A. domesticus* iflavirus (AdIV).

AdDNV is the most well-known disease affecting farmed crickets, resulting in millions of dollars of losses due to multiple large-scale outbreaks in Europe and North America over 35 years (Styer and Hamm, 1991; Szelei *et al.*, 2011). AdDNV spreads via fecal-oral transmission or cannibalism (Weissman *et al.*, 2012) and can lead to high mortality (up to 100%) of farmed crickets (Liu *et al.*, 2011; Szelei *et al.*, 2011). Elimination of this virus from a facility and its cricket population is very difficult once it has been contaminated as the pathogen can survive on the cuticle of the insects for months without causing any apparent harm (Weissman *et al.*, 2012).

The virus affects the last three nymphal instars and/or the emerging adult. Infected adult females lived a maximum of 14 days, compared to 30–40 days for uninfected females (Liu *et al.*, 2011; Szelei *et al.*, 2011). The disease paralyzes crickets prior to death: the digestive tract does not contract, and they may lay on their backs, paralyzed, for several days before finally dying, report-

edly of septicemia (Maciel-Vergara *et al.*, 2021; Maciel-Vergara and Ros, 2017). Individuals can be observed with a swollen abdomen and liquified inner tissue (Maciel-Vergara *et al.*, 2021). Death is slow (Szelei *et al.*, 2011) and therefore of significant welfare concern.

Beyond mortality, sublethal symptoms include reduced body size and body fat content, fewer eggs laid, and reductions in jumping and general activity (Szelei *et al.*, 2011). This may reflect poor welfare in the form of diminished nutritional status due to difficulty in obtaining food, poor health, and reduced behavioral expression. *Acheta domesticus* appears to be much more susceptible to AddNV than *G. assimilis* or *G. sigillatus* (Weissman *et al.*, 2012).

CrPV is considered another major pathogen for farmed *A. domesticus* crickets alongside AddNV (Fernandez-Cassi *et al.*, 2019). This is potentially due to CrPV's significantly higher host range of more than 35 species including crickets and other orthopterans, meaning most farmed cricket species are likely to be susceptible. However, according to Maciel-Vergara *et al.* (2021) and Maciel-Vergara and Ros (2017), there are no reports of the impact of CrPV in commercial cricket operations.

CrIV has been described in all three species of crickets with both covert and symptomatic infections (Table 2). In symptomatic populations, CrIV can cause high mortality, frailty, sluggish behavior, and decreased fecundity. In symptomatic infections, insects often take on an iridescent sheen caused by the viral particles' structure (Duffield *et al.*, 2021). Some invertebrate iridoviruses, including lizard-cricket iridovirus (Liz-CrIV), are also known to cause diseases in reptiles and amphibians, representing a potential welfare and health concern for exotic animals kept as pets (Papp *et al.*, 2014; Stöhr *et al.*, 2016).

AdIV has been isolated from both insects and their frass, in both rearing facilities and wild populations (de Miranda *et al.*, 2021a,b). The virus causes deformation in the wings of infected juvenile *A. domesticus*, without causing any apparent effect on their ability to mate or survive as adults (Eilenberg *et al.*, 2015). Indeed, both the wild and commercial AdIV-infected crickets were reported to be without mortality or behavioural symptoms (de Miranda *et al.*, 2021b).

There is little study of pathogenic bacteria in farmed crickets outside product safety reports (e.g. for human consumption). *Escherichia coli* has been found to be pathogenic in *A. domesticus* under laboratory conditions and direct inoculation (Fernandez-Cassi *et al.*, 2019). Cricket mortality rates were 85% and 93.3% at

48- and 72-hours post-injection, respectively; loss of appetite led to weight loss and eventually death (Reginald *et al.*, 2021). *Gryllus assimilis* is able to detect *E. coli* contaminated feed and avoid it, suggesting individuals may recognize *E. coli* as a pathogen (Aleknavičius *et al.*, 2022).

Further, some strains of *Serratia liquefaciens* and *S. marcescens* can be lethal for *A. domesticus* crickets. Males experience greater mortality (42.7%) than females (3.7%) when infected with *Serratia liquefaciens*, although precise cause of mortality and other welfare impacts are not reported (Gray, 1998). Captive rearing conditions can change disease resistance in insect populations, likely through artificial selection, reduced genetic diversity and environmental conditions such as population density: for instance, *S. marcescens* proliferated faster in *G. sigillatus* under laboratory conditions (Letendre *et al.*, 2022). Finally, *Rickettsiella grylli* causes behavioral fever in *A. domesticus*, suggesting pathogenicity – however welfare impacts are undescribed (Adamo, 1998; Cordaux *et al.*, 2007).

Fungi can be common in insect farms, especially when humidity is high. Given that most entomopathogenic fungi are necrotrophic (requiring the death of the host in order to complete their life cycle), fungal infections are expected to be detrimental for insect welfare. Most entomopathogenic fungi infect the host by weakening and then penetrating the cuticle, with a variety of sublethal impacts on behavior and physiology prior to death. *Beauveria bassiana* and *Metarhizium anisopliae* are fungi that affect several orders of insects, including Orthoptera (Inglis *et al.*, 1996). Both species can cause behavioral fever in the hosts as a response to the infection (Adamo, 1998). *M. anisopliae* can infect *A. domesticus* in rearing facilities, causing mortality (Fernandez-Cassi *et al.*, 2019; Zimmermann, 2007). although the symptoms of infection and percentage of mortality are not reported (still, *Metarhizium* sp. infections are serious enough to warrant reporting by producers; Eilenberg *et al.*, 2015). The broad host tropism of this fungus suggests all farmed cricket species are likely susceptible. The control of fungal species can be costly, including sealing the facilities, cleaning, quarantine, and re-population.

Pathogenic protists (Gregarines) are unicellular eukaryotic parasites that infect the body cavities and tissues of larger animals, typically in a host-specific manner (Leander *et al.*, 2003). Sublethal infections of Gregarines are common in crickets, resulting in decreased body size, weight loss, reduced lifespan and fecundity, even if they are not immediately lethal to the individ-

TABLE 2 Viruses isolated from or tested on crickets, ordered alphabetically by viral family. For some viruses, there are still gaps in the information about host range, transmission and symptoms

Viral family	Virus species (abbreviation)	Species reported to be affected	Transmission	Pathology	References
Baculoviridae	<i>G. bimaculatus</i> nudivirus (GbNV) ¹	<i>G. bimaculatus</i> , <i>Gryllus campestris</i> , <i>Teleogryllus oceanicus</i> , and <i>Teleogryllus commodus</i>	Oral by ingestion of dead or moribund individuals	Decrease of body size, progressively uncoordinated and lethargic	Huger, 1985; Wang <i>et al.</i> , 2007
Dicistroviridae	Cricket paralysis virus (CrPV)	<i>A. domesticus</i> (broad host range in Orthoptera)	Unknown	Paralysis of the hind legs. Exact cause of death is not reported but is potentially thirst and starvation.	Eilenberg <i>et al.</i> , 2015; Maciel-Vergara <i>et al.</i> , 2021; Wilson <i>et al.</i> , 2000
Iflaviridae	<i>A. domesticus</i> Iflavirus (AdIV)	<i>A. domesticus</i>	Unknown	Deformation in the wings of infected juveniles, without causing any apparent effect on their ability to mate or survive	de Miranda <i>et al.</i> , 2021a, 2021b; Eilenberg <i>et al.</i> , 2015
Iridoviridae	Insect iridescent virus type 6 (IIV-6) ²	<i>A. domesticus</i>	Unknown	Lethal for young instars. The infection is manifested by hypertrophy and bluish iridescence of the affected fat body cells	Bertola and Mutinelli, 2021; de Miranda <i>et al.</i> , 2021b; Jakob <i>et al.</i> , 2002; Kleespies <i>et al.</i> , 1999; Papp and Marschang, 2019
Iridoviridae	Cricket iridovirus (CrIV)	<i>A. domesticus</i> , <i>G. assimilis</i> and <i>G. sigillatus</i>	Orally	High mortalities and reduction in fecundity; in <i>G. sigillatus</i> a strong putrid odor, abnormal hemolymph, sluggish behavior, increased cuticle and tissue frailty, underdeveloped/absent ovaries in some adult females	<i>A. domesticus</i> (Bertola and Mutinelli, 2021; Jakob <i>et al.</i> , 2002) <i>G. assimilis</i> (Jakob <i>et al.</i> , 2002) <i>G. sigillatus</i> (Duffield <i>et al.</i> , 2021; Oppert <i>et al.</i> , 2020)

TABLE 2 (Continued)

Viral family	Virus species (abbreviation)	Species reported to be affected	Transmission	Pathology	References
Parvoviridae	<i>A. domesticus</i> densovirus (AdDNV)	<i>A. domesticus</i> . <i>G. sigillatus</i> ³	Fecal-oral or cannibalism. Affects the last three nymphal instars and/or the emerging adult affected.	Prior to death: digestive tract fails to contract; crickets lay on their backs, paralyzed, for days before dying, reportedly of septicemia (swollen abdomen, liquified inner tissue). Sublethal symptoms include reduced body size and fat content, fewer eggs laid, reductions in general activity level	<i>A. domesticus</i> (Duffield <i>et al.</i> , 2021; Liu <i>et al.</i> , 2011; Maciel-Vergara <i>et al.</i> , 2021; Maciel-Vergara and Ros, 2017; Szelei <i>et al.</i> , 2011; Weissman <i>et al.</i> , 2012) <i>G. sigillatus</i> (Duffield <i>et al.</i> , 2021; Weissman <i>et al.</i> , 2012)
Parvoviridae	<i>A. domesticus</i> mini ambidensovirus (AdMADV)	<i>A. domesticus</i>	Unknown	Unknown	Pham <i>et al.</i> , 2013a
Not classified (possible new family or genus)	<i>A. domesticus</i> volvovirus (AdVVV)	<i>A. domesticus</i> and <i>G. assimilis</i>	Unknown	Unknown	<i>A. domesticus</i> (Fernandez-Cassi <i>et al.</i> , 2019; Pham <i>et al.</i> , 2013a) <i>G. assimilis</i> (Pham <i>et al.</i> , 2013b)
Not classified	bait-cricket virus (BCV)	<i>A. domesticus</i>	Unknown	Unknown	Valles <i>et al.</i> , 2016
Not classified	bait-cricket virus (BCV)	<i>A. domesticus</i>	Unknown	Unknown	Valles <i>et al.</i> , 2016
Not classified yet. Possibly a new family. Proposed name by authors: Solinviviridae	<i>Solenopsis invicta</i> virus 3 (SINV-3)	<i>S. invicta</i> - <i>A. domesticus</i> (only carried the virus but not affected by it)	Oral by ingestion of dead or moribund individual (crickets consumed ants)	In ants: alters worker behavior, preventing distribution of solid food to larvae, resulting in larval mortality and impaired queen health.	Bertola and Mutinelli, 2021; Porter <i>et al.</i> , 2016; Valles <i>et al.</i> , 2016

- 1 This virus had been isolated from both *G. bimaculatus* and reptilian hosts. Further data is needed to assess if this virus could represent a health threat to reptiles kept as pets, which are often fed live crickets.
- 2 The virus has not been reported as affecting *A. domesticus*, *G. assimilis* or *G. sigillatus* yet. But given the record of other species of crickets that are affected, it may also be able to infect the three species we focus on in this report.
- 3 This species can be infected but no reported mortality by breeders so far.

uals (Adamo, 1998; Reyes Villanueva, 2004; though see Oppert *et al.*, 2020) suggesting reduced health and welfare.

Beyond disease agents, crickets may also be affected by a variety of nematoidan (nematoda or nematomorpha) parasites. Entomopathogenic nematodes are microscopic, wormlike animals that parasitize insects, generally with a broad host range. As necrotrophic parasites, there is a high likelihood of mortality for infected insects (Agrios, 2009). Most entomopathogenic nematodes carry bacterial symbionts. In the case of family Heterorhabditidae and Steinernematidae, they carry *Photorhabdus* and *Xenorhabdus*, respectively; these symbionts are generally the agents responsible for cricket mortality following nematode infection (da Silva *et al.*, 2000; Eilenberg *et al.*, 2015; Li *et al.*, 2007; Shapiro-Ilan *et al.*, 2009).

Nematodes of these two families may enter through the anus, mouth, spiracles, or (in the case of Heterorhabditidae) breaches in the cuticle. The bacterial symbiont is released into the hemolymph where they consume host resources to replicate (Ciche, 2007; Ferreira and Malan, 2014). Alongside resource consumption, the release of bacterial toxins results in poisoning, septicemia, and death of the insect within a few days (Cranshaw and Zimmerman, 2013). The nematode can then finish its lifecycle, and spread to other hosts after the body wall of the dead insect ruptures. Nematode infection represents a welfare concern through reduced health (domain 3) and a slow death.

It is demonstrated that *A. domesticus* show resistance to *Steinernema scarabaei*, *S. feltiae* and *H. bacteriophora* nematodes by successful recognition and encapsulation of the parasites (>94%) by the immune system. In the lab, *A. domesticus* can act as an intermediate host for the nematode *Abbreviata antarctica*, representing a potential concern for lizards fed crickets (King *et al.*, 2013). Finally, *Gryllus assimilis* may host *Cephalobium microbivorum* nematodes. Although information on this nematode species is scarce, it has been reported that large numbers inside the cricket can be detrimental to the individual (Ackert and Wadley, 1921) because infected males produce smaller spermatophores, however the research does not clarify the effects of the smaller spermatophores in the mating process (Luong *et al.*, 2005). The number of nematodes needed to cause an adverse effect in the crickets is not clear; papers report from one to 133 nematodes without giving details of mortality or other welfare effects (Ackert and Wadley, 1921; Luong *et al.*, 2005).

Finally, the hairworm species *Paragordius varius* has been reported to affect some *Gryllus* spp. and *A. domesticus*. *P. varius* has a freshwater, free-living adult stage; development occurs inside the body cavity of the definitive cricket host, where the hairworms consume host resources and reduce host growth (Anaya and Bolek, 2021), thus, representing a threat to welfare. The adult hairworm then manipulates its host to search for shallow water, allowing them to break through the cricket's body wall and enter the water to complete their life cycle; this often results in the death of the cricket via drowning (Poinar and Weissman, 2004).

Encounters with *P. varius* do not guarantee infection: only 31.6% of exposed crickets were infected in lab trials. Infected crickets can survive releasing their hairworms, however their longevity is reduced compared to controls (73 vs 86 days). Furthermore, only 50% of infected female crickets that survived were able to produce eggs compared to control crickets (Anaya and Bolek, 2021). Other hairworms can also attack crickets. For instance, *Gordius robustus* is commonly reported in the family Tettigonidae, and under laboratory conditions can also infect species of the genus *Gryllus* (Poinar and Weissman, 2004).

Given all the welfare and economic productivity concerns that have already manifested with pathogenic microbes and parasites, producers should take population health very seriously.

Recommendation 6a (domain 3: health). Follow best practice guidelines for hygiene for each facility type, which include regular disease monitoring and management protocols

The prevention and management of disease agents and parasites are a critical part of protecting animal welfare and economic productivity in cricket farms. Importantly, many diseases of farmed insects likely remain uncharacterized and under-reported – especially those that cause sub-lethal, chronic infections. These infections can become lethal at scale when exacerbated by some additional stressor, therefore regular monitoring for any signs of diseases (such as lethargy, color changes, bloating, paralysis, reduced food consumption, mortality, reduced growth or fecundity, etc.; Table 2 'Pathology' column) is an important part of facility management.

In response to massive, economically costly outbreaks, producers and academics have already begun to develop many best practices for managing disease, for instance: Eilenberg *et al.* (2015, 2018), Joosten *et al.* (2020), and the International Platform of Insects as Food and Feed (2022). Best practices will partially

depend on facility design for crickets. For instance, if the facility is closed, the primary risk of pathogen introduction is through the feed or oviposition substrates; if the facility is semi-open/open, pathogens can be introduced by feed, oviposition substrates, air, rain, or other insects entering facilities.

In both cases, after introduction, spread may be facilitated by contaminated equipment, rearing conditions, and high stocking densities. In the first case, sterilizing feed and substrates may go a long way to prevent outbreaks. In the second case, much more active facility management may be required to avoid disease. Other practices for managing diseases include: frequent equipment sterilization, quarantine for symptomatic or newly introduced individuals, regularly cleaning dead insects/detritus/debris, and appropriate training of staff to maintain high hygiene standards as well as monitor for disease.

Recommendation 6b (domain 2: environment; domain 3: health). *Keep humidity low (50% RH for adult *A. domesticus*, 70% for juveniles) and refresh damp oviposition substrates every 24 hours to minimize pathogen development*

Too high humidity is known to cause mortality in several cricket species (Miech *et al.*, 2016), potentially due to increasing fungal pathogen or nematode proliferation. Given the generalizability of the relationship between high humidity and mortality across many Orthopterans, higher humidity is only recommended for the earliest instars, with only 50% humidity for later instars and adults (though species-specific data is needed for *G. assimilis* and *G. sigillatus*). Nematodes are often sensitive to humidity for survival (Shapiro-Ilan *et al.*, 2009), allowing producers to reduce risks of nematode infection through careful control of abiotic conditions (Li *et al.*, 2007).

There is no published literature on the optimal levels of oviposition substrate moisture in any of the three focal species. Some substrates may initially seem preferable because they retain moisture for longer, and could thus reduce labor associated with refreshing substrates (e.g. peat moss retention is better than sand; Clifford, 1985). However, a moist substrate risks potential fungal proliferation (see 'Interspecific interactions') and consequential adverse effects on health/welfare; refreshing the oviposition substrate every 24 hours can minimize this risk (Bascuñán-García *et al.*, 2010).

Recommendation 6c (domain 2: environment; domain 3: health). *Keep stocking densities low to help prevent or manage diseases, especially AdDNV or diseases that spread via cannibalism*

High stocking densities can increase the likelihood of outbreaks by exacerbating spread through mechanisms like fecal-oral transmission and cannibalism, allowing disease to proliferate in a large, confined population. Managing stocking densities has been an especially important factor in controlling cricket diseases in the past (Eilenberg *et al.*, 2015). As there are no AdDNV-resistant *A. domesticus* strains, producers in Germany and America have reported successfully rearing this species without AdDNV outbreaks by reducing stocking densities (and avoiding cricket importation to reduce the likelihood of bringing the disease into uncontaminated rearing facilities; Szelei *et al.*, 2011). Managing stocking densities proactively may also help prevent or manage future diseases in the industry.

Recommendation 6d (domain 3: health). *In cases of untreatable disease, crickets should be isolated, humanely depopulated, and disposed of using appropriate biosafety protocols as soon as possible*

Producers will need to make determinations about the health of their population; to avoid prolonged suffering, producers should isolate and humanely depopulate (see concern 7) their animals as soon as possible in cases of untreatable disease.

Additionally, diseased crickets – especially in open/semi-open facility designs – pose a danger to the welfare of wild insect species. Escaped individuals, or untreated contaminated materials, can transmit disease to naive wild populations, minimally presenting a danger to small numbers of wild individuals – but possibly resulting in outbreaks that could further insect biodiversity declines. Traditional vertebrate livestock have both suffered from, and contributed to the spread of, diseases in wild populations (e.g. the 2022/2023 avian flu crisis in farmed poultry and wild birds; Klaassen and Wille, 2023). Appropriate biosecurity measures for containing insects within facilities and preventing entry from wild insects, as well as disposal protocols for contaminated materials/animals, will help reduce the risk of wild-farmed disease transmission in both directions.

Concern 7. Inhumane slaughter or depopulation

Barrett *et al.* (2022, 2023) described in-depth the lethal progression of different slaughter methods used for insect larvae, and rank each method as more or less likely to be humane. In addition, the authors cover the

effects of the current anesthetic options for insects (Barrett *et al.*, 2022), and the possibility of electrical stunning (Barrett *et al.*, 2023). However, the information in their reviews on yellow mealworm and black soldier fly larvae may not translate in all cases when applied to larger-bodied, adult crickets, due to their increased body mass and mobility. Therefore, in this review we briefly cover the relevant information from these prior works, with a slightly extended discussion where results may differ for crickets.

Insects currently sit outside legislation stipulating humane slaughter for vertebrate livestock (Bear, 2021). No regulations, standard operating procedures (SOPs) and very little scientific literature exist to guide producers in implementing humane slaughter of insects (Bear, 2019). Humane slaughter occurs when death is instantaneous (no time to register any pain or distress) or the animal is rendered unconscious (i.e. anesthetized or “stunned”) until death ensues (so they are unable to feel any pain or distress; RSPCA, 2021). There are no legal definitions for an instantaneous death – but it could reasonably be taken to mean within one second.

As discussed in the ‘Industry Practices’ section, reported slaughtering methods for crickets are boiling (Reverberi, 2020), immersion in pressurized steam (Tatarova, 2017), immersion in hot, non-boiling water (e.g. 60 °C; Vandeweyer *et al.*, 2018), drowning in non-boiling water (Fernandez-Cassi *et al.*, 2019; Miech, 2018), freezing in freezers (i.e. in air, 24 hours at –18 °C; Bear, 2019; Fernandez-Cassi *et al.*, 2019), heating (presumably in air; Fernandez-Cassi *et al.*, 2019), shredding (Bear, 2019), and asphyxiation via carbon dioxide (Singh *et al.*, 2020). Boiling is reported to be the most common method of cricket slaughter in Thailand (Reverberi, 2020), compared to freezing in air being most common for mass-produced crickets in Europe and North America (van Huis *et al.*, 2013; Reverberi, 2020).

Anesthesia and electrical stunning are both ways to inactivate an animal’s nervous system prior to slaughter and may be used to make the process of slaughter/depopulation more humane. For crickets, cold and, less commonly, carbon dioxide are the only forms of anesthesia reported to be used by some producers, in the absence of scientific guidelines for insect welfare during slaughter (Bear, 2019).

Improved insect welfare during slaughter may sometimes be linked to improved product quality. For example, Caligiani *et al.* (2019) report that longer killing times are associated with an increase in the metabolism of energy reserves (metabolism of triglycerides into acylglycerol and free fatty acids) in black soldier fly larvae,

presumably due to stress, which promotes oxidation and in turn reduces product quality.

Recommendation 7a (domain 3: health). Develop standard operating procedures (SOPs) for slaughter and depopulation that are instantaneous

Blanching (or boiling), shredding, and freezing in liquid nitrogen have thus far generally been recommended as the most humane methods for killing insects due to their rapidity (Bear, 2019; IPIFF, 2019; Barrett *et al.*, 2022; Erens *et al.*, 2012; Sindermann *et al.*, 2021; Spranghers *et al.*, 2021); however, further research is needed to determine whether these methods produce an instantaneous (and therefore humane) death. For non-instantaneous methods, research is needed on both the duration of the slaughter process and severity of suffering inflicted, which together comprise the determine the degree of animal welfare compromise (e.g. Rioja-Lang *et al.*, 2020).

Consideration of crickets’ high mobility and, depending on wing morph, ability to fly should be taken into account when designing a humane slaughter process, so that a state of distress is not prolonged through movement away from the source of slaughter. Maintaining an instantaneous SOP can be achieved by confining individuals or via anesthetizing/stunning.

Recommendation 7b (domain 3: health). At this time, avoid freezing in air as a method of anesthesia, depopulation, or slaughter. Freezing should never be used prior to heating

Freezing in air has been anecdotally reported as a humane slaughter method in the literature, as it is expected to anesthetise the insect prior to death; importantly, however, because cold affects the ability of the muscles to function, it is not known for many insects at what temperature the body is immobilised vs the nervous system is shut down (actual anesthesia). For instance, when freezing was applied to three species of cockroaches, movement ceases prior to the cessation of electrical activity in the intact nervous system (Anderson and Mutchmor, 1968). However, the cessation of movement and electrical activity in the nervous system may be more closely coordinated in other insect species (e.g. locusts; Srithiphaphirom *et al.*, 2019); more research is required for the focal species of this review.

Further, recent literature has challenged the claim that the process of freezing itself is humane (e.g. Barrett *et al.*, 2022, 2023) given that some insects have cold nociceptors (Turner *et al.*, 2016). Further, extreme cold has been shown to result in fitness deficits, providing an

adaptive advantage to insects that experience the stimulus as noxious (as in the case of high heat). For instance, exposure of male *A. domesticus* crickets to 0 °C for 6 h resulted in less successful mating, even one week after exposure (Chipchase *et al.*, 2021).

Behavioural evidence also supports the idea that cold may not provide a humane experience for insects. Crickets, and many other insects, are observed to attempt to escape cold areas prior to succumbing to chill coma (Barrett, pers. comm.). Indeed, one UK cricket producer reported using CO₂ to anaesthetise crickets as they believed cold anaesthesia was likely to be “stressful” for the insects, because “crickets will go out of their way to find somewhere warm, and they get quite distressed if they’re cold” (Bear, 2019). Preference data is a useful welfare indicator (Dawkins, 2008), with avoidance suggesting a stimulus that is not conducive to good welfare. Accordingly, invertebrate veterinarians have recommended the use of an inhalation anaesthetic prior to any freezing-based slaughter/depopulation methods (Murray, 2012).

Cold anaesthesia is reportedly achieved in the industry by placing containers of crickets in the fridge (at or below 4 °C) and then freezer (−18 °C), or directly into the freezer (Bear, 2019). Chilled insects will eventually reach the onset of a reversible “chill coma” (critical thermal minimum, or CT_{min}), characterized by immobility and unresponsiveness (MacMillan and Sinclair, 2011). *Acheta domesticus* CT_{min} occurs at body temperatures below 10 °C (see Figure 3), although acclimation may shift this as much as 2 °C (Lachenicht *et al.*, 2010; Morales-Ramos *et al.*, 2018). It is unclear, however, whether this immobility represents a loss of consciousness as, for instance, the body temperature where tactile information is no longer transmitted from the body to the brain is approximately 4 °C in *Acheta domesticus* (see Figure 3; Morrissey and Edwards, 1979).

Further, although chill coma shuts down the CNS and therefore represents anaesthesia, the experience leading up to loss of responsiveness may involve some degree of suffering. Chill coma onset occurs when *body* temperature reaches CT_{min}, not *ambient* temperature. No empirical work has been conducted to establish time to reach CT_{min} under industry slaughter conditions, but it is likely to take at least several seconds to minutes in the best of cases (Barrett *et al.*, 2022). One UK cricket producer reports freezing crickets to take a couple of hours until death (Bear, 2019). During this time, the insect may be experiencing negative affective states associated with nociceptive cold.

Irrespective of suffering caused by the process of freezing, the reversible nature of chill coma means that cold anaesthesia is unsuitable for use prior to heat-based slaughter methods. Insects may regain CNS functioning upon heating, and due to a lower starting temperature, time to death will be prolonged.

Recommendation 7c (domain 3: health). For methods that are not instant but must still be used, implement stunning/anaesthesia protocols prior to slaughter/depopulation

For product safety, quality, or characteristic reasons, or due to workflow/equipment constraints, producers may want to use slaughter/depopulation methods that will never be instantaneous. For instance, heating in air or immersion in non-boiling water are not likely to be instant under most, mass-rearing SOPs. Producers intent on using these methods should implement stunning or anaesthesia protocols prior to slaughter or depopulation.

Electrical stunning can incapacitate the nervous system within a second. As this instantaneous insensibility increases the humaneness of slaughter, electrical stunning is widely used prior to the slaughter of vertebrate livestock (e.g. chickens; Kettlewell and Hallworth, 1990) and may also be used for invertebrate livestock (e.g. decapod crustaceans; Conte *et al.*, 2021). As described in Barrett *et al.* (2023) no studies have tested electrical stunning for farmed insects, though electrical stunning is known to incapacitate insect nervous systems at smaller scales (Harris, 2016; Packer and Brady, 1990). Many design considerations are important for mass-/multi-animal stunning systems to ensure uniformity across animals (Kettlewell and Hallworth, 1990); designs for crickets should take into account body size variation, ungrounded movement capabilities (flight), as well as the number of insects to be stunned.

Prolonged stunning of invertebrates can also be used to slaughter them (Conte *et al.*, 2021), though studies would be needed to ascertain how consistently insects were slaughtered using particular stunning equipment designs and protocols. If electrical stunning could be used to reliably and humanely kill insects, the typical slaughter methods listed above would become post-slaughter processing methods instead, making their humaneness irrelevant. Importantly, producers should not attempt to implement electrical stunning in their facilities until equipment designed for use with their animals has been designed and demonstrated empirically to be effective at scale.

Cold anesthesia has reportedly been used for crickets (but, as described above, cold may not be a humane anesthetic choice). Continuous exposure to CO₂ is another commonly used anesthetic for insects (Williams, 1946; Poinapen *et al.*, 2017). However, carbon dioxide is known to cause distress in vertebrate animals (e.g. Sindhoj, 2021; Underwood and Anthony, 2020). Immersion of conscious animals into 100% CO₂ is deemed unacceptable for vertebrate animals, while gradual exposure to increasing concentrations is expected to be less likely to cause distress (Underwood and Anthony, 2020). The European Food Safety Authority (2004) has concluded that CO₂ stunning of vertebrate livestock is not conducive to good animal welfare. It is unknown whether CO₂ also causes suffering in insects, but many veterinary specialists do not recommend using CO₂ as a humane method of insect anesthesia (Gooley and Gooley, 2021; Gunkel and Lewbart, 2007; Murray, 2012). Still, prior to the development of other insect anesthetics (such as halogenated ethers), CO₂ or cold may be producers only options for trying to incapacitate their crickets prior to slaughter (Gooley and Gooley, 2021; MacMillan *et al.*, 2017).

Concern 8. Lack of climate control in semi-open rearing facilities or inappropriate climate control in closed rearing facilities

Temperature is a key condition in the environmental domain (domain 2) of animal welfare (Mellor, 2015). As is the case for most insects, *A. domesticus*, *G. assimilis*, and *G. sigillatus* are poikilotherms, meaning their body temperatures are close to, and highly influenced by, environmental temperatures. Because temperature is an environmental parameter that affects a multitude of physiological processes in an insect's body (e.g. water loss, metabolism, oxygen and CO₂ transport), there are many important interactions that cannot be covered in their entirety here. For instance, limiting an insect's water intake lowers the maximum temperature it can tolerate (Johnson and Stahlschmidt, 2020). The recommendations we provide in this section are therefore more general guidelines and assume that nutritional, hydration, and housing needs have been simultaneously met.

To ensure good welfare, it is important to consider both acute and sustained effects of ambient temperatures for all life stages and behaviours. Semi-open rearing facilities lack climate control, which may cause crickets to experience (1) extreme high or low temperatures; (2) significant temporal variation in temperatures; or (3) too high or low humidities; all of these may

negatively welfare. For instance, in the wild, *G. assimilis* populations are negatively correlated with relative humidity (range 54–80%; Khan *et al.*, 2011), suggesting that open/semi-open facilities lacking climate control may often have too-high humidity for optimal welfare as they experience ambient humidity (Orinda *et al.*, 2017). Further, if closed facilities do not regulate their temperatures and relative humidity carefully, they may face similar welfare challenges as semi-open facilities.

Chronic exposure to high temperatures during rearing can decrease developmental time (Clifford and Woodring, 1990), generally at the cost of limiting maximum adult size and/or increased mortality risk. Although nymphal survival to adulthood appears to be similarly high in *A. domesticus* (76–80%) when reared at 28, 33, and 35 °C (Ghouri and McFarlane, 1958), at least one study reported that adult *A. domesticus* housed at 33 °C experienced more than twice as much mortality (60–70%) as those kept at 25 or 29 °C (Lachenicht *et al.*, 2010). Further, there appears to be no significant gains in metabolic efficiency, or growth efficiency, when increasing rearing temperature above 30 °C (Clifford and Woodring, 1990).

Optimal rearing temperatures for *G. sigillatus* are similar to those of *A. domesticus* (Figure 3), with temperatures of 28 to 35 °C yielding more than 75% survival to adulthood and highest average survival (87%) observed at 33 °C (Ghouri and McFarlane, 1958). Of that subset of temperatures, 33 to 35 °C supports fastest maturation (Ghouri and McFarlane, 1958) which could improve productivity without compromising welfare (though further research is needed on both counts).

Adult *G. assimilis* appear to have greater capacity to survive in consistently hot environments than *A. domesticus* (Figure 3). Adult male *G. assimilis* housed at temperatures up to 34 °C experienced little to no mortality after four days, while females survived well at even higher temperatures (Centeno Filho *et al.*, 2022, 2023). *G. assimilis* nymphs are reported to be reared successfully at 26 ± 4 °C (Whattam and Bertram, 2011), 26 ± 1 °C (Limberger *et al.*, 2021), and 27 ± 1 °C (Centeno Filho *et al.*, 2022, 2023). Far cooler temperatures in *G. assimilis* appear to be optimal for oviposition (Figure 3). Recent work showed the greatest numbers of eggs laid per female between 25 and 29 °C, with oviposition environments as cold as 20 °C or as warm as 34 °C yielding significant declines in oviposition (Centeno Filho *et al.*, 2023; Hermansa *et al.*, 2022).

There is evidence that limiting temporal fluctuations in temperature can provide a less stressful thermal rearing environment. A recent study found significantly ele-

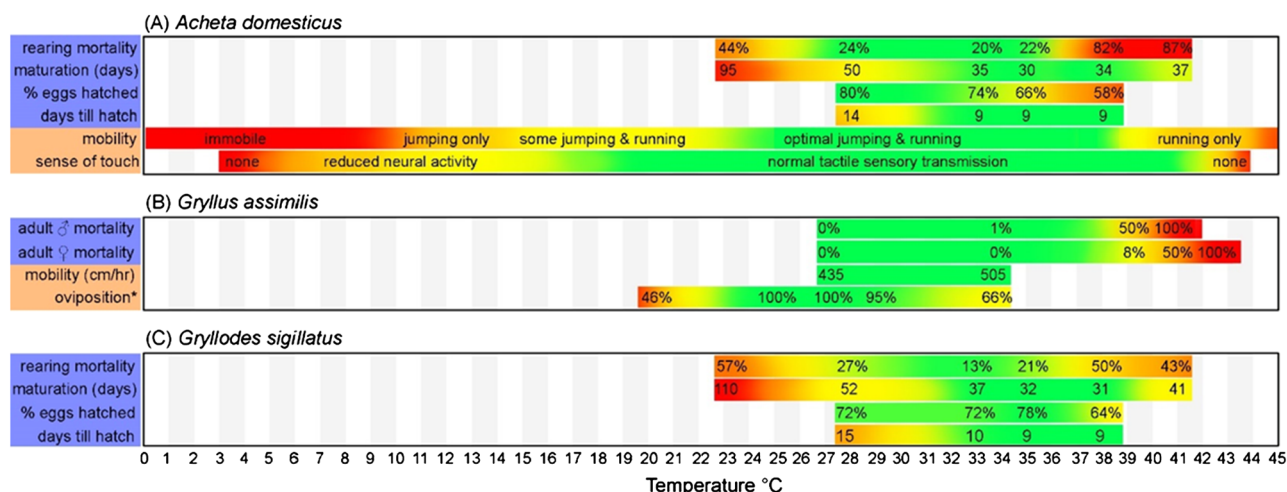


FIGURE 3 Summary of literature on the effects of long-term (blue) and short-term (peach) thermal exposure on performance of three cricket species. The color gradient from green to red indicates relative low impairment to high impairment (respectively) in each performance metric. (A) Effects of temperature on rearing mortality (Ghouri and McFarlane, 1958), maturation time (Ghouri and McFarlane, 1958), egg hatching success (McFarlane *et al.*, 1959), mobility (Morales-Ramos *et al.*, 2018), and tactile sensory transmission (Morrissey and Edwards, 1979) in *A. domesticus*. (B) Effects of temperature on 4-day adult mortality (Centeno Filho *et al.*, 2022, 2023), mobility (Centeno Filho *et al.*, 2022), and oviposition *shown as % relative to highest study values (Centeno Filho *et al.*, 2023; Hermansa *et al.*, 2022) in *G. assimilis*. (C) Effects of temperature on rearing mortality (Ghouri and McFarlane, 1958), time to maturation (Ghouri and McFarlane, 1958), egg hatching success (McFarlane *et al.*, 1959), and egg incubation time (McFarlane *et al.*, 1959) in *G. sigillatus*.

vated mortality rates in adult male *G. assimilis* housed in environments fluctuating from 27 to 34 °C (8.9% mortality after 4 days) compared to males of the same species kept at constant temperatures of either 27 °C (0% mortality) or 34 °C (1.43% mortality) (Centeno Filho *et al.*, 2022; females were less affected: Centeno Filho *et al.*, 2023). These data suggest open rearing facilities in areas with highly variable climates, across hours or days, could be particularly stressful for *G. assimilis*.

Finally, despite many records of the 'right' RH for rearing crickets, very few studies actually directly test the effects of this variable, meaning that very little information is available on optimal humidities for cricket welfare. As with other animals, proper hydration has positive effects on cricket growth, reproductive output, and survival. Restricting water access to 4 hours a day significantly reduced growth rate (0.8 mg/day) and survival (27%) compared to 24-hour access (2.2 mg/day and 61% survival; McCluney and Date, 2008). *Gryllodes sigillatus* females given *ad libitum* access to water produce significantly more nymphs than females subjected to water stress (Ivy *et al.*, 1999). For this reason, crickets are typically given *ad libitum* access to a hydration source (Gangwere, 1960); however, very low RHs (perhaps those below 40-50%) can still drive desiccation stress even with access to water.

Recommendation 8a (domain 2: environment).

Crickets should be reared in closed, climate-controlled facilities wherever natural temperatures exceed maximum rearing temperatures in Table 3.

Temperatures should be kept close to the recommended rearing temperatures listed in Table 3.

Recommendation 8b (domain 2: environment; domain 4: behaviour). Limiting temperature fluctuations over time while providing constant spatial thermal gradients from 27-32/33 °C could improve welfare for adult *G. assimilis*

As described above, fluctuations in temperature over time can reduce survival in adult *G. assimilis*. Although temporal variation in temperature can be stressful, providing a simultaneous gradient of temperatures across the rearing environment enables individuals to behaviourally thermoregulate. Opportunities to behaviourally thermoregulate can be important for animal health: crickets infected with bacterial or fungal pathogens may move into hotter areas to attempt to kill the pathogen (Eilenberg *et al.*, 2015). Providing thermal gradients in the rearing containers to allow crickets to 'self-medicate' in this way (e.g. as has been suggested for fish, also poikilotherms; Huntingford *et al.*, 2020) and could also provide an easy way to assess poor health/welfare by observing the prevalence of crickets in 'hotter' areas of their environment. Finally, this allows individuals

TABLE 3 Maximum, minimum, and recommended rearing temperature ranges for farmed crickets

Species	Life stage	Minimum rearing temperature	Maximum rearing temperature	Recommended rearing temperatures
<i>Acheta domesticus</i>	juvenile	28 °C	35 °C	32-33 °C
<i>Acheta domesticus</i>	adult	25-29 °C	32 °C	27-29 °C
<i>Gryllobates sigillatus</i>	juvenile	28 °C	36 °C	33 °C
<i>Gryllobates sigillatus</i>	adult	n.d.	34 °C	27-31 °C
<i>Gryllus assimilis</i>	juvenile	n.d.	n.d.	26-27 °C (known range; but greater range likely)
<i>Gryllus assimilis</i>	adult	n.d.	>34 °C	27-33 °C

*n.d. = not enough data to make a recommendation.

to exercise agency and choose the environment they want, also conducive to good welfare (Dawkins, 2008; Špinka and Wemelsfelder, 2018). It is likely that providing access to thermal gradients would be beneficial to crickets of all species and life stages – however, further research is required to extend this recommendation beyond adult *G. assimilis*.

Recommendation 8c (domain 1: nutrition; domain 2: environment). *RH should be kept high enough that there is no observable clustering of crickets around water sources*

Clustering of crickets around water sources may also occur due to desiccation stress at lower humidities. The lowest humidity at which no clustering of crickets around a water source was observed was 50% RH (Ghouri and McFarlane, 1958b), though (Attard, 2015) found no ‘ill effects’ at humidities lower than 20-40%. Observing crickets for clustering behavior can inform producers that they must adjust either the number of provided water sources (to keep them accessible to all individuals) or the RH of their rearing facilities.

Concern 9. Inappropriate climate control during live transport

Another important welfare consideration is shipping temperature. Due to the asymmetry of thermal performance and the underlying temperature-dependent physiological processes that govern it, sending an animal into heat coma (exceeding CT_{max}) often incurs greater irreparable damage than sending an animal into cold coma (dropping below CT_{min} ; Angilletta Jr., 2009). Although surviving heat coma is possible and surviving cold coma is common (Morrisey and Edwards, 1979), temperatures high or low enough to render immobility should be avoided due to the risk of both acute and long-

term health consequences (Chipchase *et al.*, 2021) and the welfare insults of constraining behavior and subjecting animals to the discomfort/pain of an aversive temperature.

Recommendation 9a (domain 2: environment; domain 4: behaviour): *aim for shipping temperatures slightly lower than those we report here as recommended for rearing. Temperatures should always be kept below the maximum rearing temperature and above CT_{min}*

It is common to ship insects in containers with cool packs or warm packs depending on region and time of year, targeting temperatures slightly below typical rearing temperatures (Enkerlin and Quinlan, 2002; Casada *et al.*, 2007). Although information on shipping microclimates for farmed crickets is not available, producers may currently aim for temperatures just below target rearing or housing temperatures (Table 3), in order to avoid the significant consequences of heat coma.

Chilled insects will eventually reach the onset of a reversible “chill coma” (critical thermal minimum, or CT_{min}), characterized by immobility and unresponsiveness (MacMillan and Sinclair, 2011) – this should be avoided. *Acheta domesticus* are generally rendered immobile at body temperatures below 10 °C, although acclimation may shift their critical temperatures as much as 2 °C (Lachenicht *et al.*, 2010; Morales-Ramos *et al.*, 2018). As there are no CT_{min} estimates for the other focal species, we recommend staying above 16 °C at all times as a buffer. In cases of significant chilling, crickets will experience normal to slightly impaired mobility (Morales-Ramos *et al.*, 2018), which may also impact welfare by behavioural restriction (domain 4) due to an imposed suboptimal environment (domain 2).

Small temperature loggers can be packed into shipped insect containers to check shipping tempera-

ture fluctuations (Enkerlin and Quinlan, 2002; Casada *et al.*, 2007). Two commonly used devices for this purpose are HOBO® Temperature Loggers (Onset Computer Corporation, Bourne, MA, USA) and iButtons (Maxim Integrated, San Jose, CA, USA). These data may help producers adjust their shipping methodologies accordingly.

To avoid exposure to adverse microclimates within the shipping container, cooling packs or heating packs should not come into direct contact with insects, but rather should be separated by some form of air gap such as packing foam (Casada *et al.*, 2007).

Concern 10. Stress associated with vibrations during transport

Vibrations induce anti-predator behaviors in crickets (Dambach, 1989). Octopamine levels increased in *Gryllus texensis* crickets subjected to vibrations in their rearing container (Adamo and Baker, 2011). Provisioning cricket rearing environments with hiding places *may* mitigate these effects – in a subsequent study, crickets exposed to the same mock predator spent more time under a cardboard shelter compared to controls (Adamo *et al.*, 2013). Repeated exposure to the mock vibrating predator increased basal octopamine levels (Adamo and Baker, 2011), similar to the effects of chronic stress on the basal levels of vertebrate stress hormones.

Exposure to vibrations during transportation is likely to be a welfare concern, potentially contributing to high mortality. Producers report that live transport, generally used when shipping crickets as pet feed, frequently causes mortality: “even when packed with care, the physical stress of travel can shorten the overall lifespan of the crickets” (Josh’s Frogs, n.d.; Barrett, pers. comm.). Mazurkiewicz *et al.* (2013) claim that crickets “are usually weak and soon die of exhaustion [from] conditions of transportation” when imported. This phenomenon may be called ‘shipping sickness’ in the industry and producers may even include extra crickets to cover those expected to die in the delivery process (e.g. 20% extra provided by Cricket King, n.d.). Given that most motorized transportation methods will include significant vibrations, crickets may experience significant stress during transport.

Recommendation 10a (domain 2: environment).

Minimize frequency/duration of vibrations through changes in transport method or packing materials

Adding padding that minimizes vibrations experienced by crickets in transport, or speeding up shipping options to reduce the duration of crickets’ experience of vibrations, may reduce stress and improve welfare. For

instance, producers may consider shipping insects by air with appropriate climate support, to reduce the frequency and duration of major bumps and vibrations experienced.

Concern 11. Poor ventilation leading to hypoxia or CO₂ accumulation

Farmed insects produce carbon dioxide (CO₂), ammonia (NH₃), methane (CH₄), and nitrous oxide (N₂O) gasses as a result of excretion and metabolism. When proper ventilation is not provided at high densities, these gasses may accumulate and cause hypoxic conditions (e.g. oxygen levels below normoxia, ~21%). Hypoxic conditions can reduce growth rates, increase mortality, induce desiccation stress by causing insect spiracles to remain open, change behaviors, and more (e.g. Cancino and López-Arriaga, 2016; Klok and Harrison, 2009; Nicolas and Sillans, 1989).

Hypoxic conditions can be especially detrimental to juveniles that are close to molting into the next instar. This is because growth in body mass (and thus oxygen demand) outpaces the respiratory system’s ability to supply oxygen to tissues throughout the instar – causing functional hypoxia as the insect approaches its molt (Harrison *et al.*, 2018). Oxygen limitation in the environment would only contribute to the detrimental effects of hypoxia for these juveniles; ventilation should thus be carefully monitored at both the juvenile and adult stage.

Beyond low oxygen levels, high levels of some gasses may be detrimental in and of themselves. Atmospheric CO₂ levels are rising due to anthropogenic causes (currently, around 0.041%), where CO₂ can increase desiccation stress and mortality, and reduce growth (Nicolas and Sillans, 1989). In the long term, rising atmospheric levels may make it more challenging to regulate CO₂ levels, especially given the high amount of CO₂ produced by the insects themselves.

Recommendation 11a (domain 2: environment).

Continuously monitor gas flow and oxygen/CO₂ concentrations throughout rearing containers, as this may be affected by cricket age, container depth, and spatial arrangements

In truly open rearing systems, ventilation is high and hypoxia is unlikely to be a concern. However, closed rearing systems – particularly where covers are placed on individual rearing containers to maintain darkness, humidity, or temperature and prevent escape – are more likely to generate hypoxic conditions and ventilation will be needed.

CO₂ production may be especially high for some cricket adults, which are capable of flight (a very metabolically expensive activity). *Ad libitum* feeding and increased activity both increase insect metabolic rates and thus gas production. Nymphs had higher gas production during the daytime (Oonincx *et al.*, 2010), meaning that producers should monitor and take special care to avoid hypoxia during the day part of day-night cycling. Producers that increase the surface area for rearing crickets in a container but provide the same amount of atmospheric gas volume (e.g. through the use of egg cartons) may be especially likely to cause hypoxic conditions by increasing stocking density per volumetric unit of atmosphere. Further, egg cartons could reduce air movement, especially deeper in the rearing container, making it especially important to monitor gas levels through the rearing pen.

Recommendation 11b (domain 2: environment). *Use breathable covers, instead of closed lids, to prevent escape and improve ventilation in smaller containers. Use active ventilation in larger containers or where gas flow is poor*

For small containers, breathable mesh covers may be enough to provide good oxygen and CO₂ levels in all areas upon monitoring. For larger containers, active ventilation may be needed to force the turnover of atmospheric gases throughout the space. An 11.18 L/min ventilation rate was used for the nymphs, reared at 28 °C, in a 265 L chamber (the number of individuals was not provided, but the body mass of all nymphs combined was 0.96 kg at the start of the experiment). This ventilation rate kept CO₂ levels at ~1% during the experiment and could serve as a guide for producers looking to set their own ventilation rates (Oonincx *et al.*, 2010).

Concern 12. Stress and fear associated with the use of light and handling during harvesting or maintenance

Stress can be defined as a disturbance in an animal's normal physiological and/or psychological equilibrium in response to an event (a stressor), leading to activation of appropriate regulatory mechanisms (a stress response), in order to cope with the stressor and re-establish internal equilibrium (Hogg *et al.*, 2018; Kollack-Walker *et al.*, 2000). Stress responses include enhanced energy supply, increased muscle performance, increased sensory perception, and behavioral modification (Roeder, 2005). Avoidance of stressors is linked to fear in sentient animals (Veissier and Boissy, 2007); what an animal finds negatively reinforcing (something they choose to avoid – here, light), is

used to represent negative emotional states in vertebrate animals (Dawkins, 2008).

Handling may be perceived as a predation threat by insects, inducing stress responses (Davenport and Evans, 1984). In *A. domesticus*, handling induced a hyperglycaemic and hyperlipaemic response (an increase of sugar and lipids in the plasma to mobilize energy resources). Levels of the neurotransmitter octopamine increased rapidly in response to handling and decreased rapidly when the stressor was removed (Woodring *et al.*, 1988, 1989). Octopamine is called the insect “fight or flight” hormone (Orchard, 1982; Roeder, 1999) or stress hormone (Adamo and Baker, 2011). Although there are no data on handling-associated stress in *G. assimilis* and *G. sigillatus*, the widespread association between handling and octopamine release in insects (Bailey *et al.*, 1984; Harris and Woodring, 1992; Orchard *et al.*, 1981) would suggest that they may also experience stress when handled.

Other disturbances can also induce stress: for instance, overhead shadows (e.g. of humans or equipment) can be perceived as a predation threat (as in *Drosophila* fruit flies; Gibson *et al.*, 2015), as can vibrations (see concern 13). As crickets are intensely photophobic, and exhibit negative phototaxis, light may also induce stress; however, light is routinely used in harvesting or maintenance.

Recommendation 12a (domain 2: environment). *Avoid the use of a fear response, e.g. avoidance of targeted direct light, for controlling cricket motion. If light will be used, reduce open space and provide enough shelters nearby for all crickets to hide*

Notably, harvesting may use light to induce fear-linked escape behaviors in crickets (Kyllönen and Manzanares, 2022); these responses are likely to be accompanied by fear in sentient animals (Veissier and Boissy, 2007). Cricket farmers may turn on lights intermittently when servicing rearing units to induce the crickets to attempt to hide and thus leave feeders and waterers (Big Cricket Solutions, 2019a). Crickets are known to find light aversive; utilising a fear response for routine activities is problematic for welfare. Negative phototaxis occurs in both adults and nymphs, though nymphs may be more intensely photophobic than adults (Big Cricket Solutions, 2019a).

Importantly, escapable stress may not always be bad for animal welfare; enabling the ability to escape from a stressor may even promote a positive affective state (for example as shown in sheep, Doyle *et al.*, 2010), thereby improving welfare. However, this net positive effect is

dependent on the animal's ability to truly 'escape' and on the magnitude of the acute stressor itself.

If light is to be used, providing adequate nearby hiding space e.g. the rearing matrices, presents the possibility to mitigate this concern, if it provides space for all crickets to rapidly escape this stressor by hiding. This may also reduce potential fear associated with light itself. If crickets are not provided with shelters in order to escape the light, this presents a potentially inescapable stressor – and thus a net welfare harm. The stressful or fear-inducing effects of light may thus also interact with spatial arrangements. For instance, crickets find open space aversive: brighter illumination correlated with higher velocities as crickets crossed an open arena (Vossen *et al.*, 2023). Crickets also preferred resting places that reduced their contact with objects, but were found to be more motivated by photophobia than by reducing their contact with other surfaces (e.g. like the insides of tubes; Kieruzel, 1976). Taken altogether, these data suggest that true escape may require adequate shelters and not just escape from the direct beam into an open arena.

Further, if the acute fear or stress caused by the light is sufficiently severe, even the opportunity to escape may not be sufficient to result in overall better welfare (especially compared to alternative maintenance or harvesting SOPs that do not use fear responses). More research will be needed to determine the severity of the fear response induced by light in crickets.

Recommendation 12b (domain 2: environment).

Reduce the frequency and duration of handling events

Frequent disturbances in farming operations may subject crickets to *chronic stress*. Chronic stress reduced weight gain, decreased feeding, and enhanced weight loss after food deprivation in adult female *G. texensis*; further, repeated exposure to the mock vibrating predator increased basal octopamine levels (Adamo and Baker, 2011), similar to the effects of chronic stress on the basal levels of vertebrate stress hormones. Long-term elevation of stress hormone levels may be a common response to sub-optimal environments in animals, and therefore may represent a potential welfare outcome measure in insects. Reducing the frequency and duration of handling events can reduce the likelihood of chronic stress.

Concern 13: inappropriate light cycling or light types in closed production systems

All three focal species are nocturnal (Ackert and Wadley, 1921; Kieruzel, 1976; Nowosielski and Patton, 1963; Ushi-

rogawa *et al.*, 1997) with distinct circadian rhythms. *Acheta domesticus* has increased activity in the night period and highest activity immediately after dusk (Kieruzel, 1976; Ushirogawa *et al.*, 1997); daylight hours are spent mostly hidden in dark shelters, although some daytime activity is observed in all three species (Ackert and Wadley, 1921; Kieruzel, 1976; Ushirogawa *et al.*, 1997). Mating behavior, such as stridulatory activity and production of the spermatophore in males, is also under circadian control in many cricket species (Tomioka, 2014, but see Clifford and Woodring, 1990). Preservation of appropriate dark and light periods may be important for normal reproductive behavior and output at 'night' (Tomioka, 2014).

While semi-open production systems use the natural day/light cycle, closed systems can set their own light/dark cycles and potentially disrupt cricket circadian rhythms. Although there are reports that exposure to 24 hours of light optimizes productivity (Collavo *et al.*, 2005; Nakagaki and Defoliart, 1991), it is not clear how frequently this practice occurs in industry. Typical reports for *A. domesticus* are 8:16, 10:14, or 12:12 L:D (Big Cricket Solutions, 2019a; Kyllönen and Manzanares, 2022; Mott, 2016; Sens, n.d.; Vandeweyer *et al.*, 2018). There is little literature on light cycles used for the other two focal species, though a 14:10 L:D cycle has been reported for *G. assimilis* crickets in a laboratory setting (Pacheco and Bertram, 2014).

Further, semi-open cricket production systems are likely to rely largely on natural light; closed cricket facilities are likely to use artificial light. The effect of natural light on cricket production and welfare parameters appears not to have been investigated. Natural light has been shown to be important for welfare in vertebrate species by increasing expression of positive behaviours (domain 4), such as increased foraging and exploration in broiler chickens, potentially due to increased variability in intensity and a wider range of wavelengths in natural light compared to artificial light (de Jong and Gunnink, 2019).

Recommendation 13a (domain 2: environment; domain 4: behaviour). Avoid permanent light and use light cycles that promote the species' circadian rhythm

Disruption of the circadian rhythm can lead to poor welfare through abnormal behaviour (domain 4) and disease (domain 3; e.g. Shirasu-Hiza *et al.*, 2007); changes in circadian rhythm have been used as a welfare indicator in vertebrate species (Rhodes *et al.*, 2022). Some unnatural light cycles can disrupt circadian rhythms in crickets, resulting in sublethal stress effects even with-

out differences in growth or development outcomes (Patton, 1978). Permanent darkness did not affect circadian rhythms in *G. sigillatus* (Abe *et al.*, 1997; Ushirogawa *et al.*, 1997), but *A. domesticus* reared under 24 hour darkness showed no rhythm in octopamine and activity levels (Woodring *et al.*, 1988). *Grylloides sigillatus* crickets exposed to 24 hr light showed arrhythmic behavior, even when the light was dim (Ushirogawa *et al.*, 1997).

Exposure to permanent light is likely to be aversive to crickets since they are photophobic, prefer dark resting places, and avoid open, illuminated spaces (Kieruzel, 1976). In support of constant illumination being a stressor, dopamine and serotonin levels were higher in *A. domesticus* crickets kept under constant illumination for three days than in those kept under 12:12 L:D conditions (Germ and Kral, 1995). Stress has been found to increase serotonin levels (Harris and Woodring, 1992; Hirashima and Eto, 1993) and dopamine levels (Gruntenko *et al.*, 2000) in other insects, which may be linked to reduced welfare and even resulting in mortality (Matsumoto *et al.*, 2003). Further, studies show that light stress compromised immune system function in field crickets (Durrant *et al.*, 2020).

Recommendation 13b (domain 2: environment; domain 4: behaviour). If lights must be turned on during dark parts of the cycle, use red lights as long as it does not interfere with worker safety

A US cricket farming consultancy reports crickets to be red-blind, and states that using red lights whilst working reduces stress on the crickets (Big Cricket Solutions, 2019a). Little information is available on the color vision of the three focal species; however, many insects are unable to see red, and *G. bimaculatus* only have photoreceptors sensitive to blue, green and UV light (Frolov *et al.*, 2014). Using red lights will avoid disrupting crickets' circadian rhythm and can reduce stress compared to white lights (see concern 11).

Recommendation 13c (domain 3: physical health). Provide at least 4 hours of UVB light exposure when using artificial lights for rearing

Data on rearing *A. domesticus* crickets under LED lights on an 8:16 L:D cycle did not affect growth or survival (Psarianos *et al.*, 2022). However, survival increased 16.7% with exposure of 4 hours of UVB radiation during the light period. This may be due to an increase in vitamin D, due to de novo synthesis during exposure to UVB light. Exposure to UVB light for 8 or 12 hours a day resulted in increases of vitamin D concentration in *A.*

domesticus and *G. sigillatus* crickets, although the effect on survivability and welfare were not investigated (Kienzle and Kappen, 2022; Oonincx *et al.*, 2018). Vitamin D is known to be important for hormonal functioning in vertebrates and therefore future research should assess if increased concentrations improve health in insects (Oonincx *et al.*, 2018).

Concern 14. Exposure to dead conspecifics

Acheta domesticus shows strong avoidance of “necromones” (chemical cues of death) in their environment. The crickets strongly avoided both shelters and open areas treated with body extracts of dead conspecifics, oleic acid, or linoleic acid, identified as the active chemical fractions of necromones causing necrophobic responses in other insects (Aksenov and David Rollo, 2017). Females showed decreasing aversion and males increasing aversion with time. Active avoidance is taken as a general fear response and therefore an expression of a negative mental state in vertebrates (e.g. Meuser *et al.*, 2021). If avoidance is likewise associated with negative affect in insects, then exposure to dead conspecifics in cricket farms is likely to result in negative welfare.

Recommendation 14a (domain 2: environment).

Frequently monitor for, and remove, dead crickets

Producers may consider a monitoring program that checks bins for dead crickets every few days; in bins with stacked egg carton matrixes, it may be very difficult, if not impossible, for producers to efficiently engage in monitoring for and removal of dead insects. Minimally, producers can consider monitoring for and removing dead insects any time they change nymphs to a new rearing bin; if excessive mortality is noted during these events, producers may consider if there's cause to humanely depopulate their stock.

In single-layer set-ups or concrete pens, producers may be able to monitor for and remove dead crickets more easily – for instance, as they engage in regular maintenance activities (food/water replacement) and live crickets naturally move away from the human activity leaving only dead or injured crickets behind to be removed.

Concern 15. Lack of appropriate enrichment opportunities

Environmental enrichment refers to enriching the environment, and thereby the life, of a captive animal through provision of resources or management practices “necessary for optimal psychological and physi-

ological well-being” beyond those needed for survival (Shepherdson, 1998). Enrichments can also be defined as something an animal wants (as a species or an individual; Dawkins, 2008), introducing choice into an animal’s environment, catering to individual preference, and allowing for a sense of agency (Špinka and Wemelsfelder, 2018). Enrichments may be physical, social, nutritional, cognitive or sensory, with the aim of improving animal welfare through meeting basic biological needs, stimulating positive welfare, or reducing behavioral restriction. Therefore, enrichments must be validated by welfare outcome measures, such as physical and mental health, positive and negative behaviors, and animal preference for and utilization of the enrichment (Taylor *et al.*, 2023).

Importantly, environmental enrichment can significantly improve the welfare of *all* animals that have welfare. A recent review in fish and aquatic invertebrates – animals, like insects, historically thought to be insentient and often farmed in barren environments – found that environmental enrichment positively improved their welfare (Zhang *et al.*, 2022). Research on environmental enrichment for farmed crickets to avoid the negative welfare impacts of rearing in barren environments is, therefore, crucial.

Recommendation 15a (domain 2: environment; domain 4: behaviour). Provide vertical space within rearing containers for climbing

Most producers already provide climbing substrates for crickets, in the form of rearing matrices: the use of vertical space within rearing containers can increase the surface area available to crickets while supporting natural behaviors and microclimate choice (Mott, 2016). Cardboard rearing matrices are commonly used, though technological innovation is ongoing: Takuya and Giovanni (2022) developed a polyhedron scaffolding structure which allowed “comfortable movement” – ease of climbing – within the 3D space. This resulted in a more even distribution of crickets throughout the module compared to egg trays, where crowding can occur in the spaces between the egg cups. Even distribution may decrease negative behavioral interactions that occur due to crowding and therefore improve welfare.

Recommendation 15b (domain 2: environment; domain 4: behaviour). Provide resting places that protect crickets from light

Whether an enrichment is something that an animal wants can be determined by assessing motivation to access specific enrichments (behavioral demand),

use/interaction over time, and preferences (Taylor *et al.*, 2023). Crickets are known to have resting place preferences, for example choosing opaque over transparent tubes (Kieruzel, 1976) and choose the darkest areas possible in which to rest (Tatarova, 2017). Egg cartons, or other opaque materials, are frequently used by producers to provide crickets with preferred shelters.

Recommendation 15c (domain 1: nutrition; domain 2: environment). Consider providing novel sensory enrichments, particularly in dietary choice

Most studies that have examined the use of environmental enrichment in crickets have studied the effects of enrichment on learning, memory, and neurobiology. Mallory *et al.* (2016) investigated the provision of environmental enrichment for *A. domesticus* (a piece of egg carton for shelter, sticks for climbing, access to conspecifics, and supplemental fruits and grains). Crickets with access to these enrichments as young adults performed better on a memory task than individuals without any enrichments. Crickets with only the nutritional enrichment did not differ from crickets in the condition with all four enrichments, suggesting that a varied diet may explain a large portion of the environmentally-induced memory variation. This may be because crickets evolved as generalist omnivores; therefore, dietary variation may be a common part of their natural lifestyle.

The role of environmental enrichment in cricket learning and memory is linked to neurogenesis. New neurons are produced in the brains of *A. domesticus* crickets throughout their life, mainly in the mushroom bodies (integrative learning and memory centers). The rate of neurogenesis is at least partially controlled by environmental cues. Adult *A. domesticus* crickets reared in an enriched sensory environment consisting of odors, hiding places, conspecifics, and extra space showed increased neuroblast proliferation compared to those reared in barren environments (Cayre *et al.*, 2007). Scotto-Lomassese *et al.* (2000) observed similar results by enriching female cricket housing with hiding spaces, aromatic plant branches, different types of textured surface, exposure to the sound and smell of males, and distributing food and water to encourage exploration of the 3D space. Neurogenesis was in direct response to olfactory and visual stimuli, rather than being mediated via hormonal control (Scotto-Lomassese *et al.*, 2002).

Crickets that lacked neurogenesis also exhibited delayed learning and reduced memory retention (Cayre *et al.*, 2007; Scotto-Lomassese *et al.*, 2003). This suggests that an enriched environment is important for optimiz-

ing learning, and therefore the ability to adapt to and cope with the environment, which may reduce suffering; this neurogenesis thus serves as a partial validation of the above enrichments as welfare improvements for crickets. However, it should be noted that negative events can also enhance brain development: *A. domesticus* males which engaged in fights showed increased survival of newborn neurons than males that did not fight (Ghosal *et al.*, 2009).

Finally, Cayre *et al.* (2007) reports that the effect of environmental stimulation on neurogenesis decreased when insects were reared for long periods in enriched conditions, and so was subject to habituation. By contrast, neurogenesis could be enhanced by a sudden change of environment, suggesting that novelty is important in maintaining the benefit of environmental enrichment (as observed in vertebrates, e.g. Trickett *et al.*, 2009).

We summarize all aforementioned welfare concerns and recommendations in Table 4.

6 Future possible welfare concerns for crickets and recommendations

We also identified concerns that do not currently appear to be prevalent in the industry, but may become more prevalent as the industry continues to grow.

Future Concern 1. Novel waste feeds that contain contaminants/pathogens or are nutritionally inadequate, resulting in poor physical health

Contaminants are a concern for farmed insects both for their welfare and for product safety. Feed is considered the principal route of exposure to environmental contaminants (van der Fels-Klerx *et al.*, 2018). Some novel food waste streams may be more likely to contain pathogenic microbes/parasites or environmental pollutants than currently utilized feeds. For instance, the risk of infection with *Cryptosporidium* spp., *Gregarine* spp., *Balantidium* spp., *Entamoeba* spp., *Steinernema* spp., *Gordiidea*, *Hammerschmidtella designi* and *Acaridae* was higher in insects fed kitchen discards than commercial diets (Gałęcki and Sokół, 2019). Environmental pollutants such as pesticides or *Bt* toxins may be more likely in agricultural waste streams, while pharmaceutical pollution might be more common in waste streams sourced from manufacturing or human dwelling areas. More research is needed on the types of pollutants (which may include pharmaceuticals, dioxins, heavy metals, microplastics, Cry/Bt toxins, mycotoxins, pesti-

cides/herbicides/fungicides, etc.; see Supplemental File S1) found in different waste streams and their effects on cricket welfare and product safety.

Some pollutants/pathogens may be detectable to and thus avoided by some cricket species. This can reduce the negative effects of the contaminant on physical health, but will likely result in malnutrition. For instance, *G. assimilis* crickets show avoidance of food contaminated with *Escherichia coli*, due to detection of a volatile compound produced by the bacteria, identified as indole (Aleknavičius *et al.*, 2023). However, *A. domesticus* crickets did not show avoidance of feed contaminated with *E. coli*, and neither *G. assimilis* nor *A. domesticus* avoided of food contaminated with *Pseudomonas aeruginosa* or *Bacillus subtilis* (Aleknavičius *et al.*, 2023). *G. bimaculatus* crickets show conditioned taste aversion of food paired with injection of a toxic solution, suggesting they could learn to avoid eating toxin-contaminated feed (Lyu and Mizunami, 2022). Generally, crickets are considered undiscerning feeders (Mott, 2016), and so may risk consuming contaminated feed detrimental to their health or welfare.

Further, a push for less expensive and more sustainable diets has led many in the industry to look into agricultural by-products, consumer food waste, and other forms of underutilized plant biomass (such as weeds, grasses, or algae) as novel sources of feed (Cortes Ortiz *et al.*, 2016; Miech, 2018). When these alternative feed sources are used with animal welfare in mind, they may also lead to economic improvements for farmers, illustrating that economically viable farming and good animal welfare need not be mutually exclusive (Dawkins, 2017). However, diets that heavily focus on using high proportions of only a few agricultural by-products (e.g. mono-streams) may overlook the importance of a diverse diet and balanced nutrition to increase yields and improve welfare (Harsányi *et al.*, 2020; Sorjonen *et al.*, 2022; Van Peer *et al.*, 2021). Further, many waste products have very low-density nutrition and are largely composed of fibers that may be difficult for insects to digest (Straub *et al.*, 2019), resulting in poor welfare on these substrates.

Recommendation 1a (domain 3: health). *Avoid feeding crickets on waste streams that contain compounds known to be detrimental to their health or welfare, such as certain pesticides or toxins. Avoid feeding crickets novel pollutant-contaminated waste streams until they have been tested for safety and synergistic impacts*

Novel waste streams may contain pollutants such as toxins or pathogenic microbes that can act independently,

TABLE 4 Summary of welfare concerns and recommendations for farmed crickets

Welfare concern	Recommendations
Intentional or accidental dietary restriction (quantity) leading to hunger	A. Provide adequate quantities of nutrition at all times during juvenile rearing. Avoid fasting whenever possible. If needed for product safety reasons, fasting should be limited to 24 hours. B. To avoid accidental dietary restriction caused by inaccessible feed, diets should be ground to a 20 mesh particle size for juveniles.
Inappropriate macronutrient or micronutrient provisioning, leading to malnutrition	A. Provide minimally 20%, preferably closer to 30–35% protein, in cricket diets. B. Supplement cricket diets with heterogenous materials that increase opportunities for nutrient self-selection and improve welfare via enrichment. C. For crickets fed plant-based diets, supplement vitamin B-12 using brewer's yeast or pumpkin in feed. Consider micronutrient supplementation using phosphorus, vitamin C, sterol, and manganese, for all crickets.
Too high stocking densities	A. Rear <i>A. domesticus</i> at densities near 0.09 crickets/cm ² ; do not use densities above 0.4 crickets/cm ² . <i>G. assimilis</i> can be reared at densities up to 0.37 ml crickets/L space, and possibly at slightly higher densities than this.
Macropterous morphs in closed facilities	A. Monitor density and temperature to avoid producing macropterous individuals in wing-polymorphic species (<i>A. domesticus</i> and <i>G. sigillatus</i>). B. Provide opportunities for dispersal-like behaviours in <i>G. assimilis</i> .
Aggression, injury, and cannibalism	A. Reduce fasting period length to a maximum of 24 hours and provide adequate food and water during rearing. B. Maintain stocking densities below the species-specific recommendations designed to avoid self-thinning behaviour and aggression. Monitor levels of aggression and cannibalism. C. Monitor populations for live individuals being cannibalized (or otherwise grossly injured), which should be removed and humanely euthanize.
Mortality or reductions in physical health due to disease or parasitism	A. Follow best practice guidelines for hygiene for each facility type, which include regular disease monitoring and management protocols. B. Keep humidity low (50% RH for adult <i>A. domesticus</i> , 70% for juveniles) and refresh damp oviposition substrates every 24 hours to minimize pathogen development. C. Keep stocking densities low to help prevent or manage diseases, especially AddNV or diseases that spread via cannibalism. D. In cases of untreatable disease, crickets should be isolated, humanely depopulated, and disposed of using appropriate biosafety protocols as soon as possible.
Inhumane slaughter or depopulation	A. Develop standard operating procedures (SOPs) for slaughter and depopulation that are instantaneous. B. At this time, avoid freezing in air as a method of anesthesia, depopulation, or slaughter. Freezing should never be used prior to heating. C. For methods that are not instant but must still be used, implement stunning/anesthesia protocols prior to slaughter/depopulation.
Lack of climate control in semi-open rearing facilities or inappropriate climate control in closed rearing facilities	A. Crickets should be reared in closed, climate-controlled facilities wherever natural temperatures exceed maximum rearing temperatures in Table 3. Temperatures should be kept close to the recommended rearing temperatures. B. Limiting temperature fluctuations over time while providing constant spatial thermal gradients from 27 – 32/33 °C could improve welfare for adult <i>G. assimilis</i> . C. RH should be kept high enough that there is no observable clustering of crickets around water sources.

TABLE 4 (Continued)

Welfare concern	Recommendations
Inappropriate climate control during live transport	Aim for shipping temperatures slightly lower than those reported here as recommended for rearing in Table 3. Temperatures should always be kept below the maximum rearing temperature and above CT _{min} .
Stress associated with vibrations during transport	Minimize frequency/duration of vibrations through changes in transport method or packing materials.
Poor ventilation leading to hypoxia or CO ₂ accumulation	A. Continuously monitor gas flow and oxygen/CO ₂ concentrations throughout rearing containers, as this may be affected by cricket age, container depth, and spatial arrangements. B. Use breathable covers, instead of closed lids, to prevent escape and improve ventilation in smaller containers. Use active ventilation in larger containers or where gas flow is poor.
Stress associated with the use of light and handling during harvesting or maintenance	A. Avoid the use of a fear response, e.g. targeted direct light, for controlling cricket motion. If light will be used, reduce open space and provide shelters nearby. B. Reduce the frequency and duration of handling events.
Inappropriate light cycling or light types in closed production systems	A. Avoid permanent light and use light cycles that promote the species' circadian rhythm. B. If lights must be turned on during dark parts of the cycle, use red lights as long as it does not interfere with worker safety. C. Provide at least 4 hours of UVB light exposure when using artificial lights for rearing.
Exposure to dead conspecifics	Frequently monitor for, and remove, dead crickets.
Lack of appropriate enrichment opportunities	A. Provide vertical space within rearing containers for climbing. B. Provide resting places that protect crickets from light. C. Consider providing novel sensory enrichments, particularly in dietary choice.

or synergistically, to have lethal or sublethal welfare impacts.

For instance, some crops have been genetically engineered to express *Bacillus thuringiensis* proteins (called Cry proteins or Bt toxins) as a pesticide (Then and Bauer-Panskus, 2017). These toxins are generally taxon-specific and dissolve gut cell membranes, paralyzing the digestive system and starving the insect; sepsis from bacterial infection of the hemolymph may also be possible (Hilbeck *et al.*, 2018; Ibrahim *et al.*, 2010). *Acheta domesticus* crickets fed a diet of 90% Bt-positive corn leaves mixed with commercial cricket food showed drastically increased mortality (94%) compared to those fed 25% or 0% Bt-positive diet (22% mortality). Dumont *et al.* (2016) observed that crickets fed the 90% Bt-positive diet increased their food consumption, potentially in response to increased hunger levels (alternately, it is possible corn leaves had reduced nutrition or had a preferred taste compared to the commercial diet). As

time-to-death and mortality are dose-dependent, a non-lethal dose of Bt could simply decrease the ability to digest food and detrimentally affect hunger status.

These examples suggest that waste streams from Orthopteran-targeting *Bt* crops should never be used to feed crickets, and that pesticides should be individually tested for their welfare impacts to determine if agricultural waste contaminated with those pesticides can be used in diets. Producers should avoid waste streams that contain known harmful compounds, even if welfare impacts are sublethal. Further, producers should test any novel pollutants for lethal and sublethal welfare impacts prior to deploying those compounds at scale; effects of concentration should also be considered (e.g. a pollutant found not to be harmful at a low concentration may still be harmful at higher concentrations). Finally, producers should test for synergistic impacts of pollutants – for example, contamination with heavy metals has also been found to increase pathogenic bac-

teria in the gut microbiome of *Eucrotettix oculus* grasshoppers, negatively impacting physical health (Li *et al.*, 2021).

Recommendation 1b (domain 3: health). Pre-process novel waste feeds to reduce disease-causing agents and pollutants

Pre-processing can help to reduce welfare risks associated with the use of novel substrates related to disease, toxicity, or inadequate nutrition. For instance, freezing may kill certain parasites in feed (e.g. inactivation of certain *Cryptosporidium* oocysts at -20°C for 24+ hours; Fayer and Nera, 1996) while heating to $60\text{--}90^{\circ}\text{C}$ can inactivate toxic Cry proteins (Ujváry, 2010).

Recommendation 1c (domain 1: nutrition). When using novel waste streams, test their nutritional composition and digestibility. Consider using animal preference data as a mechanism to rapidly search for high-welfare waste streams

As discussed in current concerns 1 and 2, appropriate macronutrients and micronutrients, as well as accessible/digestible feed, are important components of animal welfare. Some high-quality food waste ingredients may be an appropriate substitution for ingredients in current feed as long as the overall macronutrient levels are maintained. However, other ingredients of poor quality (such as unprocessed post-consumer organic waste, manure, straw from wheat, maize, or rice, and various weed plants) may result in high mortality, lower body weights, and poor welfare (Lundy and Parrella, 2015; Miech, 2018; Straub *et al.*, 2019).

For instance, a 1:9 ratio of chicken feed to vegetable waste, garden waste, horse manure, and cattle manure decreased survival from 94.7% in controls to 77.67%, 68%, 64.33%, and 54.67%, respectively. Further, only vegetable waste and the control diet could support juvenile growth between days 30 and 45; this resulted in much smaller crickets in, for instance, the manure treatments compared to the control. These mixed diets were shown to contain significantly less protein and carbohydrates, as well as less of key micronutrients like phosphorus, compared to the control diet (Harsányi *et al.*, 2020).

Novel diets can also suffer from digestibility challenges. For instance, increased neutral detergent fiber and low levels of non-fiber carbohydrates in novel storable livestock feed materials also led to lower digestibility and slower growth for crickets. Crickets in these treatments had significantly higher consumption, despite their slower growth, as a reaction to the

low density of nutrients in the feed. This reaction may be indicative of hunger when provided an inadequate nutrient density which may be common in many waste streams (Straub *et al.*, 2019).

Given these and other examples, most crickets are unlikely to be successfully reared economically or with high welfare on mono streams, given their unsuitable moisture content, texture, low nutrient density, micronutrient deficiency, or macronutrient imbalances (Van Peer *et al.*, 2021). Given the variability of novel waste streams and the fact that many have already performed poorly for cricket growth and welfare, producers should analyze these variables prior to using them at scale in their facilities. Further, producers may find that compound diets that mix side streams could improve key deficiencies (Van Peer *et al.*, 2021), supporting cricket growth, health, and welfare.

Animal preference data can be useful to determine which nutritional substrates are best for animal welfare (though it should be clear that animal preferences are likely to be life stage dependent, potentially experience- and condition-dependent, and may also reflect short-term priorities over long-term health and wellbeing). Comparing cricket preferences for novel waste streams and standard commercial diets could help producers quickly identify the most promising novel substrates for cricket welfare. Further, Morales-Ramos *et al.* (2020) found that the most profitable diet formulation closely matched the self-selected, preferred macronutrient ratio of *A. domesticus* and was primarily composed of agricultural by-products while still producing high growth rates. This illustrates that considering animal welfare may also lead to economic benefits for farmers (Barrett and Adcock, 2023).

Future Concern 2. Accidental/artificial selection events or genetic modification that result in behavioral restriction or poor physical health

Selective breeding and genetic manipulation can pose a fundamental challenge to the integrity of an animal (Roeckensberg *et al.*, 2018). For practical applications, this requires that any changes should still allow for proper health and development (domain 3) and the expression of natural behavior (domain 4).

Selective breeding is likely to emerge first in the industry (prior to genetic manipulation), and always bears a risk of off-target effects and inbreeding depression (Jensen *et al.*, 2018). Even though crickets have been bred globally for decades, no particular strains with specific properties for commercial production have been established yet, indicating that breeding efforts are

currently mostly limited to a small scale within farms (Nakajima and Ogura, 2022). Crickets generally produce a large number of offspring in a short time frame, which will make selective breeding more efficient than in other animal production systems (Eriksson and Picard, 2021).

Even before the wide-scale application of Crispr/Cas9, other genome editing techniques such as Zinc finger nucleases and TAL effector nucleases had been established in crickets, demonstrating the possibility to knock-out genes related to cuticle tanning (Watanabe *et al.*, 2012). Since then, crickets have proven to be highly suitable for a range of gene editing techniques, including Crispr/Cas9 (Horch *et al.*, 2017). A recent study not only demonstrated the possibility for knocking genes out, but also for inserting genes of interest, such as genes encoding for fluorescent markers, into a cricket (Matsuoka *et al.*, 2021). No off-target effect, in which genes other than the one of interest had been altered through the Crispr/Cas9 machinery, have been reported so far; however, it is noteworthy that in other insects target mutagenesis of a certain gene caused unexpected developmental effects (Lin *et al.*, 2023). Until now, all of these approaches have only been used for research purposes.

Recommendation 2a (domain 3: health). Avoid artificial selection efforts with small breeding populations

Insect producers might initially be tempted to select a low number of specific individuals for breeding efforts to achieve specific characteristics (e.g. faster growth, larger body size, increased fecundity), which can generate a population of very uniform insects that produce a high-yield under certain environmental conditions very rapidly. However, such an approach will almost inevitably lead to high inbreeding and a high chance of genetic drift, which will threaten the health and well-being of these populations (Jensen *et al.*, 2018). This seems especially important in the context of known global disease threats to the cricket farming industry. Immunity related genetic traits in crickets have been found to be highly complex and are often sex and pathogen specific (Letendre *et al.*, 2022), suggesting that a certain degree of genetic heterogeneity will be required to maintain resistance to different pathogens.

Further, inbred *G. sigillatus* females (23 generations of full-sibling mating) were less likely to mount outbred males compared to outbred females mounting inbred males; inbred males were less likely than outbred males to transfer a spermatophore to females (Sakaluk *et al.*, 2019). Inbreeding also depressed female fitness, leading to fewer offspring with longer developmental

times. Therefore, large and diverse breeding populations should be used, protecting economic productivity and individual animal welfare.

Modern molecular tools may be used to safeguard the genetic diversity and robustness of a particular population to reach certain breeding goals more quickly without necessarily threatening the health of the animals (Eriksson and Picard, 2021). Six full or partial genomes, as well as many transcriptomic datasets, are already available for different members of the *Gryllidae* (Mito *et al.*, 2022; Nakamura *et al.*, 2022), with some studies explicitly aiming to provide resources for crickets as food and feed.

Recommendation 2b (domain 3: health; domain 4: behaviour). Test each novel strain/modification for adverse physiological or behavioural welfare impacts across the entire life cycle, prior to employing at scale

Notably, selective breeding programs that have not considered welfare have produced negative welfare outcomes in other livestock rearing industries (e.g. fast growing chicken lines and lameness; Kestin *et al.*, 2001). In all cases, research on the economic productivity and product quality of breeding and genome editing should be complemented by behavioral and physiological studies of animal welfare.

In principle, the selection of strains with certain favorable traits could be neutral or even beneficial to the welfare of the animals, as long as the selection programs do not hamper species-specific development, health, and behavior in a way that harms welfare. For instance, of great interest are genes which might improve resistance to pathogens, the nutritional properties of crickets, or their suitability for mass rearing in general (Kataoka *et al.*, 2020; Nakajima and Ogura, 2022; Oppert *et al.*, 2020; Ylla *et al.*, 2021). Some of these aims, such as increased pathogen resistance or lower cannibalism rates, might ultimately improve cricket welfare. However, other potential breeding goals, such as flightlessness or a less selective courtship behavior, might hinder the expression of normal welfare-promoting phenotypes.

The process of selectively breeding or genetic engineering insects according to human needs, does provide a potential challenge with regards to the welfare of an animal and further research will be required in defining the ethical boundaries of this process (Roeckensberg *et al.*, 2018). Additionally, it should be ensured that selective breeding or genetic engineering is not used to fix problems of insufficient animal husbandry such as overcrowding or malnutrition, and that the animal's per-

spective is taken into account when formulating novel selection goals (de Graeff *et al.*, 2019).

Recommendation 2c (domain 3: health; domain 4: behaviour). Monitor for the accidental selection of traits expected to harm welfare, such as increased aggression

Insect populations on farms are evolving under the unique selective pressures of their environments, as created by producers; therefore, it is possible to accidentally select for traits that may harm welfare. For instance, after 105 generations in captivity, farmed populations of male *A. domesticus* were more aggressive than wild-caught males – despite predictions that they would be less aggressive due to selection from living in dense conditions with unlimited access to food and shelter. Furthermore, commercial males engaged in more energy-intensive and higher risk (of injury/death) aggressive behaviors such as mandible flaring, aggressive chirping and grappling, compared to more antennal fencing in wild males (which is energetically inexpensive and low risk; Hack, 1997a,b; Olzer *et al.*, 2019).

The authors suggested this could be due to increased population densities on farms leading to increased perceived (but not actual) competition for resources, and/or an increased ability to monopolize females; in fact, the presence of valued resources may remove the effect of high density on reducing aggression. Chemical cues from females increase aggression in male *A. domesticus* (Buena and Walker, 2008; Otte and Cade, 1976), with variation in male aggression associated with dominance and mating status (Stevenson and Rillich, 2013). The inclusion of burrows (used as calling sites) and females resulted in an increased incidence of aggression at all densities in laboratory populations of *Gryllus bimaculatus* males (Simmons, 1986). *A. domesticus* males also fight over burrow residency (Hack, 1997a). Therefore, increased competition for abundant females or other resources in high population densities could lead to increased aggression, injury, and/or death.

Aggression amongst farmed crickets could be assessed by monitoring songs, as mentioned in concern 5, recommendation 5b. This example illustrates that it is possible to accidentally select for traits that may harm cricket survival, health, or welfare over time. Monitoring traits indicative of negative welfare in each generation can provide producers with an ‘early warning’ if selection is favoring traits that harm welfare in their facility.

Future Concern 3. Novel diseases, particularly for *G. sigillatus* and *G. assimilis*

Although *G. sigillatus* and *G. assimilis* have proven less susceptible than *A. domesticus* to one of the most devastating farmed cricket diseases (AdDNV), mass-rearing animals at high densities is likely to cultivate novel diseases over time.

Recommendation 3a (domain 3: health). Proactively manage and monitor all species for disease

Producers may monitor *A. domesticus* more diligently for diseases or rear this species at lower density given the costly outbreaks of AdDNV that have occurred. Proactively monitoring all cricket species for signs of disease and coordinating efforts to manage diseases early on as they appear (see Eilenberg *et al.*, 2018), as well as following the stocking density and hygiene best practices guides referenced in concern 6, can help ensure all cricket species are reared with good health and welfare. This may be essential for preventing the emergence and spread of novel diseases in *G. sigillatus* and *G. assimilis* (and *G. bimaculatus*).

Recommendation 3b (domain 3: health). Avoid moving livestock between facilities where possible

Cricket farms commonly report a plethora of pathogenic agents from viruses to bacteria and fungi (Eilenberg *et al.*, 2015). The movement of livestock from one facility to another can facilitate disease transmission between those populations, and potentially facilitate the emergence of new pathogenic agents in species less susceptible to the current major cricket diseases (e.g. *G. sigillatus* and *G. assimilis*). Indeed, insect farms that received shipments of insects from other farms more often had higher rates of *Nosema* spp., *Isospora* spp., *Cryptosporidium* spp., *Entamoeba* spp., *Cestoda*, *Pharyngodon* spp., *Gordius* spp., *Physaloptera* spp., *Thelastoma* spp. and *H. diesigni* than farms that did not receive any livestock from other facilities (Gałęcki and Sokół, 2019). Further, producers that have successfully avoided AdDNV outbreaks in their facilities avoided the importation of any crickets into their rearing facilities (Szelei *et al.*, 2011).

Future Concern 4. The expanding list of farmed orthopteran species

There are already a large number of farmed orthopteran species, including the crickets covered herein, many other crickets, and locusts (van Huis and Tomberlin, 2016). Although these species are all in the order Orthoptera, we might reasonably expect there to be differences in the conditions most likely to safeguard their

welfare (after all, both cattle and pigs are in the order Artiodactyla). Given the difficulty of studying insect welfare, a rapid expansion in the number of species of crickets or locusts farmed could make it difficult to accurately characterise the different welfare needs of all these species (Barrett and Fischer, 2023).

Recommendation 4a. When considering farming a new cricket species, begin collaborating early with ethologists, physiologists, pathologists, and welfare scientists

Best practice should include consideration of livestock welfare, not simply productivity; however, in the early days of a new venture, it is often hard to keep welfare in view with so many competing demands. External scientists can help R & D teams document behaviours, identify diseases, and develop welfare improvements as soon as they know those species are being farmed at scale. However, if producers do not reach out to them to partner with them, they are unlikely to be aware new ventures exist (Barrett and Fischer, 2023). Academic-industry partnerships, like NSF CEIF (Tomberlin *et al.*, 2022) are key to building these connections and reducing communication delays.

Rather than delay the study of a novel species' welfare until it is farmed by many companies at significant scale, contacting scientists early means a thorough understanding of the species and what improves individual's welfare will be much further along as the industry grows. Further, these welfare improvements may also help improve productivity, especially in novel species that are poorly studied (Barrett and Adcock, 2023), advancing the economical farming of novel species more quickly and with better welfare.

Future Concern 5. Highly automated facilities introducing aversive electrical fields or shocks

In adult cockroaches, electrostatic deflection of mechanoreceptors on the antennae appears to be important for sensing and avoiding electrical fields, while in flies and other insects the wings may be used (Hunt *et al.*, 2005; Newland *et al.*, 2008, 2015). In crickets, electrical information may be sensed with their cercal hairs (Palmer *et al.*, 2021); the use of the antennae or wings has not been tested.

Importantly, many insects find static electrical fields aversive, seeking to avoid contact (England and Robert, 2022). Proximity to these electrical fields can even increase octopamine (stress hormone) levels in the brain, at least in flies (Newland *et al.*, 2015). At this time it is unknown how/if crickets respond to electrical fields

of different magnitudes, adversely or otherwise (and, at least in some ants, there is evidence for electrical field preference, making it important to test the effects on each species; England and Robert, 2022).

Electrical shocks are known to be highly aversive to crickets, even disrupting fitness-beneficial activities like mating (Jaffe *et al.*, 1992; Sakai *et al.*, 1991).

Recommendation 5a (domain 2: environment; domain 3: health). Reduce electrical equipment's proximity to live crickets. Reduce the use of materials that may produce shocks

Many pieces of equipment used for rearing may end up producing static electrical fields, especially in highly automated facilities; even plastic containers used to rear insects may end up charged when moved, through tribo-electrification (Newland *et al.*, 2015). Producers should seek to avoid the use of materials that are likely to produce shocks; in the case that these materials are best used for cost, hygiene, or other reasons, producers should take care to minimize the likelihood of shocks to insects in their handling SOPs.

7 Urgent future research directions in cricket welfare

Insect pain/sentience research is in its infancy, and there are significant gaps in our understanding of Orthopteran sentience-relevant neurobiology and behavior across development (Gibbons *et al.*, 2022); therefore, further research is needed on Orthopteran neurobiology and behavior. Data validating possible behavioral (e.g. prevalence of hiding, aggression, escape, etc.) or physiological (e.g. prevalence of macropterous morphs in wing polymorphic species, octopamine levels, etc.) indicators of welfare in farmed settings would also be valuable for helping producers assess the welfare of their populations.

We found many current and future welfare concerns for farmed crickets in our review of the literature, just using metrics like survival, growth rate, reproduction, and presence of disease. However, there were many cases where there was not enough data to determine the impact of a variable on cricket welfare across all species (this review, and see Supplemental File S1). The following questions are most essential for improving our understanding of farmed cricket welfare, and can be considered urgent future research directions (in no particular order):

- *Dietary restriction* – High feed costs can encourage producers to limit provided food, in an effort to promote higher feed utilization efficiency. However, at high stocking densities this can cause resource competition when crickets cannot all adequately access food resources. New methods of distributing food resources may reduce competition while improving feed utilization; developing these methods of reducing dietary restriction is an urgent research direction.
- *Climate conditions across development* – Research in *A. domesticus* suggests that optimal temperature and RH may vary for juvenile vs adult crickets. While more research on the interaction of these variables is needed for all species, very little data on the optimal climate conditions for rearing *G. sigillatus* or *G. assimilis* exists. Studies of cricket welfare under different temperatures and RH would be beneficial for all species. Further, investigations of microclimate choice by providing a gradient of temperatures/humidities for crickets to self-select their optimal climate conditions, would be beneficial to determine the effects of this practice on welfare and economic productivity.
- *Stocking densities* – Optimal stocking densities for promoting animal welfare appear to be species-specific, and potentially life-stage or sex-specific (as they relate to male mating behavior, for example). Studies are needed that determine the optimal stocking density for these species' welfare, taking into account development stage and sex. Studies should also assess the welfare impacts of too-high stocking densities, especially as relates to macropterous morphs, disease, resource limitation, aggression, injury, and cannibalism. Finally, studies should consider if there are different impacts of volumetric vs aerial stocking density, such that producers could make better use of 3D space in providing additional habitat for their crickets, simultaneously improving welfare through lower stocking densities and increasing productivity.
- *Disease* – Viral, bacterial, fungal, and other pathogens/parasites can be prevalent in farmed cricket facilities. Studies are needed that determine the most common pathogens/parasites and their prevalence, virulence, methods of transmission, and any welfare impacts. Studying other stressors that may lead to lethal outbreaks when chronic, sublethal infections are present will also be important to proactively managing these diseases.
- *Humane slaughter and depopulation* – Research is needed to produce standard operating procedures that are humane (kill or stun instantly) when conducted at industry scale, with low error rates, for slaughter and depopulation events; product quality and safety effects should be considered alongside welfare. Further, protocols for the safe disposal of material associated with a depopulation event (e.g. failure to thrive insects and any associated substrates) are needed to avoid biosecurity risks to the surrounding environment. The need for research to establish the humaneness of boiling and freezing, two of the most commonly used methods for crickets, is particularly urgent.
- *Anesthesia and stunning protocols* – Some necessary slaughter methods may never be humane, even with the best possible standard operating procedure. In these cases, researchers should develop stunning/anesthesia protocols that can be used prior to the use of that slaughter method. These protocols should be tested for their efficacy in large batches, the length of time unconsciousness will last, any dose-dependent negative effects on welfare of the anesthesia or stunning protocol itself, and any effects on product quality or safety. Research should include time to reach CT_{min} under industry slaughter conditions and the ambient temperature and other conditions needed to achieve anesthesia by freezing in air as humanely as possible. The use of inhalation anesthetics (e.g. halogenated ethers), and electrical stunning, which could be used as a “stun-to-kill” method, are also of particular interest for future research.
- *Cannibalism* – Cannibalism may be caused by resource limitation (food, water) or too-high stocking densities (or both). Researchers should study mechanisms to reduce cannibalism – for example, providing different spatial arrangements that change volumetric/aerial stocking densities, providing more distributed water stations, or increasing protein content in the diet. Although it may also be possible to artificially select against cannibalistic behavior (as studies have demonstrated it is possible to select for cannibalistic behavior in insects), genetic management of cannibalistic behaviour may potentially mask, and therefore prevent the improvement of, sub-optimal environments (e.g. high densities) that typically lead to cannibalism.
- *Gas levels, ammonia, and ventilation* – Studies are needed that determine the welfare impacts of high ammonia or other gas levels (for instance, on their respiratory system) and how to best ventilate pens/containers of different designs.

- *Stress associated with behavioral restriction in macropterous morphs* – In *G. sigillatus* and *A. domesticus*, wing polymorphism can occur when stocking densities are too high – indicating individuals are getting ready to disperse. In *G. assimilis*, all individuals are macropterous. However, dispersal behaviors are prevented in closed farming systems, which represents a behavioral restriction. Studies should assess if this behavioral restriction impacts cricket welfare; if it does, studies should look into mechanisms for reducing macropterous morphs in populations of these species that are wing-polymorphic (e.g. through changing spatial arrangements or reducing stocking densities) and providing opportunities for behavioral expression in *G. assimilis*, as all individuals are macropterous. Further, studies should look into the use of wing shedding behaviors around day 20 post-eclosion in *A. domesticus* for monitoring the impact of stocking density on welfare, as higher wing shedding rates may suggest that densities are not so high the insects retain their wings to disperse away from the stressful conditions.
 - *Shipping sickness* – Crickets that are transported live often experience what is colloquially referred to as ‘shipping sickness’, where populations experience high mortality in transit and immediately upon arrival to their destination. The causes of shipping sickness, which may relate to climate, vibration-induced stress, feed availability, etc., should be investigated and remedied, reducing product waste and improving animal welfare and economic productivity. Different species may also experience different mortality rates during transport; if some species are more resistant to transport-related mortality, the industry should live-ship those species preferentially.
 - *Dietary preferences and composition for G. assimilis and G. sigillatus* – Very little information is available on the dietary preferences or appropriate macronutrient ratios for crickets outside of *A. domesticus*. Further research on the effect of nutrition on welfare in these species is needed.
 - *Effect of non-lethal insecticide concentrations in cricket feed* – there is a current lack of research on the welfare impact of non-lethal doses of insecticides in cricket diets. These are substances that have been designed to be poisonous to insects, and which have been identified as present in farmed insects via their diets.
- There are also many interesting, but less-urgent, research areas that would improve our understanding of farmed cricket welfare (in no particular order):
- *Fasting period length* – We do not recommend fasting crickets prior to slaughter; if it must be done, we recommend the shortest currently utilized time window of 24 hours. However, research should determine at what time point fasting actually induces negative welfare for crickets, which may be assessed through mortality, changes in behavior (e.g. increased food searching or cannibalism), or possibly rising hormone levels related to hunger.
 - *Acoustic monitoring for farmed cricket welfare* – Welfare is already monitored using acoustic equipment in some vertebrate species (Winship and Jones, 2023). Breeding crickets produce many distinct noises; it is plausible that acoustic monitoring devices could be developed that help producers more easily and autonomously assess the welfare of their populations.
 - *Reducing water restriction* – High stocking densities can reduce crickets’ access to water, even when provided *ad libitum*. Delivery methods that make water more accessible, while avoiding drowning, when stocking densities are high should be designed.
 - *Feed contaminants* – As the waste products crickets are fed become more expansive, novel contaminants (mycotoxins, pesticides, heavy metals) may be introduced at levels that are bad for cricket welfare (or, if bioaccumulation occurs, the welfare of organisms that are fed these crickets). Research is needed to determine the welfare impacts of a variety of common pesticides/herbicides/heavy metals, etc. that may find their way into cricket feed, at a variety of concentrations. Producers can then test for and adjust feed contaminant levels as feed arrives on site, for the welfare of their crickets and for product safety reasons.
 - *Antibiotic use* – It is infrequently reported that producers use antibiotics when rearing crickets. Antibiotics may also be found in certain waste feeds (for example, vertebrate livestock manure) where these are fed to insects. Although a very minor concern at this time, antibiotic use should be administered by veterinarians (e.g. as in honey bees) to reduce the likelihood of antimicrobial resistance developing. Antibiotics should be tested for their welfare impacts as well, as they may harm the gut microbiome – and thus the health and nutrient status of the animal.
 - *Accidental and artificial selection* – The welfare impacts of artificial selection efforts should be assessed routinely before any new strains or lines are used at industrial scales. Researchers should also study mechanisms for producers to avoid conditions

that lead to accidental selection for traits expected to have a detrimental impact on welfare (e.g. cannibalistic behavior).

- *Dominance hierarchies* – In some cricket species, males in wild populations control territories and establish dominance hierarchies. Large population sizes in breeding stock populations can challenge the ability of crickets to express these natural intraspecific behaviors. Researchers should look into if the increased aggression often observed in captive populations is linked to these spatial challenges and behavioral restrictions, and mechanisms producers might use to overcome these welfare challenges/restrictions.
- *Physical injury* – Insects respond in unusual ways to physical injury (but see Gibbons and Sarlak, 2020). For instance, Orthopterans may consume their own abdomen when injured (Eisemann *et al.*, 1984) – but male crickets find injuries to the epiphallus so unpleasant that it leads to prolonged disruption of mating behavior (Sakai *et al.*, 1991). Determining if there is variation in the welfare effects of injuries of varying severity, delivered to different body parts (perhaps, related to the distribution of nociceptors), or in different contexts would help producers determine when it is necessary to euthanize an injured animal on their farms.
- *Micronutrient/vitamin/probiotic supplementation* – Dietary supplements may improve the health, growth, survival, reproductive outcomes, and therefore the welfare of insects. Further studies on the welfare impacts of supplementation with sterol, manganese, phosphorus, vitamin B12, and vitamin C could be beneficial. Possible probiotics that might promote insect health should be tested, such as *Lactobacillus*, *Saccharomyces*, *Streptococcus*, and *Bacillus*. The benefits of specific probiotics in restoring gut health after treatment with antibiotics should also be researched.
- *Critical temperatures* – Determination of the critical thermal maxima and minima of these species across developmental stages would be useful for informing humane slaughter practices.
- *Oviposition substrate preference, and stress associated with behavioral restriction* – No research has demonstrated oviposition substrate preference or preferred moisture level in the substrate. Further, no research has tested if denying oviposition behavior through substrate removal prior to harvest induces stress in female crickets via behavioral restriction.
- *Positive behavior ethograms* – Current cricket ethograms, in the wild or captivity, are largely focused on negative behaviors like aggression. More complete ethograms that include neutral and positive behaviors will help producers better assess the overall welfare of their population and provision for positive welfare.
- *Enrichments* – There are many potential enrichments that may be provided for crickets, for example: hiding places, climbing infrastructure, increased space, access to conspecifics, plant odors, and a varied diet. However, these enrichments must be tested to determine whether they *actually* improve cricket welfare. Preference/use tests, willingness to pay assays, and other behavioral studies may lend insight into the benefits of these different enrichments for cricket welfare.
- *Positive female welfare from the male* *G. sigillatus* spermatophylax – Female *G. sigillatus* are provided a spermatophylax mating gift, made of water and amino acids, by males to aid in spermatophore retention. Female *G. sigillatus* can exert choice in mate selection by discarding the spermatophore; this occurs in approximately 25% of all matings (Gershman *et al.*, 2012; Sakaluk, 1984). Male diet determines the successful formation of an appealing spermatophylax in *G. assimilis*. An appetizing spermatophylax could be investigated as a possible positive welfare opportunity (pleasurable experience) for female crickets.
- *Disturbance-associated stress* – Lights and handling can induce stress in crickets; if this disturbance is frequently experienced, crickets may develop chronic stress. Researchers should study what frequencies/durations of light and handling are stressful to insects and if there is any possibility of habituating insects to disturbance to reduce the associated stress.
- *Natural light, artificial lights and UVB* – Researchers should test if there are any welfare impacts of using natural light, vs different solar-simulating, LED or other artificial light sources used in closed facilities. Further, researchers should test if the increase in vitamin D that occurs after UVB exposure in crickets is significant to their welfare.

8 Conclusion

As the IAFF continues to scale, research on the welfare of crickets across their lifespan will grow increasingly important to ensure species-specific conditions for optimal rearing are met. This review accumulates the cur-

rently available research on the welfare of *A. domesticus*, *G. sigillatus*, and *G. assimilis* crickets; however, as the 'Urgent Future Research Directions' section suggests, there are many important gaps in our knowledge.

Particularly important moving forward is transparent collaboration between producers, with hands-on knowledge of the conditions and practices used when rearing these mini-livestock, and welfare scientists, with theoretical frameworks for evaluating and assessing livestock welfare (Barrett and Fischer, 2023). Insect welfare aligns with IAFF industry interests in economic productivity, product diversification, and ethical resource stewardship, and can improve consumer acceptance of industry practices (Barrett and Adcock, 2023). By sharing the results of these collaborations widely, producers can support consumer perceptions of their entire industry as ethical leaders in animal agriculture, encouraging greater integration of IAFF products into the global food chain.

Supplementary material

<https://doi.org/10.6084/m9.figshare.25189142>

Acknowledgements

Thanks to Alexander Haverkamp for contributing information on Genetics & Selection to this report.

Conflict of interest

Elizabeth Rowe and Meghan Barrett report a relationship with Rethink Priorities that includes: consulting or advisory.

Funding statement

Rethink Priorities provided funding to all authors for the researching, writing, and/or editing of this work.

References

- Abdourhamane, I.M. and Petek, M., 2022. Health-based welfare indicators and fear reaction of slower growing broiler compared to faster growing broiler house in free range and conventional deep litter housing systems. *Journal of Applied Animal Welfare Science*. <https://doi.org/10.1080/10888705.2022.2100221>
- Abe, Y., Ushirogawa, H. and Tomioka, K., 1997. Circadian locomotor rhythms in the cricket, *Gryllobates sigillatus*. I. Localization of the pacemaker and the photoreceptor. *Zoological Science* 14: 719-727.
- Ackert, J.E. and Wadley, F.M., 1921. Observations on the distribution and life history of *Cephalobium microbivorum* Cobb and of its host, *Gryllus assimilis* Fabricius. *Transactions of the American Microscopical Society* 40: 97.
- Adamo, S., 2019. Is it pain if it does not hurt? On the unlikelihood of insect pain. *The Canadian Entomologist* 151: 685-695.
- Adamo, S.A., 1998. The specificity of behavioral fever in the cricket *Acheta domesticus*. *The Journal of Parasitology* 84: 529.
- Adamo, S.A., 2016. Do insects feel pain? A question at the intersection of animal behaviour, philosophy and robotics. *Animal Behaviour* 118: 75-79.
- Adamo, S.A. and Baker, J.L., 2011. Conserved features of chronic stress across phyla: the effects of long-term stress on behavior and the concentration of the neurohormone octopamine in the cricket, *Gryllus texensis*. *Hormones and Behavior* 60: 478-483.
- Adamo, S.A. and Hoy, R.R., 1995. Agonistic behaviour in male and female field crickets, *Gryllus bimaculatus*, and how behavioural context influences its expression. *Animal Behaviour* 49: 1491-1501.
- Adamo, S.A., Kovalko, I. and Mosher, B., 2013. The behavioural effects of predator-induced stress responses in the cricket (*Gryllus texensis*): the upside of the stress response. *Journal of Experimental Biology* 216: 46084614.
- Agrios, G.N., 2009. Plant pathogens and disease: general introduction. In: Schaechter, M. (ed.) *Encyclopedia of microbiology* (third edition). Academic Press, Oxford, UK, pp. 613-646.
- Aksenov, V. and David Rollo, C., 2017. Necromone death cues and risk avoidance by the cricket *Acheta domesticus*: effects of sex and duration of exposure. *Journal of Insect Behavior* 30: 259-272.
- Albers, J. and Reichert, M.S., 2022. Personality affects individual variation in olfactory learning and reversal learning in the house cricket, *Acheta domesticus*. *Animal Behaviour* 191: 1-13.
- Aleknavičius, D., Lukša, J., Strazdaitė-Žielenė, Ž. and Servienė, E., 2022. The bacterial microbiota of edible insects *Acheta domesticus* and *Gryllus assimilis* revealed by high content analysis. *Foods* (Basel, Switzerland) 11: 1073.
- Aleknavičius, D., Markaitytė, E., Budiene, J., Blazyte-Cereskiene, L., Stanevičienė, R., Mozūraitis, R. and Servienė, E., 2023. Can crickets recognise bacterially con-

- taminated feed? *Gryllus assimilis* odour perception of *Escherichia coli*. *Journal of Insects as Food and Feed* 9: 947-954. <https://doi.org/10.3920/JIFF2022.0123>
- Alexander, R.D., 1961. Aggressiveness, territoriality, and sexual behavior in field crickets (Orthoptera: Gryllidae). *Behaviour* 17: 130-223.
- Alexander, R.D., 1962. Evolutionary change in cricket acoustical communication. *Evolution* 16: 443-467.
- Alexander, R.D., 1968. Life cycle origins, speciation, and related phenomena in crickets. *The Quarterly Review of Biology* 43: 1-41.
- Alexander, R.D. and Walker, T.J., 1962. Two introduced field crickets new to eastern United States (Orthoptera: Gryllidae). *Annals of the Entomological Society of America* 55: 90-94.
- Anaya, C. and Bolek, M.G., 2021. Is there life after parasitism? Survival, longevity, and oogenesis in *Acheta domesticus* (Orthoptera: Gryllidae) infected with the hairworm, *Paragordius varius* (Phylum: Nematomorpha). *Parasitology Research* 120: 2333-2342.
- Anderson, R.L. and Mutchmor, J.A., 1968. Temperature acclimation and its influence on the electrical activity of the nervous system in three species of cockroaches. *Journal of Insect Physiology* 14: 243-251.
- Angilletta Jr., M.J., 2009. Thermal adaptation: a theoretical and empirical synthesis. Oxford University Press, Oxford, UK. Available at: <https://doi.org/10.1093/acprof:oso/9780198570875.001.1>
- Anonymous, 2022. Why are my crickets eating each other? (5 possible reasons). Available at: <https://furwingsandscalythings.com/why-are-my-cricket-eating-each-other/>
- arachnoboards, 2021. Help! My crickets are EATING each other! Available at: <https://arachnoboards.com/threads/help-my-cricket-eating-each-other.349642/>.
- Archer, C.R., Zajitschek, F., Sakaluk, S.K., Royle, N.J. and Hunt, J., 2012. Sexual selection affects the evolution of lifespan and ageing in the decorated cricket *Gryllodes Sigillatus*: sexual selection and the evolution of ageing. *Evolution* 66: 3088-3100.
- Armstrong Crickets Georgia, n.d. The original cricket est. 1947. Available at: <https://www.armstrongcricket.com/blogs/the-original-cricket-est-1947>
- Arnqvist, G. and Nilsson, T., 2000. The evolution of polyandry: multiple mating and female fitness in insects. *Animal Behaviour* 60: 145-164.
- Attard, L., 2015. An overview of the development and evaluation of a gut-loading diet for feeder crickets formulated to provide a balanced nutrient source for insectivorous amphibians and reptiles. Available at: <http://titag.org/2015/2015papers/attardcrickets.pdf>.
- Bailey, B.A., Martin, R.J. and Downer, R.G.H., 1984. Haemolymph octopamine levels during and following flight in the American cockroach, *Periplaneta americana* L. *Canadian Journal of Zoology* 62: 19-22.
- Baringer, B. and Hamilton, T., 2020. Testing the payoff asymmetry hypothesis: effects of short-term isolation on territorial aggression behavior among individuals of *Gryllodes sigillatus* (Banded Cricket). <https://doi.org/10.13140/RG.2.2.34933.93921>
- Barrett, M., 2024. Insects. In: Clutton, E. and Musk, G. (eds.) Pain and suffering in farmed animals: first steps in better understanding and management. Springer.
- Barrett, M. and Adcock, S.J.J., 2023. Animal welfare science: an integral piece of sustainable insect agriculture. *Journal of Insects as Food and Feed*. <https://doi.org/10.1163/23524588-20230126>
- Barrett, M., Chia, S.Y., Fischer, B. and Tomberlin, J.K., 2022. Welfare considerations for farming black soldier flies, *Hermetia illucens* (Diptera: Stratiomyidae): a model for the insects as food and feed industry. *Journal of Insects as Food and Feed* 9: 119-148. <https://doi.org/10.3920/JIFF2022.0041>
- Barrett, M. and Fischer, B., 2023. Challenges in farmed insect welfare: beyond the question of sentience. *Animal Welfare* 32: e4-e4.
- Bascuñán-García, A.P., Lara, C. and Córdoba-Aguilar, A., 2010. Immune investment impairs growth, female reproduction and survival in the house cricket, *Acheta domesticus*. *Journal of Insect Physiology* 56: 204-211.
- Bateman, P.W. and Fleming, P.A., 2006. Increased susceptibility to predation for autotomized house crickets (*Acheta domestica*). *Ethology* 112: 670-677.
- Bateman, P.W. and MacFadyen, D.N., 1999. Mate guarding in the cricket *Gryllodes sigillatus*: influence of multiple potential partners. *Ethology* 105: 949-957.
- Bawa, M., Songsermpong, S., Kaewtapee, C. and Chanput, W., 2020. Effect of diet on the growth performance, feed conversion, and nutrient content of the house cricket. *Journal of Insect Science* 20: 10.
- Bear, C., 2019. Approaching insect death: understandings and practices of the UK's edible Insect farmers. *Society & Animals* 27: 751-768.
- Bear, C., 2021. Making insects tick: responsibility, attentiveness and care in edible insect farming. *Environment and Planning E: Nature and Space* 4: 1010-1030.
- Behrens, W., Hoffmann, K.H., Kempa, S., Gäßler, S. and Merkel-Wallner, G., 1983. Effects of diurnal thermoperiods and quickly oscillating temperatures on the development and reproduction of crickets, *Gryllus bimaculatus*. *Oecologia* 59: 279-287. <https://doi.org/10.1007/BF00378849>
- Bentham, J., 1789. An introduction to the principles of morals and legislation. Oxford University Press, Oxford, UK.

- Bertola, M. and Mutinelli, F., 2021. A systematic review on viruses in mass-reared edible insect species. *Viruses* 13: 2280.
- Bertram, S.M., Loranger, M.J., Thomson, I.R., Harrison, S.J., Ferguson, G.L., Reifer, M.L., Corlett, D.H. and Gowaty, P.A., 2016. Linking mating preferences to sexually selected traits and offspring viability: good versus complementary genes hypotheses. *Animal Behaviour* 119: 75-86.
- Bertram, S.M., Loranger, M.J., Thomson, I.R., Harrison, S.J., Ferguson, G.L., Reifer, M.L., Corlett, D.H. and Gowaty, P.A., 2017. Choosy males in Jamaican field crickets. *Animal Behaviour* 133: 101-108.
- Bertram, S.M. and Rook, V., 2012. Relationship between condition, aggression, signaling, courtship, and egg laying in the field cricket, *Gryllus assimilis*. *Ethology* 118: 360-372.
- Bertram, S.M., Rook, V.L.M., Fitzsimmons, J.M. and Fitzsimmons, L.P., 2011. Fine- and broad-scale approaches to understanding the evolution of aggression in crickets. *Ethology* 117: 1067-1080.
- Bertram, S.M., Whattam, E.M., Visanuvimol, L., Bennett, R. and Lauzon, C., 2009. Phosphorus availability influences cricket mate attraction displays. *Animal Behaviour* 77: 525-530.
- Big Cricket Solutions, 2019a. Environmental ranges. Available at: <https://bigcricketsolutions.com/knowledge-base/environmental-ranges/>
- Big Cricket Solutions, 2019b. Cricket farming: the basics. Available at: <https://bigcricketsolutions.com/knowledge-base/knowledge-base-article/>
- Birch, J., 2017. Animal sentience and the precautionary principle. *Animal Sentience*: 2. <https://doi.org/10.51291/2377-7478.1200>
- Birch, J., 2022. The search for invertebrate consciousness. *Noûs* 56: 133-153.
- Brambell, F.W.R., 1965. Report of the technical committee to enquire into the welfare of animals kept under intensive livestock husbandry systems. H.M.S.O., London, UK.
- Breedinginsects.com, n.d. New low maintenance cricket breeding methods. Available at: <https://www.breedinginsects.com/better-cricket-breeding-methods/#1561508311082-37dcc4ee-d65d>.
- Broder, E.D., Gallagher, J.H., Wikle, A.W., Welsh, G.T., Zonana, D.M., Firneno, T.J. and Tinghitella, R.M., 2023. A well-studied parasitoid fly of field crickets uses multiple alternative hosts in its introduced range. *Evolutionary Ecology* 37: 477-492.
- Broom, D., 2019. Sentience. In: Choe, J.C. (ed.) *Encyclopaedia of animal behaviour*. Academic Press, London, UK.
- Brown, W.D., Chimenti, A.J. and Siebert, J.R., 2007. The payoff of fighting in house crickets: motivational asymmetry increases male aggression and mating success. *Ethology* 113: 457-465.
- Brown, W.D., Smith, A.T., Moskalik, B. and Gabriel, J., 2006. Aggressive contests in house crickets: size, motivation and the information content of aggressive songs. *Animal Behaviour* 72: 225-233.
- Browning, H., 2020. The natural behavior debate: two conceptions of animal welfare. *Journal of Applied Animal Welfare Science* 23: 325-337.
- Buena, L.J. and Walker, S.E., 2008. Information asymmetry and aggressive behaviour in male house crickets, *Acheta domesticus*. *Animal Behaviour* 75: 199-204.
- Buyfeedercrickets, n.d. *Acheta Domesticus* crickets. Available at: <https://buyfeedercrickets.com/products/live-crickets?variant=34152239171>
- Cadena-Castañeda, O.J., 2011. A new genus of cricket near to *Miogryllus* and *Kazuemba* from the Colombian Atlantic coast and the first report of *Grylloides sigillatus* from Colombia (Orthoptera: Gryllidae: Gryllinae: Modicogryllini). *Zootaxa* 3126: 55.
- Caligiani, A., Marseglia, A., Sorci, A., Bonzanini, F., Lolli, V., Maistrello, L. and Sforza, S., 2019. Influence of the killing method of the black soldier fly on its lipid composition. *Food Research International* 116: 276-282.
- Cancino, J. and López-Arriaga, F., 2016. Effect of hypoxia and its repercussions in packing pupae of the parasitoid *Diachasmimorpha longicaudata* (Hymenoptera: Braconidae) for shipment. *Biocontrol Science and Technology* 26: 665-677.
- Caparros Megido, R., Desmedt, S., Blecker, C., Béra, F., Haubruge, É., Alabi, T. and Francis, F., 2017. Microbiological load of edible insects found in Belgium. *Insects* 8: 12. <https://doi.org/10.3390/insects8010012>
- Casada, M.E., Ram, M.S. and Flinn, P.W., 2007. Thermal design of shipping containers for beneficial insects. *Applied Engineering in Agriculture* 24: 63-70.
- Cayre, M., Scotto-Lomassese, S., Malaterre, J., Strambi, C. and Strambi, A., 2007. Understanding the regulation and function of adult neurogenesis: contribution from an insect model, the house cricket. *Chemical Senses* 32: 385-395.
- Centeno Filho, B.L., Limberger, G.M., Esteves, K.P., Fonseca, D.B. da and Maciel, F.E., 2023. Mortality, metabolic rate, and oviposition of *Gryllus* (*Gryllus*) *assimilis* (Fabricius, 1775) (Orthoptera: Gryllidae) females under constant and fluctuating warm temperatures. *Journal of Thermal Biology* 114: 103574.
- Centeno Filho, B.L., Limberger, G.M., Fonseca, D.B. da and Maciel, F.E., 2022. Impact of warmer constant and fluctuating temperatures in the male Jamaican field cricket, *Gryllus assimilis* (Fabricius, 1775) (Orthoptera: Gryllidae). *Journal of Thermal Biology* 105: 103145.

- Chen, Q., Wen, M., Li, J., Zhou, H., Jin, S., Zhou, J.-J., Wang, Y. and Ren, B., 2019. Involvement of heat shock protein 40 in the wing dimorphism of the house cricket *Acheta domestica*. *Journal of Insect Physiology* 114: 35-44.
- Chipchase, K.M., Enders, A.M., Jacobs, E.G., Hughes, M.R. and Killian, K.A., 2021. Effect of a single cold stress exposure on the reproductive behavior of male crickets. *Journal of Insect Physiology* 133: 104287.
- Cicche, T., 2007. The biology and genome of *Heterorhabditis bacteriophora*. WormBook: The online review of *C. elegans* biology. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK19785/> (Accessed July 22, 2023).
- Clifford, C., 1985. The biology of egg production in the house cricket, *Acheta domestica* L. LSU Historical Dissertations and Theses. 4046. https://repository.lsu.edu/gradschool_disstheses/4046
- Clifford, C.W. and Woodring, J.P., 1990. Methods for rearing the house cricket, *Acheta domestica* (L.), along with baseline values for feeding rates, growth rates, development times, and blood composition. *Journal of Applied Entomology* 109: 1-14.
- Collavo, A., Glew, R., Huang, Y., Chuang, L., Bosse, R. and Paoletti, M., 2005. House cricket small-scale farming. In: Paoletti, M.G. (ed.) *Ecological implications of minilivestock*. CRC Press, Boca Raton, FL, USA, pp. 533-558.
- Conte, F., Voslarova, E., Vecerek, V., Elwood, R.W., Coluccio, P., Pugliese, M. and Passantino, A., 2021. Humane slaughter of edible decapod crustaceans. *Animals* 11: 1089. <https://doi.org/10.3390/ani11041089>
- Cordaux, R., Paces-Fessy, M., Raimond, M., Michel-Salzat, A., Zimmer, M. and Bouchon, D., 2007. Molecular characterization and evolution of arthropod-pathogenic *Rickettsiella* bacteria. *Applied and Environmental Microbiology* 73: 5045-5047.
- Cortes Ortiz, J.A., Ruiz, A.T., Morales-Ramos, J.A., Thomas, M., Rojas, M.G., Tomberlin, J.K., Yi, L., Han, R., Giroud, L. and Jullien, R.L., 2016. Insect mass production technologies. In: Dossey, A.T., Morales-Ramos, J.A. and Rojas, M.G. (eds.) *Insects as sustainable food ingredients: production, processing and food applications*. Academic Press, Cambridge, MA, USA, pp. 153-201.
- Cottrell, T.E. and Yeargan, K.V., 1998. Effect of pollen on *Coleomegilla maculata* (Coleoptera: Coccinellidae) population density, predation, and cannibalism in sweet corn. *Environmental Entomology* 27: 1402-1410.
- Cranshaw, W.S. and Zimmerman, R., 2013. Insect parasitic nematodes. In: *Insect series | Home and garden. Fact sheet no. 5.573*. Colorado State University Extension. Available at: <https://www.extension.colostate.edu/docs/pubs/insect/05573.pdf>.
- Cricket King, n.d. Cricket king. Available at: <https://www.cricketking.com.au/>.
- Crocker, K.C. and Hunter, M.D., 2018. Social density, but not sex ratio, drives ecdysteroid hormone provisioning to eggs by female house crickets (*Acheta domestica*). *Ecology and Evolution* 8: 10257-10265.
- da Rosa Machado, C. and Thys, R.C.S., 2019. Cricket powder (*Gryllus assimilis*) as a new alternative protein source for gluten-free breads. *Innovative Food Science & Emerging Technologies* 56: 102180.
- da Silva, C., Dunphy, G.B. and Rau, M.E., 2000. Interaction of hemocytes and prophenoloxidase system of fifth instar nymphs of *Acheta domestica* with bacteria. *Developmental & Comparative Immunology* 24: 367-379.
- Dalos, J., Royauté, R., Hedrick, A.V. and Dochtermann, N.A., 2022. Phylogenetic conservation of behavioral variation and behavioral syndromes. *Journal of Evolutionary Biology* 35: 311-321.
- Dambach, M., 1989. Vibrational responses. In: Huber, F. (ed.) *Cricket behavior and neurobiology*. Cornell University Press, Ithaca, NY, USA, pp. 178-197.
- Davenport, A.P. and Evans, P.D., 1984. Stress-induced changes in the octopamine levels of insect haemolymph. *Insect Biochemistry* 14: 135-143.
- Dawkins, M.S., 1990. From an animal's point of view: motivation, fitness, and animal welfare. *Behavioral and Brain Sciences* 13: 1-9.
- Dawkins, M.S., 2008. The science of animal suffering. *Ethology* 114: 937-945.
- Dawkins, M.S., 2017. Animal welfare with and without consciousness. *Journal of Zoology* 301: 1-10.
- de Farias-Martins, F., Sperber, C.F., Albeny-Simões, D., Breaux, J.A., Fianco, M. and Szinwelski, N., 2017. Forest litter crickets prefer higher substrate moisture for oviposition: evidence from field and lab experiments. *PLOS ONE* 12: e0185800.
- de Goede, D.M., Erens, J., Kapsomenou, E. and Peters, M., 2013. Large scale insect rearing and animal welfare. In: Röcklinsberg, H. and Sandin, P. (eds.) *The ethics of consumption*. Wageningen Academic Publishers, Wageningen, The Netherlands, pp. 236-243.
- de Graeff, N., Jongsma, K.R., Johnston, J., Hartley, S. and Breddenoord, A.L., 2019. The ethics of genome editing in non-human animals: a systematic review of reasons reported in the academic literature. *Philosophical Transactions of the Royal Society B* 374: 20180106.
- de Jong, B. and Nikolik, G., 2021. No longer crawling: insect protein to come of age in the 2020s. RaboResearch, Rabobank, Utrecht, the Netherlands.
- de Jong, I.C. and Gunnink, H., 2019. Effects of a commercial broiler enrichment programme with or without natural

- light on behaviour and other welfare indicators. *Animal* 13: 384-391.
- de Miranda, J.R., Granberg, F., Low, M., Onorati, P., Semberg, E., Jansson, A. and Berggren, Å., 2021a. Virus diversity and loads in crickets reared for feed: implications for husbandry. *Frontiers in Veterinary Science* 8: 642085.
- de Miranda, J.R., Granberg, F., Onorati, P., Jansson, A. and Berggren, Å., 2021b. Virus prospecting in crickets – discovery and strain divergence of a novel iflavivirus in wild and cultivated *Acheta domesticus*. *Viruses* 13: 364.
- DeGrazia, D., 1996. Taking animals seriously: mental life and moral status. Cambridge University Press, Cambridge, UK.
- Delvendahl, N., Rumpold, B.A. and Langen, N., 2022. Edible insects as food – insect welfare and ethical aspects from a consumer perspective. *Insects* 13: 121. <https://doi.org/10.3390/insects13020121>
- Dizor, G., 2022. Do crickets eat each other? Available at: <https://whatanimalseat.com/do-crickets-eat-each-other/>
- Dochtermann, N.A. and Nelson, A.B., 2014. Multiple facets of exploratory behavior in house crickets (*Acheta domesticus*): split personalities or simply different behaviors? *Ethology* 120: 1110-1117.
- Doyle, R.E., Fisher, A.D., Hinch, G.N., Boissy, A. and Lee, C., 2010. Release from restraint generates a positive judgement bias in sheep. *Applied Animal Behaviour Science* 122: 28-34.
- Duffield, K.R., Hunt, J., Sadd, B.M., Sakaluk, S.K., Oppert, B., Rosario, K., Behle, R.W. and Ramirez, J.L., 2021. Active and covert infections of cricket iridovirus and *Acheta domesticus* densovirus in reared *Gryllobates sigillatus* crickets. *Frontiers in Microbiology* 12. <https://doi.org/10.3389/fmicb.2021.780796>
- Dumont, M., Kim, S., Ozbas, M. and Rivera, M., 2016. Bt Toxin effects on cricket mortality. *Journal of Biological Sciences* 2: 9-13.
- Durrant, J., Green, M.P. and Jones, T.M., 2020. Dim artificial light at night reduces the cellular immune response of the black field cricket, *Teleogryllus commodus*. *Insect Science* 27: 571-582.
- Durst, P.B. and Hanboonsong, Y., 2015. Small-scale production of edible insects for enhanced food security and rural livelihoods: experience from Thailand and Lao People's Democratic Republic. *Journal of Insects as Food and Feed* 1: 25-31.
- EFSA, 2015. Risk profile related to production and consumption of insects as food and feed. *EFSA Journal* 13: 4257-4257.
- Eilenberg, J. and Jensen, A.B., 2018. Prevention and management of diseases in terrestrial invertebrates. In: Hajek, A.E. and Shapiro-Ilan, D.I. (eds.) *Ecology of invertebrate diseases*. John Wiley & Sons Inc., Hoboken, NJ, USA, pp. 495-526.
- Eilenberg, J., van Oers, M.M., Jensen, A.B., Lecocq, A., Maciel-Vergara, G., Santacoloma, L.P.A., van Loon, J.J.A. and Hesketh, H., 2018. Towards a coordination of European activities to diagnose and manage insect diseases in production facilities. *Journal of Insects as Food and Feed* 4: 157-166.
- Eilenberg, J., Vlask, J.M., Nielsen-LeRoux, C., Cappellozza, S. and Jensen, A.B., 2015. Diseases in insects produced for food and feed. *Journal of Insects as Food and Feed* 1: 85-168.
- Eisemann, C.H., Jorgensen, W.K., Merritt, D.J., Rice, M.J., Cribb, B.W., Webb, P.D. and Zalucki, M.P., 1984. Do insects feel pain? – A biological view. *Experientia* 40: 164-167.
- England, S.J. and Robert, D., 2022. The ecology of electricity and electroreception. *Biological Reviews* 97: 383-413.
- Enkerlin, W.R. and Quinlan, M.M., 2002. Development of an international standard to facilitate the transboundary shipment of sterile insects. In: *Proceedings of 6th International Fruit Fly Symposium*. Stellenbosch, South Africa, pp. 203-212.
- EntoCube, n.d. Farming. Available at: <https://entocube.com/en/farming/>
- Entomofarms, 2016. Farming crickets. Available at: <https://entomofarms.com/farming-crickets/>
- Erens, J., Es van, S., Haverkort, F., Kapsomenou, E. and Luijben, A., 2012. A bug's life: large-scale insect rearing in relation to animal welfare pp 57. Wageningen University & Research, Wageningen, The Netherlands.
- Eriksson, T. and Picard, C.J., 2021. Genetic and genomic selection in insects as food and feed. *Journal of Insects as Food and Feed* 7: 661-682.
- Eurocrickets, n.d. Live crickets. Available at: <https://eurocrickets.com/production-frozen-crickets-cricket-flour/live-crickets-live-feeder-insects-feeding-insects/?v=79cball85463>
- Farm Animal Welfare Council, 2009. Farm animal welfare in Great Britain: past, present and future. Available at: https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/319292/Farm_Animal_Welfare_in_Great_Britain_-_Past_Present_and_Future.pdf
- Fayer, R. and Nerad, T., 1996. Effects of low temperatures on viability of *Cryptosporidium parvum* oocysts. *Applied and Environmental Microbiology* 62: 1431-1433.
- Fernandez-Cassi, X., Supeanu, A., Jansson, A., Boqvist, S. and Vagsholm, I., 2018. Novel foods: a risk profile for the house cricket (*Acheta domesticus*). *EFSA Journal* 16: e16082. <https://doi.org/10.2903/j.efsa.2018.e16082>
- Fernandez-Cassi, X., Supeanu, A., Vaga, M., Jansson, A., Boqvist, S. and Vagsholm, I., 2019. The house cricket

- (*Acheta domesticus*) as a novel food: a risk profile. *Journal of Insects as Food and Feed* 5: 137-157.
- Ferreira, T. and Malan, A.P., 2014. Xenorhabdus and Photorhabdus, bacterial symbionts of the entomopathogenic nematodes Steinernema and Heterorhabditis and their in vitro liquid mass culture: a review. *African Entomology Entomological Society of Southern Africa* 22: 1-14.
- Finke, M.D., 2002. Complete nutrient composition of commercially raised invertebrates used as food for insectivores. *Zoo Biology* 21: 269-285.
- Finke, M.D., Dunham, S.U. and Cole, J.S., 2004. Evaluation of various calcium-fortified high moisture commercial products for improving the calcium content of crickets, *Acheta domesticus*. *Journal of Herpetological Medicine and Surgery* 14: 17-20.
- Finke, M.D., Dunham, S.U. and Kwabi, C.A., 2005. Evaluation of four dry commercial gut loading products for improving the calcium content of crickets, *Acheta domesticus*. *Journal of Herpetological Medicine and Surgery* 15: 7-12.
- Fischer, B., 2021. *Animal ethics: a contemporary introduction*. Routledge, New York, NY, USA.
- Fischer, B. and Sandall, E., 2023. If adult insects matter, how much do juveniles matter? Rethink Priorities. <https://rethinkpriorities.org/publications/how-much-do-juveniles-matter>
- Fontanetti, C.S. and Zefa, E., 2000. Morphological characterization of the proventriculus of *Gryllus assimilis* Fabricius (Orthoptera, Gryllidae). *Revista Brasileira de Zoologia* 17: 193-198.
- Fréchette, M., Lefavre, D. and Frechette, M., 1995. On self-thinning in animals. *Oikos* 73: 425-425.
- Gałęcki, R. and Sokół, R., 2019. A parasitological evaluation of edible insects and their role in the transmission of parasitic diseases to humans and animals. *PLOS ONE* 14: e0219303.
- Gangwere, S.K., 1961. A monograph on food selection in Orthoptera. *Transactions of the American Entomological Society* 87: 67-230.
- Germ, M. and Kral, K., 1995. Influence of visual deprivation on levels of dopamine and serotonin in the visual system of house crickets, *Acheta domesticus*. *Journal of Insect Physiology* 41: 57-63.
- Ghosal, K., Gupta, M. and Killian, K.A., 2009. Agonistic behavior enhances adult neurogenesis in male *Acheta domesticus* crickets. *Journal of Experimental Biology* 212: 2045-2056.
- Ghouri, A.S.K., 1961. Home and distribution of the house cricket *Acheta domesticus* L. *Nature* 192: 1000-1000.
- Ghouri, A.S.K. and McFarlane, J.E., 1958a. Occurrence of a macropterous form of *Grylodes sigillatus* (Walker) (Orthoptera: Gryllidae) in laboratory culture. *Canadian Journal of Zoology* 36: 837-838.
- Ghouri, A.S.K. and McFarlane, J.E., 1958b. Observations on the development of crickets. *The Canadian Entomologist* 90: 158-165.
- Gibbons, M., Crump, A., Barrett, M., Sarlak, S., Birch, J. and Chittka, L., 2022. Chapter three – Can insects feel pain? A review of the neural and behavioural evidence. In: Jurenka, R. (ed.) *Advances in insect physiology*. Academic Press, Boca Raton, FL, USA, pp. 155-229. <https://doi.org/10.1016/bs.aiip.2022.10.001>
- Gibbons, M. and Sarlak, S., 2020. Inhibition of pain or response to injury in invertebrates and vertebrates. *Animal Sentience*: 29. <https://doi.org/10.51291/2377-7478.1649>
- Gibson, W.T., Gonzalez, C.R., Fernandez, C., Ramasamy, L., Tabachnik, T., Du, R.R., Felsen, P.D., Maire, M.R., Perona, P. and Anderson, D.J., 2015. Behavioral responses to a repetitive visual threat stimulus express a persistent state of defensive arousal in *Drosophila*. *Current Biology* 25: 1401-1415.
- Gjerris, M., Gamborg, C. and Röcklinsberg, H., 2016. Ethical aspects of insect production for food and feed. *Journal of Insects as Food and Feed* 2: 101-110.
- Gooley, Z.C. and Gooley, A.C., 2021. Metabolic effects of anesthetics (cold, CO₂, and isoflurane) and captivity conditions in isolated honey bee (*Apis mellifera*) foragers under different ambient temperatures. *Journal of Apicultural Research* 62: 1052-1060.
- Gray, D.A., 1997. Female house crickets, *Acheta domesticus*, prefer the chirps of large males. *Animal Behaviour* 54: 1553-1562.
- Gray, D.A., 1998. Sex differences in susceptibility of house crickets, *Acheta domesticus*, to experimental infection with *Serratia liquefaciens*. *Journal of Invertebrate Pathology* 71: 288-289.
- Gruntenko, N.E., Khlebodarova, T.M., Vasenkova, I.A., Sukhanova, M.J., Wilson, T.G. and Rauschenbach, I.Y., 2000. Stress-reactivity of a *Drosophila melanogaster* strain with impaired juvenile hormone action. *Journal of Insect Physiology* 46: 451-456.
- Gunkel, C. and Lewbart, G.A., 2007. Invertebrates. In: West, G., Heard, D. and Caultkett, N. (eds.) *Zoo animal and wildlife immobilization and anesthesia*. Blackwell Publishing, Ames, IA, USA, pp. 147-158.
- Gutiérrez, Y., Fresch, M., Ott, D., Brockmeyer, J. and Scherber, C., 2020. Diet composition and social environment determine food consumption, phenotype and fecundity in an omnivorous insect. *Royal Society Open Science* 7: 200100-200100.
- Hack, M.A., 1997a. The energetic costs of fighting in the house cricket, *Acheta domesticus* L. *Behavioral Ecology* 8: 28-36.

- Hack, M.A., 1997b. Assessment strategies in the contests of male crickets, *Acheta domesticus* (L.). *Animal Behaviour* 53: 733-747.
- Halloran, A., Megido, R.C., Oloo, J., Weigel, T., Nsevolo, P. and Francis, F., 2018. Comparative aspects of cricket farming in Thailand, Cambodia, Lao People's Democratic Republic, Democratic Republic of the Congo and Kenya. *Journal of Insects as Food and Feed* 4: 101-114.
- Halloran, A., Roos, N., Flore, R. and Hanboonsong, Y., 2016. The development of the edible cricket industry in Thailand. *Journal of Insects as Food and Feed* 2: 91-100.
- Hanboonsong, Y., Jamjanya, T. and Durst, P.B., 2013. Six-legged livestock: edible insect farming, collection and marketing in Thailand. *FAO, Regional Office for Asia and the Pacific, Bangkok, Thailand*.
- Harris, J., 2016. Chapter 8: insect light traps. In: Heeps, J. (ed.) *Insect management for food storage and processing*. Elsevier, Amsterdam, the Netherlands, pp. 55-66.
- Harris, J.W. and Woodring, J., 1992. Effects of stress, age, season, and source colony on levels of octopamine, dopamine and serotonin in the honey bee (*Apis mellifera* L.) brain. *Journal of Insect Physiology* 38: 29-35.
- Harrison, J.F., Greenlee, K.J. and Verberk, W.C.E.P., 2018. Functional hypoxia in insects: definition, assessment, and consequences for physiology, ecology, and evolution. *Annual Review of Entomology* 63: 303-325.
- Harrison, S.J., Raubenheimer, D., Simpson, S.J., Godin, J.-G.J. and Bertram, S.M., 2014. Towards a synthesis of frameworks in nutritional ecology: interacting effects of protein, carbohydrate and phosphorus on field cricket fitness. *Proceedings of the Royal Society B: Biological Sciences* 281: 20140539.
- Harsányi, E., Jushász, C., Kovács, E., Huzsvai, L., Pintér, R., Fekete, G., Varga, Z.I., Aleksza, L. and Gyuricza, C., 2020. Evaluation of organic wastes as substrates for rearing *Zophobas morio*, *Tenebrio molitor*, and *Acheta domesticus* larvae as alternative feed supplements. *Insects* 11: 604.
- Hermansa, N., Kustra, K., Trela, M. and Lis, M., 2022. Influence of ambient temperature on reproduction of the red-headed cricket (*Gryllus assimilis*). *Science, Technology and Innovation* 15: 38-44.
- Hertel, T.W., 2015. The challenges of sustainably feeding a growing planet. *Food Security* 7: 185-198.
- Hilbeck, A., Defarge, N., Bohn, T., Krautter, M., Conradin, C., Amiel, C., Panoff, J.-M. and Trtikova, M., 2018. Impact of antibiotics on efficacy of cry toxins produced in two different genetically modified Bt maize varieties in two Lepidopteran herbivore species, *Ostrinia nubilalis* and *Spodoptera littoralis*. *Toxins* 10: 489-489.
- Hirashima, A. and Eto, M., 1993. Effect of stress on levels of octopamine, dopamine and serotonin in the American cockroach (*Periplaneta americana* L.). *Comparative Biochemistry and Physiology Part C: Comparative Pharmacology* 105: 279-284.
- Hogg, D.W., Husic, M. and Lovejoy, D.A., 2018. Corticotropin-releasing hormone peptide family. In: Huhtaniemi, I. and Martini, L. (eds.) *Encyclopedia of endocrine diseases*, volume 3. Academic press, Boca Raton, FL, USA, pp. 16-26.
- Hölldobler, B. and Wilson, E.O., 1990. *The ants*. Belknap Press of Harvard University Press, Cambridge, MA, USA.
- Horch, H.W., Mito, T., Popadic, A., Ohuchi, H. and Noji, S., 2017. *The cricket as a model organism*. Springer, Tokyo, Japan.
- Houslay, T.M., Hunt, J., Tinsley, M.C. and Bussière, L.F., 2015. Sex differences in the effects of juvenile and adult diet on age-dependent reproductive effort. *Journal of Evolutionary Biology* 28: 1067-1079.
- Huger, A.M., 1985. A new virus disease of crickets (Orthoptera: Gryllidae) causing macronucleosis of fatbody. *Journal of Invertebrate Pathology* 45: 108-111.
- Hunt, E.P., Jackson, C.W. and Newland, P.L., 2005. "Electropellancy" behaviour of *Periplaneta americana* exposed to friction charged dielectric surfaces. *Journal of Electrostatics* 63: 853-859.
- Hunt, J., Brooks, R., Jennions, M.D., Smith, M.J., Bentsen, C.L. and Bussière, L.F., 2004. High-quality male field crickets invest heavily in sexual display but die young. *Nature* 432: 1024-1027.
- Huntingford, F., Rey, S. and Quaggiotto, M.-M., 2020. Behavioural fever, fish welfare and what farmers and fishers know. *Applied Animal Behaviour Science* 231: 105090. <https://doi.org/10.1016/j.applanim.2020.105090>
- Ibrahim, M.A., Griko, N., Junker, M. and Bulla, L.A., 2010. *Bacillus thuringiensis*. *Bioengineered Bugs* 1: 31-50.
- Inglis, G.D., Johnson, D.L. and Goettel, M.S., 1996. Effects of temperature and thermoregulation on mycosis by *Beauveria bassiana* in grasshoppers. *Biological Control* 7: 131-139.
- IPIFF, 2019. Ensuring high standards of animal welfare in insect production. Available at: <https://ipiff.org/wp-content/uploads/2019/02/Animal-Welfare-in-Insect-Production.pdf>
- IPIFF, 2022. IPIFF missions. Available at: <https://ipiff.org/about-ipiff/>
- Ismail, I.I., 1978. Biological and ecological studies on the house cricket, *Acheta domesticus* L. (Orthopt., Gryllidae) in Giza region, Egypt. *Zeitschrift für Angewandte Entomologie* 85: 230-235.
- Ivy, T.M., Johnson, J.C. and Sakaluk, S.K., 1999. Hydration benefits to courtship feeding in crickets. *Proceedings of the Royal Society B: Biological Sciences* 266: 1523.

- Ivy, T.M. and Sakaluk, S.K., 2005. Polyandry promotes enhanced offspring survival in decorated crickets. *Evolution* 59: 152-159.
- Jaffe, K., Blau, S. and Zabala, N., 1992. Amino acid levels during learning and memory consolidation of an aversive conditioning task in crickets. *Pharmacology Biochemistry and Behavior* 43: 205-214.
- Jakob, N.J., Kleespies, R.G., Tidona, C.A., Müller, K., Gelderblom, H.R. and Darai, G., 2002. Comparative analysis of the genome and host range characteristics of two insect iridoviruses: *Chilo iridescent* virus and a cricket iridovirus isolate. *Journal of General Virology Microbiology Society* 83: 463-470.
- Jensen, K., Kristensen, T.N., Heckmann, L.H. and Sørensen, J.G., 2018. Breeding and maintaining high-quality insects. In: van Huis, A. and Tomberlin, J.T. (eds.) *Insects as food and feed: from production to consumption*. Wageningen Academic Publishers, Wageningen, the Netherlands, pp. 175-198.
- Jobin, L.J., 1961. The nymphal instars of certain American species of *Gryllus* (Orthoptera: Gryllidae). McGill University. Available at: <https://escholarship.mcgill.ca/concern/theses/c82l9p62t>.
- Johnson, D.J. and Stahlschmidt, Z.R., 2020. City limits: heat tolerance is influenced by body size and hydration state in an urban ant community. *Ecology and Evolution* 10: 4944-4955.
- Jonsson, T., 2017. Metabolic theory predicts animal self-thinning. *Journal of Animal Ecology* 86: 645-653.
- Joosten, L., Lecocq, A., Jensen, A.B., Haenen, O., Schmitt, E. and Eilenberg, J., 2020. Review of insect pathogen risks for the black soldier fly (*Hermetia illucens*) and guidelines for reliable production. *Entomologia Experimentalis et Applicata* 168: 432-447.
- Josh's Frogs, n.d. How to store feeder insects. Available at: <https://www.joshsfrogs.com/catalog/blog/2020/08/how-to-store-feeder-insects/>
- Judge, K.A. and Bonanno, V.L., 2008. Male weaponry in a fighting cricket. *PLOS ONE* 3: e3980.
- Kataoka, K., Minei, R., Ide, K., Ogura, A., Takeyama, H., Takeda, M., Suzuki, T., Yura, K. and Asahi, T., 2020. The draft genome dataset of the Asian cricket *Teleogryllus occipitalis* for molecular research toward entomophagy. *Frontiers in Genetics*: 11. <https://doi.org/10.3389/fgene.2020.00470>
- Kestin, S.C., Gordon, S., Su, G. and Sørensen, P., 2001. Relationships in broiler chickens between lameness, liveweight, growth rate and age. *Veterinary Record* 148: 195-197.
- Kettlewell, P.J. and Hallworth, R.N., 1990. Electrical stunning of chickens. *Journal of Agricultural Engineering Research* 47: 139-151.
- Kienzle, M. and Kappen, K., 2022. Exposure to UVB light increases the vitamin D concentration in two species of feeder crickets (*Acheta domesticus* and *Gryllodes sigillatus*).
- Kieruzel, M., 1976. The selection of shelter place by the house cricket. *Acta neurobiologiae experimentalis* 36: 561-580.
- Killian, K.A., Snell, L.C., Ammarell, R. and Crist, T.O., 2006. Suppression of escape behaviour during mating in the cricket *Acheta domesticus*. *Animal Behaviour* 72: 487-502.
- King, C., Jones, H.I. and Tay, C.Y., 2013. Arthropod intermediate hosts of *Abbreviata antarctica* (Nematoda: Physalopteridae) in Australia. *The Journal of Parasitology* 99: 708-711.
- Kirkden, R.D. and Pajor, E.A., 2006. Using preference, motivation and aversion tests to ask scientific questions about animals' feelings. *Applied Animal Behaviour Science* 100: 29-47.
- Klaassen, M. and Wille, M., 2023. The plight and role of wild birds in the current bird flu panzootic. *Nature Ecology and Evolution* 7: 1541-1542.
- Kleespies, R.G., Tidona, C.A. and Darai, G., 1999. Characterization of a new iridovirus isolated from crickets and investigations on the host range. *Journal of Invertebrate Pathology* 73: 84-90.
- Klein, C. and Barron, A.B., 2016. Insects have the capacity for subjective experience. *Animal Sentience*: 1. <https://doi.org/10.51291/2377-7478.1113>
- Klobučar, T. and Fisher, D.N., 2023. When do we start caring about insect welfare? *Neotropical Entomology* 52: 5-10.
- Klok, C.J. and Harrison, J.F., 2009. Atmospheric hypoxia limits selection for large body size in insects. *PLOS ONE* 4: e3876.
- Knutsson, S. and Munthe, C., 2017. A virtue of precaution regarding the moral status of animals with uncertain sentience. *Journal of Agricultural and Environmental Ethics* 30: 213-224.
- Kollack-Walker, S., Heidi, E.W.D. and Akil, H., 2000. Central stress neurocircuits. In: Fink, G. (ed.) *Encyclopedia of stress*. Academic Press, Boca Raton, FL, USA, pp. 414-423.
- Kortsmit, Y., van der Bruggen, M., Wertheim, B., Dicke, M., Beukeboom, L.W. and van Loon, J.J.A., 2023. Behaviour of two fly species reared for livestock feed: optimising production and insect welfare. *Journal of Insects as Food and Feed* 9: 149-169.
- Kyllönen, T. and Manzanares, A., 2022. Insect production and operations management. In: Veldkamp, T., Claeys, J., Haenen, O.L.M., van Loon and Spranghers, T. (eds.) *The basics of edible insect rearing*. Wageningen Academic Publishers, Wageningen, the Netherlands.
- Lachenicht, M.W., Clusella-Trullas, S., Boardman, L., Le Roux, C. and Terblanche, J.S., 2010. Effects of acclimation temperature on thermal tolerance, locomotion performance and

- respiratory metabolism in *Acheta domesticus* L. (Orthoptera: Gryllidae). *Journal of Insect Physiology* 56: 822-830.
- Lambert, H., Elwin, A. and D'Cruze, N., 2021. Wouldn't hurt a fly? A review of insect cognition and sentience in relation to their use as food and feed. *Applied Animal Behaviour Science* 243: 105432-105432.
- Leander, B.S., Clopton, R.E. and Keeling, P.J., 2003. Phylogeny of gregarines (Apicomplexa) as inferred from small-subunit rDNA and β -tubulin. *International Journal of Systematic and Evolutionary Microbiology* 53: 345-354.
- Letendre, C., Duffield, K.R., Sadd, B.M., Sakaluk, S.K., House, C.M. and Hunt, J., 2022. Genetic covariance in immune measures and pathogen resistance in decorated crickets is sex and pathogen specific. *Journal of Animal Ecology* 91: 1471-1488.
- Li, X.-D., Xin, L., Rong, W.-T., Liu, X.-Y., Deng, W.-A., Qin, Y.-C. and Li, X.-L., 2021. Effect of heavy metals pollution on the composition and diversity of the intestinal microbial community of a pygmy grasshopper (*Eucrotettix ocellatus*). *Ecotoxicology and Environmental Safety* 223: 112582-112582.
- Li, X.-Y., Cowles, R.S., Cowles, E.A., Gaugler, R. and Cox-Foster, D.L., 2007. Relationship between the successful infection by entomopathogenic nematodes and the host immune response. *International Journal for Parasitology* 37: 365-374.
- Limberger, G.M., Nery, L.E.M. and Fonseca, D.B. da, 2021. Reproductive status modulates mortality rate, lifespan and egg production, but do not the physiological aging in the field cricket *Gryllus assimilis* (Fabricius, 1775) (Orthoptera: Gryllidae). *Neotropical Entomology* 50: 237-246.
- Lin, Y.-H., Silven, J.J.M., Wybouw, N., Fandino, R.A., Dekker, H.L., Vogel, H., Wu, Y.-L., de Koster, C., Große-Wilde, E., Haring, M.A., Schuurink, R.C. and Allmann, S., 2023. A salivary GMC oxidoreductase of *Manduca sexta* re-arranges the green leaf volatile profile of its host plant. *Nature Communications* 14: 3666.
- Liu, K., Li, Y., Jousset, F.-X., Zadori, Z., Szelei, J., Yu, Q., Pham, H.T., Lépine, F., Bergoin, M. and Tijssen, P., 2011. The *Acheta domesticus* densovirus, isolated from the European house cricket, has evolved an expression strategy unique among parvoviruses. *Journal of Virology American Society for Microbiology (ASM)* 85: 10069.
- Loaiza, V., Jonas, J.L. and Joern, A., 2008. Does dietary P affect feeding and performance in the mixed-feeding grasshopper (Acrididae) *Melanoplus bivitattus*? *Environmental Entomology* 37: 333-339.
- Loranger, M.J. and Bertram, S.M., 2016a. The effect of sire dominance and aggression on fitness measures in a field cricket (*Gryllus assimilis*). *Animal Behaviour* 119: 135-142.
- Loranger, M.J. and Bertram, S.M., 2016b. The effect of male dominance on female choice in a field cricket (*Gryllus assimilis*). *Animal Behaviour* 114: 45-52.
- Low, M., Eksell, I., Jansson, A. and Berggren, Å., 2022. Viral infection changes the expression of personality traits in an insect species reared for consumption. *Scientific Reports* 12: 9503.
- Lundy, M.E. and Parrella, M.P., 2015. Crickets are not a free lunch: protein capture from scalable organic side-streams via high-density populations of *Acheta domesticus*. *PLOS ONE* 10: e0118785.
- Luong, L.T., Kortet, R. and Hedrick, A.V., 2005. Prevalence and intensity of *Cephalobium microbivorum* (Nematoda: Diplogasterida) infection in three species of *Gryllus* field crickets. *Parasitology Research* 97: 336-339.
- Lyn, J.C., Naikhwah, W., Aksenov, V. and Rollo, C.D., 2011. Influence of two methods of dietary restriction on life history features and aging of the cricket *Acheta domesticus*. *Age* 33: 509-522.
- Lyu, H. and Mizunami, M., 2022. Conditioned taste aversion in the cricket *Gryllus bimaculatus*. *Scientific Reports* 12: 9751-9751.
- Maciel-Vergara, G., Jensen, A.B., Lecocq, A. and Eilenberg, J., 2021. Diseases in edible insect rearing systems. *Journal of Insects as Food and Feed* 7: 621-638.
- Maciel-Vergara, G. and Ros, V.I.D., 2017. Viruses of insects reared for food and feed. *Journal of Invertebrate Pathology* 147: 60-75.
- MacMillan, H.A., Norgard, M., MacLean, H.J., Overgaard, J. and Williams, C.J.A., 2017. A critical test of *Drosophila* anaesthetics: isoflurane and sevoflurane are benign alternatives to cold and CO₂. *Journal of Insect Physiology* 101: 97-106.
- MacMillan, H.A. and Sinclair, B., 2011. Mechanisms underlying insect chill-coma. *Journal of Insect Physiology* 57: 12-20.
- Magara, H.J.O., Niassy, S., Ayieko, M.A., Mukundamago, M., Egonyu, J.P., Tanga, C.M., Kimathi, E.K., Ongere, J.O., Fiaboe, K.K.M., Hugel, S., Orinda, M.A., Roos, N. and Ekesi, S., 2021. Edible crickets (Orthoptera) around the world: distribution, nutritional value, and other benefits—a review. *Frontiers in Nutrition* 7. <https://doi.org/10.3389/fnut.2020.537915>
- Mahavidanage, S., Fuciarelli, T.M., Li, X. and Rollo, C.D., 2023. The effects of rearing density on growth, survival, and starvation resistance of the house cricket *Acheta domesticus*. *Journal of Orthoptera Research* 32: 25-31.
- Makkar, H.P.S., Tran, G., Heuzé, V. and Ankers, P., 2014. State-of-the-art on use of insects as animal feed. *Animal Feed Science and Technology* 197: 1-33.
- Mallard, S. and Barnard, C.J., 2004. Food stress, fluctuating asymmetry and reproductive performance in the gryl-

- lid crickets *Gryllus bimaculatus* and *Gryllodes sigillatus*. Behaviour 141: 219-232.
- Mallory, H.S., Howard, A.F. and Weiss, M.R., 2016. Timing of environmental enrichment affects memory in the house cricket, *Acheta domesticus*. PLOS ONE 11: e0152245.
- Masaki, S. and Walker, T.J., 1987. Cricket life cycles. In: Hecht, M.K., Wallace, B. and Prance, G.T. (eds.) Evolutionary biology. Plenum Publishing Corporation, New York, NY, USA, pp. 349-423.
- Masson, M.V., Tavares, W.d.S., Alves, J.M., Ferreira-Filho, P.J., Barbosa, L.R., Wilcken, C.F. and Zanuncio, J.C., 2020b. Bioecological aspects of the common black field cricket, *Gryllus assimilis* (Orthoptera: Gryllidae) in the laboratory and in Eucalyptus (Myrtaceae) plantations. Journal of Orthoptera Research 29: 83-89.
- Mathad, S.B. and McFarlane, J.E., 1967. Effect of initial rearing at 28 °C followed by 35 °C on wing development in *Gryllodes sigillatus* (Walk.). Canadian Journal of Zoology 45: 135-137.
- Matsumoto, H., Tanaka, K., Noguchi, H. and Hayakawa, Y., 2003. Cause of mortality in insects under severe stress. European Journal of Biochemistry 270: 3469-3476.
- Matsuoka, Y., Nakamura, T., Watanabe, T., Barnett, A.A., Noji, S., Mito, T. and Extavour, C.G., 2021. Establishment of CRISPR/Cas9-based knock-in in a hemimetabolous insect: targeted gene tagging in the cricket *Gryllus bimaculatus*. bioRxiv: 2021.05.10.441399.
- Mazurkiewicz, A., Tumialis, D., Pezowicz, E., Urbanski, J., Galewski, P. and Goral, K., 2013. The effect of density on the breeding optimization of the tropical house cricket *Gryllodes sigillatus* (Walker) (Orthoptera: Gryllidae). Annals of Warsaw University of Life Sciences – SGGW Animal Science 52: 135-139.
- McCluney, K.E. and Date, R.C., 2008. The effects of hydration on growth of the house cricket, *Acheta domesticus*. Journal of Insect Science 8: 32.
- McFarlane, J., Ghouri, A. and Kennard, C., 1959. Water absorption by the eggs of crickets. Canadian Journal of Zoology 37: 391-399.
- Megido, R.C., Alabi, T., Nieuw, C., Blecker, C., Danthine, S., Bogaert, J., Haubruge, É. and Francis, F., 2016. Optimisation of a cheap and residential small-scale production of edible crickets with local by-products as an alternative protein-rich human food source in Ratanakiri Province, Cambodia. Journal of the Science of Food and Agriculture 96: 627-632.
- Mellor, D., 2015. Enhancing animal welfare by creating opportunities for positive affective engagement. New Zealand Veterinary Journal 63: 3-8.
- Mellor, D., 2016. Updating animal welfare thinking: moving beyond the “five freedoms” towards “a life worth living”. Animals 6: 21-21.
- Mellor, D.J., 2017. Operational details of the five domains model and its key applications to the assessment and management of animal welfare. Animals 7: 60.
- Mellor, D.J., Hunt, S. and Gusset, M., 2015. Caring for wildlife: the world zoo and aquarium animal welfare strategy. WAZA Executive Office, Gland, Switzerland.
- Mellor, D.J. and Reid, C.S.W., 1994. Concepts of animal well-being and predicting the impact of procedures on experimental animals. Improving the well-being of animals in the research environment. In: Baker, R.M., Jenkin, G. and Mellor, D.J. (eds.) Improving the well-being of animals in the research environment. Australian and New Zealand Council for the Care of Animals in Research and Teaching, Glen Osmond, Australia, pp. 3-18.
- Meuser, V., Weinhold, L., Hillemacher, S. and Tiemann, I., 2021. Welfare-related behaviors in chickens: characterization of fear and exploration in local and commercial chicken strains. Animals 11: 679-679.
- Miech, P., 2018. Cricket farming: an alternative for producing food and feed in Cambodia. Doctoral thesis. Swedish University of Agricultural Sciences.
- Miech, P., Berggren, Å., Lindberg, J.E., Chhay, T., Khieu, B. and Jansson, A., 2016. Growth and survival of reared Cambodian field crickets (*Teleogryllus testaceus*) fed weeds, agricultural and food industry by-products. Journal of Insects as Food and Feed 2: 285-292.
- Mito, T., Ishimaru, Y., Watanabe, T., Nakamura, T., Ylla, G., Noji, S. and Extavour, C.G., 2022. Cricket: the third domesticated insect. In: Goldstein, B. and Srivastava, M. (eds.) Current topics in developmental biology. Academic Press, Boca Raton, FL, USA, pp. 291-306.
- Modanu, M., Li, L.D.X., Said, H., Rathitharan, N. and Andrade, M.C.B., 2014. Sibling cannibalism in a web-building spider: effects of density and shared environment. Behavioural Processes 106: 12-16.
- Montealegre-Z, F., Jonsson, T. and Robert, D., 2011. Sound radiation and wing mechanics in stridulating field crickets (Orthoptera: Gryllidae). Journal of Experimental Biology 214: 2105-2117.
- Montroy, K., Loranger, M.J. and Bertram, S.M., 2016. Male crickets adjust their aggressive behavior when a female is present. Behavioural Processes 124: 108-114.
- Morales-Ramos, J.A., Rojas, M.G. and Dossey, A.T., 2018. Age-dependent food utilisation of *Acheta domesticus* (Orthoptera: Gryllidae) in small groups at two temperatures. Journal of Insects as Food and Feed 4: 51-60.
- Morales-Ramos, J.A., Rojas, M.G., Dossey, A.T. and Berhow, M., 2020. Self-selection of food ingredients and agricultural by-products by the house cricket, *Acheta domesticus* (Orthoptera: Gryllidae): a holistic approach to develop optimized diets. PLOS ONE 15: e0227400.

- Morrissey, R. and Edwards, J.S., 1979. Neural function in an Alpine grylloblattid: a comparison with the house cricket, *Acheta domesticus*. *Physiological Entomology* 4: 241-250.
- Mott, G., 2016. Operational and biological considerations for the commercial production of crickets. In: van Huis, A. and Tomberlin, J. (eds.) *Insects as food and feed: from production to consumption*. Wageningen Academic Publishers, Wageningen, the Netherlands.
- Murray, M.J., 2012. Euthanasia. In: Lewbart, G.A. (ed.) *Invertebrate medicine*, 2nd edition. Wiley-Blackwell, Ames, IA, USA, pp. 441-444.
- Murtaugh, M.P. and Denlinger, D.L., 1985. Physiological regulation of long-term oviposition in the house cricket, *Acheta domesticus*. *Journal of Insect Physiology* 31: 611-617.
- Nakagaki, B.J. and Defoliart, G.R., 1991. Comparison of diets for mass-rearing *Acheta domesticus* (Orthoptera: Gryllidae) as a novelty food, and comparison of food Conversion efficiency with values reported for livestock. *Journal of Economic Entomology* 84: 891-896.
- Nakajima, Y. and Ogura, A., 2022. Genomics and effective trait candidates of edible insects. *Food Bioscience* 48: 101793-101793.
- Nakamura, T., Ylla, G. and Extavour, C.G., 2022. Genomics and genome editing techniques of crickets, an emerging model insect for biology and food science. *Current Opinion in Insect Science* 50: 100881.
- Neuman-Lee, L.A., Hopkins, G.R., Brodie, E.D. and French, S.S., 2013. Sublethal contaminant exposure alters behavior in a common insect: important implications for trophic transfer. *Journal of Environmental Science and Health, Part B* 48: 442-448.
- Newland, P.L., Al Ghamdi, M.S., Sharkh, S., Aonuma, H. and Jackson, C.W., 2015. Exposure to static electric fields leads to changes in biogenic amine levels in the brains of *Drosophila*. *Proceedings of the Royal Society B: Biological Sciences* 282: 20151198.
- Newland, P.L., Hunt, E., Sharkh, S.M., Hama, N., Takahata, M. and Jackson, C.W., 2008. Static electric field detection and behavioural avoidance in cockroaches. *Journal of Experimental Biology* 211: 3682-3690.
- Nicolas, G. and Sillans, D., 1989. Immediate and latent effects of carbon dioxide on insects. *Annual Review of Entomology* 34: 97-116.
- Nix, P.M. and Bass, M.H., 1973. Biological and toxicological notes on the house cricket. *Agricultural Experiment Station*, Auburn University, Auburn, AL, USA. Available at: <https://aurora.auburn.edu/bitstream/handle/11200/1255/0377LEAF.pdf?sequence=1&isAllowed=y> (Accessed August 2, 2023).
- Nosil, P., 2002. Food fights in house crickets, *Acheta domesticus*, and the effects of body size and hunger level. *Canadian Journal of Zoology* 80: 409-417.
- Nowosielski, J.W. and Patton, R.L., 1963. Studies on circadian rhythm of the house cricket, *Gryllus domesticus* L. *Journal of Insect Physiology* 9: 401-410.
- Nowosielski, J.W. and Patton, R.L., 1965. Life-tables for the house cricket, *Acheta domesticus* L., and the effect of intra-specific factors on longevity. *Journal of Insect Physiology* 11: 201-209.
- Oloo, J.A., Ayieko, M. and Nyongesah, J.M., 2020. *Acheta domesticus* (Cricket) feed resources among smallholder farmers in Lake Victoria region of Kenya. *Food Science & Nutrition* 8: 69-78.
- Olzer, R., Deak, N., Tan, X., Heinen-Kay, J.L. and Zuk, M., 2019. Aggression and mating behavior in wild and captive populations of the house cricket, *Acheta domesticus*. *Journal of Insect Behavior* 32: 89-98.
- Oonincx, D.G.A.B., Itterbeeck, J. van, Heetkamp, M.J.W., Brand, H. van den, Loon, J.J.A. van and Huis, A. van, 2010. An exploration on greenhouse gas and ammonia production by insect species suitable for animal or human consumption. *PLOS ONE* 5: e14445.
- Oonincx, D.G.A.B., van Keulen, P., Finke, M.D., Baines, F.M., Vermeulen, M. and Bosch, G., 2018. Evidence of vitamin D synthesis in insects exposed to UVb light. *Scientific Reports* 8: 10807-10807.
- Oonincx, D.G.A.B., van Broekhoven, S., van Huis, A. and van Loon, J.J.A., 2015. Feed conversion, survival and development, and composition of four insect species on diets composed of food by-products. *PLOS ONE* 10: e0144601.
- Oppert, B., Perkin, L.C., Lorenzen, M. and Dossey, A.T., 2020. Transcriptome analysis of life stages of the house cricket, *Acheta domesticus*, to improve insect crop production. *Scientific Reports* 10: 3471.
- Orchard, I., 1982. Octopamine in insects: neurotransmitter, neurohormone, and neuromodulator. *Canadian Journal of Zoology* 60: 659-669.
- Orchard, I., Loughton, B.G. and Webb, R.A., 1981. Octopamine and short-term hyperlipaemia in the locust. *General and Comparative Endocrinology* 45: 175-180.
- Orinda, M.A., Mosi, R.O., Ayieko, M.A. and Amimo, F.A., 2017. Effects of housing on growth performance of common house cricket (*Acheta domesticus*) and field cricket (*Gryllus bimaculatus*). *Journal of Entomology and Zoology Studies* 5: 1138-1142.
- Orinda, M.A., Oloo, J.A., Ayieko, M.A., Magara, H.J.O., Ekesi, S. and Roos, N., 2021a. Cricket rearing handbook. Services for Science and Education, UK. <https://doi.org/10.14738/eb.86.2020>

- Otte, D., 2007. Australian crickets (Orthoptera: Gryllidae). Academy of Natural Sciences of Philadelphia, Philadelphia, PA, USA.
- Otte, D. and Cade, W., 1976. On the role of olfaction in sexual and interspecies recognition in crickets (*Acheta* and *Gryllus*). *Animal Behaviour* 24: 1-6.
- Pacheco, K. and Bertram, S.M., 2014. How male sound pressure level influences phonotaxis in virgin female Jamaican field crickets (*Gryllus assimilis*). *PeerJ* 2: e437-e437.
- Packer, M.J. and Brady, J., 1990. Efficiency of electric nets as sampling devices for tsetse flies (Diptera: Glossinidae). *Bulletin of Entomological Research* 80: 43-47.
- Palmer, R.A., Chenchiah, I.V. and Robert, D., 2021. Analysis of aerodynamic and electrostatic sensing in mechanoreceptor arthropod hairs. *Journal of Theoretical Biology* 530: 110871.
- Papp, T., Spann, D. and Marschang, R.E., 2014. Development and use of a real-time polymerase chain reaction for the detection of group II invertebrate iridovirus in pet lizards and prey insects. *Journal of Zoo and Wildlife Medicine* 45: 219-227. <https://doi.org/10.1638/2012-0044.1>
- Papp, T. and Marschang, R.E., 2019. Detection and characterization of invertebrate iridoviruses found in reptiles and prey insects in Europe over the past two decades. *Viruses* 1: 600. <https://doi.org/10.3390/v11070600>
- Parajulee, M.N., Defoliart, G.R. and Hogg, D.B., 1993. Model for use in mass-production of *Acheta domesticus* (Orthoptera: Gryllidae) as food. *Journal of Economic Entomology* 86: 1424-1428.
- Parametpisit, P., Panmuang, P., Sonsilphong, A. and Soemphol, C., 2023. Experimental investigation of hybrid thermoelectric evaporative air-cooling system for crickets rearing process. *Indonesian Journal of Electrical Engineering and Computer Science* 29: 1374-1374.
- Patton, R.L., 1967. Oligidic diets for *Acheta domesticus* (Orthoptera: Gryllidae). *Annals of the Entomological Society of America* 60: 1238-1242.
- Patton, R.L., 1978. Growth and development parameters for *Acheta domesticus*. *Annals of the Entomological Society of America* 71: 40-42.
- Pener, M.P., 2016. Allergy to crickets: a review. *Journal of Orthoptera Research* 25: 91-95.
- Pham, H.T., Bergoin, M. and Tijssen, P., 2013a. *Acheta domesticus* volvoxvirus, a novel single-stranded circular DNA virus of the house cricket. *Genome Announcements* 1: e00079-13.
- Pham, H.T., Iwao, H., Bergoin, M. and Tijssen, P., 2013b. New volvoxvirus isolates from *Acheta domesticus* (Japan) and *Gryllus assimilis* (United States). *Genome Announcements* 1: e00328-13.
- Poinapen, D., Konopka, J.K., Umoh, J.U., Norley, C.J.D., McNeil, J.N. and Holdsworth, D.W., 2017. Micro-CT imaging of live insects using carbon dioxide gas-induced hypoxia as anesthetic with minimal impact on certain subsequent life history traits. *BMC Zoology* 2: 9.
- Poinar, G. and Weissman, D.B., 2004. Hairworm and nematode infections of North American Jerusalem crickets, field crickets, and katydids (Orthoptera: Stenopelmatidae, Gryllidae and Tettigoniidae). *Journal of Orthoptera Research* 13: 143-147.
- Poissonnier, L.-A., Simpson, S.J. and Dussutour, A., 2014. Observations of the "egg white injury" in ants. *PLOS ONE* 9: e112801.
- Porter, S.D., Valles, S.M. and Pereira, R.M., 2016. Scavenging crickets (Orthoptera: Gryllidae) transmit solenopsis invicta virus 3 to red imported fire ant (Hymenoptera: Formicidae) colonies. *Florida Entomologist Florida Entomological Society* 99: 811-812.
- Psarianos, M., Fricke, A., Ojha, S., Baldermann, S., Schreiner, M. and Schlüter, O.K., 2022. Effect of narrowband UV-B irradiation on the growth performance of house crickets. *Foods* 11. <https://doi.org/10.3390/foods11213487>
- Raheem, D., Carrascosa, C., Oluwale, O.B., Nieuwland, M., Saraiva, A., Millán, R. and Raposo, A., 2019. Traditional consumption of and rearing edible insects in Africa, Asia and Europe. *Critical Reviews in Food Science and Nutrition* 59: 2169-2188.
- Rapkin, J., Jensen, K., Archer, C.R., House, C.M., Sakaluk, S.K., del Castillo, E. and Hunt, J., 2018. The geometry of nutrient space-based life-history trade-offs: sex-specific effects of macronutrient intake on the trade-off between encapsulation ability and reproductive effort in decorated drickets. *The American Naturalist* 191: 421-552.
- Reginald, K., Wong, Y.R., Shah, S.M.R., Teh, K.F., Freddy Jalin, E.J. and Khan, N.A., 2021. Investigating immune responses of the house cricket, *Acheta domesticus* to pathogenic *Escherichia coli* K1. *Microbes and Infection* 23: 104876.
- Rehn, J.A.G. and Hebard, M., 1915. The genus *Gryllus* (Orthoptera) as found in America. *Proceedings of the Academy of Natural Sciences of Philadelphia* 67. Available at: <https://www.biodiversitylibrary.org/page/1708699#page/312/mode/thumb>.
- Reifer, M.L., Harrison, S.J. and Bertram, S.M., 2018. How dietary protein and carbohydrate influence field cricket development, size and mate attraction signalling. *Animal Behaviour* 139: 137-146.
- Reverberi, M., 2020. Edible insects: cricket farming and processing as an emerging market. *Journal of Insects as Food and Feed* 6: 211-220.

- Reyes Villanueva, F., 2004. Generalidades y potencialidad en biocontrol de las gregarinas entomoparásitas. *Ciencia UANL* 3: 355-360.
- Rhodes, V., Maguire, M., Shetty, M., McAloon, C. and Smeaton, A.F., 2022. Periodicity intensity of the 24 h circadian rhythm in newborn calves show indicators of herd welfare. *Sensors* 22: 5843. <https://doi.org/10.3390/s22155843>
- Riche, M. and Brown, P.B., 1996. Availability of phosphorus from feedstuffs fed to rainbow trout, *Oncorhynchus mykiss*. *Aquaculture* 142: 269-282.
- Rillich, J., Schildberger, K. and Stevenson, P.A., 2010. Octopamine and occupancy: an aminergic mechanism for intruder-resident aggression in crickets. *Proceedings of the Royal Society B: Biological Sciences* 278: 1873-1880.
- Rillich, J. and Stevenson, P.A., 2011. Winning fights induces hyperaggression via the action of the biogenic amine octopamine in crickets. *PLOS ONE* 6: e28891.
- Rioja-Lang, F., Bacon, H., Connor, M. and Dwyer, C.M., 2020. Prioritisation of animal welfare issues in the UK using expert consensus. *Veterinary Record* 187: 490-490.
- Ritchot, C. and McFarlane, J.E., 1961. The b vitamin requirements of the house cricket. *Canadian Journal of Zoology* 39: 11-15.
- Röcklinsberg, H., Gamborg, C. and Gjerris, M., 2017. Ethical issues in insect production. In: van Huis, A. and Tomberlin, J.K. (eds.) *Insects as food and feed: from production to consumption*. Wageningen Academic Publishers, Wageningen, The Netherlands.
- Roecklinsberg, H., Gamborg, C. and Gjerris, M., 2018. Ethical issues in insect production. In: van Huis, A. and Tomberlin, J.K. (eds.) *Insects as food and feed: from production to consumption*. Wageningen Academic Publishers, Wageningen, The Netherlands.
- Roeder, T., 1999. Octopamine in invertebrates. *Progress in Neurobiology* 59: 533-561.
- Roeder, T., 2005. Tyramine and octopamine: ruling behavior and metabolism. *Annual Review of Entomology* 50: 447-477.
- Roff, D.A., 1984. The cost of being able to fly: a study of wing polymorphism in two species of crickets. *Oecologia* 63: 30-37.
- Rowe, A., 2020. Insects raised for food and feed – global scale, practices, and policies. *Effective Altruism Forum*. Available at: <https://tinyurl.com/yxzf8hec>
- Royauté, R. and Dochtermann, N.A., 2017. When the mean no longer matters: developmental diet affects behavioral variation but not population averages in the house cricket (*Acheta domestica*). *Behavioral Ecology* 28: 337-345.
- RSPCA, 2021. What does the term humane killing or humane slaughter mean? RSPCA. Available at: <https://kb.rspca.org.au/knowledge-base/what-does-the-term-humane-killing-or-humane-slaughter-mean/>
- Rumbos, C.I. and Athanassiou, C.G., 2021. 'Insects as food and feed: if you can't beat them, eat them!' – To the magnificent seven and beyond. *Journal of Insect Science (Online)* United States 21:9. <https://doi.org/10.1093/jisesa/ieab019>
- Sakai, M., Taoda, Y., Mori, K., Fujino, M. and Ohta, C., 1991. Copulation sequence and mating termination in the male cricket *Gryllus bimaculatus* DeGeer. *Journal of Insect Physiology* 37: 599-615.
- Sakaluk, S.K., 1987. Reproductive behaviour of the decorated cricket, *Gryllodes supplicans* (Orthoptera: Gryllidae): calling schedules, spatial distribution, and mating. *Behaviour* 100: 202-225.
- Sakaluk, S.K., 1997. Cryptic female choice predicated on wing dimorphism in decorated crickets. *Behavioral Ecology* 8: 326-331.
- Sakaluk, S.K., Oldziej, J., Poppe, C.J., Harper, J.L., Rines, I.G., Hampton, K.J., Duffield, K.R., Hunt, J. and Sadd, B.M., 2019. Effects of inbreeding on life-history traits and sexual competency in decorated crickets. *Animal Behaviour* 155: 241-248.
- Sakaluk, S.K., Schaus, J.M., Eggert, A.-K., Snedden, W.A. and Brady, P.L., 2002. Polyandry and fitness of offspring reared under varying nutritional stress in decorated crickets. *Evolution* 56: 1999-2007.
- Sandford, F., 1987. Intra- and interspecific aggression in two species of field crickets, *Gryllus integer* and *G. alogus*. *Aggressive Behavior* 13: 149-164.
- Schmidt, E., 2015. An analysis of individual variation in behavior. Dissertation. University of North Carolina at Chapel Hill, NC, USA.
- Scotto-Lomassese, S., Strambi, C., Aouane, A., Strambi, A. and Cayre, M., 2002. Sensory inputs stimulate progenitor cell proliferation in an adult insect brain. *Current Biology: CB* 12: 1001-1005.
- Scotto-Lomassese, S., Strambi, C., Strambi, A., Aouane, A., Augier, R., Rougon, G. and Cayre, M., 2003. Suppression of adult neurogenesis impairs olfactory learning and memory in an adult insect. *The Journal of Neuroscience* 23: 9289-9296.
- Scotto-Lomassese, S., Strambi, C., Strambi, A., Charpin, P., Augier, R., Aouane, A. and Cayre, M., 2000. Influence of environmental stimulation on neurogenesis in the adult insect brain. *Journal of Neurobiology* 45: 162-171.
- Sens, n.d. Cricket farm tour – what do crickets need to thrive? Available at: <https://eatsens.com/en-gb/blogs/news/cricket-farm-tour-what-do-crickets-need-to-thrive>
- Shapiro-Ilan, D.I., Mbata, G.N., Nguyen, K.B., Peat, S.M., Blackburn, D. and Adams, B.J., 2009. Characterization of biocontrol traits in the entomopathogenic nematode *Heterorhab-*

- ditis georgiana* (Kesha strain), and phylogenetic analysis of the nematode's symbiotic bacteria. *Biological Control* 51: 377-387.
- Shepherdson, D.J., 1998. Tracing the path of environmental enrichment in zoos. Second nature: Environmental enrichment for captive animals. Smithsonian Institution, Washington DC, USA, pp. 1-14.
- Shirasu-Hiza, M.M., Dionne, M.S., Pham, L.N., Ayres, J.S. and Schneider, D.S., 2007. Interactions between circadian rhythm and immunity in *Drosophila melanogaster*. *Current Biology* 17: R353-R355.
- Simmons, L.W., 1986. Female choice in the field cricket *Gryllus bimaculatus* (De Geer). *Animal Behaviour* 34: 1463-1470.
- Sindermann, D., Heidhues, J., Kirchner, S., Stadermann, N. and Köhl, A., 2021. Industrial processing technologies for insect larvae. *Journal of Insects as Food and Feed* 7: 857-875.
- Sindhøj, E., Lindahl, C. and Bark, L., 2021. Review: potential alternatives to high-concentration carbon dioxide stunning of pigs at slaughter. *Animal* 15: 100164.
- Singer, P., 2002. *Animal liberation*. HarperCollins, New York, NY, USA.
- Singh, Y., Cullere, M., Kovitvadhi, A., Chundang, P. and Dalle Zotte, A., 2020. Effect of different killing methods on physicochemical traits, nutritional characteristics, in vitro human digestibility and oxidative stability during storage of the house cricket (*Acheta domesticus* L.). *Innovative Food Science & Emerging Technologies* 65: 102444.
- Smith, R.L. and Thomas, W., 1988. Southwestern distribution and habitat ecology of *Gryllobates supplicans*. *Bulletin of the Entomological Society of America* 34: 186-191.
- Smolinsky, A.N., Bergner, C.L., LaPorte, J.L. and Kalueff, A.V., 2009. Analysis of grooming behavior and its utility in studying animal stress, anxiety, and depression. In: Gould, T. (ed.) *Mood and anxiety related phenotypes in mMice*. *Neuromethods*, vol 42. Humana Press, Totowa, NJ, USA. https://doi.org/10.1007/978-1-60761-303-9_2
- Sørensen, J., Addison, M. and Terblanche, J., 2012. Mass-rearing of insects for pest management: challenges, synergies and advances from evolutionary physiology. *Crop Protection* 38: 87-94.
- Sorjonen, J.M., Karhapää, M., Holm, S., Valtonen, A. and Roininen, H., 2022. Performance of the house cricket (*Acheta domesticus*) on by-product diets in small-scale production. *Journal of Insects as Food and Feed* 8: 289-294.
- Sorjonen, J.M., Valtonen, A., Hirvisalo, E., Karhapää, M., Lehtovaara, V.J., Lindgren, J., Marnila, P., Mooney, P., Mäki, M., Siljander-Rasi, H., Tapio, M., Tuiskula-Haavisto, M. and Roininen, H., 2019. The plant-based by-product diets for the mass-rearing of *Acheta domesticus* and *Gryllus bimaculatus*. *PLOS ONE* 14: e0218830.
- Špinka, M. and Wemelsfelder, F., 2018. *Environmental challenge and animal agency*. CABI International, Wallingford, UK, pp. 39-55.
- Spranghers, T., Noyez, A., Schildermans, K. and De Clercq, P., 2017. Cold hardiness of the black soldier fly (Diptera: Stratiomyidae). *Journal of Economic Entomology* 110: 1501-1507.
- Strithiphaphirom, P., Lavalley, S. and Robertson, R.M., 2019. Rapid cold hardening and octopamine modulate chill tolerance in *Locusta migratoria*. *Comparative Biochemistry and Physiology Part A: Molecular & Integrative Physiology* 234: 28-35.
- Stevenson, P.A. and Rillich, J., 2013. Isolation associated aggression – a consequence of recovery from defeat in a territorial animal. *PLOS ONE* 8: e74965.
- Stöhr, A.C., Papp, T. and Marschang, R.E., 2016. Repeated detection of an invertebrate iridovirus in amphibians. *Journal of Herpetological Medicine and Surgery* 26: 54-58.
- Straub, P., Tanga, C.M., Osuga, I., Windisch, W. and Subramanian, S., 2019. Experimental feeding studies with crickets and locusts on the use of feed mixtures composed of storable feed materials commonly used in livestock production. *Animal Feed Science and Technology* 255: 114215.
- Styer, E.L. and Hamm, J.J., 1991. Report of a densovirus in a commercial cricket operation in the southeastern United States. *Journal of Invertebrate Pathology* 58: 283-285.
- Subramaniam, M. and Subramoniam, T., 1990. Mating behaviour and spermatophore transfer in *Gryllobates sigillatus* (Walker) (Orthoptera: Gryllidae). *Mitteilungen aus dem Museum für Naturkunde in Berlin. Zoologisches Museum und Institut für Spezielle Zoologie (Berlin)* 66: 65-71.
- Suckling, J., Druckman, A., Moore, C.D. and Driscoll, D., 2020. The environmental impact of rearing crickets for live pet food in the UK, and implications of a transition to a hybrid business model combining production for live pet food with production for human consumption. *The International Journal of Life Cycle Assessment* 25: 1693-1709.
- Szelei, J., Woodring, J., Goettel, M.S., Duke, G., Jousset, F.-X., Liu, K.Y., Zadori, Z., Li, Y., Styer, E., Boucias, D.G., Kleespies, R.G., Bergoin, M. and Tijssen, P., 2011. Susceptibility of north-American and European crickets to *Acheta domesticus* densovirus (AdDNV) and associated epizootics. *Journal of Invertebrate Pathology* 106: 394-399.
- Takuya, O. and Giovanni, I., 2022. Design development of cricket breeding system. In: Bruyns, G. and Wei, H. (eds.) [] *With design: reinventing design modes*. Springer Nature Singapore, Singapore, pp. 2359-2375. https://doi.org/10.1007/978-981-19-4472-7_153
- Tanga, C.M., Egonyu, J.P., Beesigamukama, D., Niassy, S., Emily, K., Magara, H.J., Omuse, E.R., Subramanian, S. and Ekesi, S., 2021. Edible insect farming as an emerging and profitable

- enterprise in east Africa. *Current Opinion in Insect Science* 48: 64-71.
- Tatarova, G., 2017. Automated cricket farming vending machine viability check. Available at: <http://gtatarova.com/assets/technicalreportprint.pdf>.
- Taylor, P.S., Schrobback, P., Verdon, M. and Lee, C., 2023. An effective environmental enrichment framework for the continual improvement of production animal welfare. *Animal Welfare* 32: e14. <https://doi.org/10.1017/awf.2023.5>
- Tennis, P.S., Koonce, J.F. and Teraguchi, M., 1977. The effects of population density and food surface area on body weight of *Acheta domesticus* (L.) (Orthoptera: Gryllidae). *Canadian Journal of Zoology* 55: 2004-2010.
- Then, C. and Bauer-Panskus, A., 2017. Possible health impacts of Bt toxins and residues from spraying with complementary herbicides in genetically engineered soybeans and risk assessment as performed by the European Food Safety Authority EFSA. *Environmental Sciences Europe* 29: 1-1.
- Thompson, P.B., 2020. Philosophical ethics and the improvement of farmed animal lives. *Animal Frontiers* 10: 21-28.
- Timberline, n.d. Crickets. Available at: <https://timberlinefresh.com/learn/insects/crickets/>
- Tomberlin, J.K., Picard, C.J., Jordan, H.R., Preyer, C., Warburton, C., Crowley, P., Zheng, R., Boulanger, F.X., Banks, I., Lefranc, M., Zorrilla, M.J., Olson, G., Aid, G., Fluker, D., Bench, B.J., Teo, M., Richards, C.S., Jones, C., Gonzalez, J. and Elijendy, M., 2022. Government and industry investment plays crucial role in further establishment, evolution, and diversification of insect agriculture: a case example from the United States. *Journal of Insects as Food and Feed* 8: 109-111.
- Tomioka, K., 2014. Chronobiology of crickets: a review. *Zoological Science* 31: 624-632.
- Top Hat Cricket Farm, n.d. Shipping. Available at: <https://www.tophatcrickets.com/shipping>.
- Trickett, S.L., Guy, J.H. and Edwards, S.A., 2009. The role of novelty in environmental enrichment for the weaned pig. *Applied Animal Behaviour Science* 116: 45-51.
- Turner, H.N., Armengol, K., Patel, A.A., Himmel, N.J., Sullivan, L., Iyer, S.C., Bhattacharya, S., Iyer, E.P.R., Landry, C., Galko, M.J. and Cox, D.N., 2016. The TRP channels Pkd2, NompC, and Trpm act in cold-sensing neurons to mediate unique aversive behaviors to noxious cold in *Drosophila*. *Current Biology* 26: 3116-3128.
- Ujváry, I., 2010. Pest control agents from natural products. In: Kreiger, R. (ed.) *Hayes' handbook of pesticide toxicology*, 3rd edition. Academic Press, London, UK, pp. 119-232.
- Underwood, W. and Anthony, R., 2020. AVMA guidelines for the euthanasia of animals: 2020 edition. Available at: <https://www.avma.org/sites/default/files/2020-02/Guidelines-on-Euthanasia-2020.pdf>
- United Nations, 2022. World population prospects 2022. Available at: https://www.un.org/development/desa/pd/sites/www.un.org.development.desa.pd/files/wpp2022_summary_of_results.pdf.
- Ushirogawa, H., Abe, Y. and Tomioka, K., 1997. Circadian locomotor rhythms in the cricket, *Gryllos sigillatus* II. Interactions between bilaterally paired circadian pacemakers. *Zoological Science Zoological Society of Japan* 14: 729-736.
- Vaga, M., Berggren, Å., Pauly, T. and Jansson, A., 2020. Effect of red clover-only diets on house crickets (*Acheta domesticus*) growth and survival. *Journal of Insects as Food and Feed* 6: 179-189.
- Valles, S.M., Oi, D.H., Becnel, J.J., Wetterer, J.K., LaPolla, J.S. and Firth, A.E., 2016. Isolation and characterization of *Nylanderia fulva* virus 1, a positive-sense, single-stranded RNA virus infecting the tawny crazy ant, *Nylanderia fulva*. *Virology* 496: 244-254.
- Van Buskirk, J., 1989. Density-dependent cannibalism in larval dragonflies. *Ecology* 70: 1442-1449.
- van der Fels-Klerx, H.J., Camenzuli, L., Belluco, S., Meijer, N. and Ricci, A., 2018. Food safety issues related to uses of insects for feeds and foods. *Comprehensive Reviews in Food Science and Food Safety* 17: 1172-1183.
- van Huis, A., 2020. Edible crickets, but which species? *Journal of Insects as Food and Feed* 6: 91-94.
- van Huis, A., 2021. Welfare of farmed insects. *Journal of Insects as Food and Feed* 7: 573-584.
- van Huis, A. and Oonincx, D.G.A.B., 2017. The environmental sustainability of insects as food and feed. A review. *Agronomy for Sustainable Development* 37: 43.
- van Huis, A. and Tomberlin, J.K. (eds.), 2016. *Insects as food and feed: from production to consumption*. Wageningen Academic Publishers, Wageningen, the Netherlands.
- van Huis, A., Van Itterbeeck, J., Klunder, H., Mertens, E., Halloran, A., Muir, G. and Vantomme, P., 2013. *Edible insects: future prospects for food and feed security*. FAO, Rome, Italy.
- Van Peer, M., Frooninckx, L., Coudron, C., Berrens, S., Álvarez, C., Deruytter, D., Verheyen, G. and van Miert, S., 2021. Valorisation potential of using organic side streams as feed for *Tenebrio molitor*, *Acheta domesticus*, and *Locusta migratoria*. *Insects* 12: 796.
- Vandeweyer, D., Wynants, E., Crauwels, S., Verreth, C., Viaene, N., Claes, J., Lievens, B. and Van Campenhout, L., 2018. Microbial dynamics during industrial rearing, processing, and storage of tropical house crickets (*Gryllos sigillatus*) for human consumption. *Applied and Environmental Microbiology* 84. <https://doi.org/10.1128/AEM.00255-18>
- Vasanth, M., 1988. Observations on the mating behaviour of *Gryllos sigillatus* (Walker) (Insecta: Orthoptera: Gryllidae). *Records of the Zoological Survey of India* 85: 411-418.

- Veissier, I. and Boissy, A., 2007. Stress and welfare: two complementary concepts that are intrinsically related to the animal's point of view. *Physiology and Behavior* 92: 429-433.
- Veldkamp, T., Claeys, J., Haenen, O., van Loon, J. and Spranghers, T., 2021. *The basics of edible insect rearing*. Wageningen Academic Publishers, Wageningen, the Netherlands. <https://doi.org/10.3920/978-90-8686-902-2>
- Villarreal, A.E., Godin, J.-G.J. and Bertram, S.M., 2018. Influence of the operational sex ratio on mutual mate choice in the Jamaican field cricket (*Gryllus assimilis*): testing the predictions of the switch point theorem. *Ethology* 124: 816-828.
- Visanuvmol, L. and Bertram, S.M., 2010. Dietary phosphorus availability influences female cricket lifetime reproductive effort. *Ecological Entomology* 35: 386-395.
- Visanuvmol, L. and Bertram, S.M., 2011. How dietary phosphorus availability during development influences condition and life history traits of the cricket, *Acheta domesticus*. *Journal of Insect Science* 11: 1-17.
- Vossen, L.E., Nilsson, E., Jansson, A. and Roman, E., 2023. Open field behavior in the house cricket (*Acheta domesticus*): effect of illumination, sex differences and individual consistency. *Journal of Insects as Food and Feed* 9: 317-324.
- Voulgari-Kokota, A., van Loon, M.S. and Bovenkerk, B., 2023. Insects as mini-livestock: considering insect welfare in feed production. *NJAS: Impact in Agricultural and Life Sciences* 95: 2191797.
- Walker, T.J., 1962. Factors responsible for intraspecific variation in the calling songs of crickets. *Evolution* 16: 407-428.
- Walker, T.J., 1980. Mixed oviposition in individual females of *Gryllus firmus*: graded proportions of fast-developing and diapause eggs. *Oecologia* 47: 291-298.
- Walker, T.J. and Sivinski, J.M., 1986. Wing dimorphism in field crickets (Orthoptera: Gryllidae: Gryllus). *Annals of the Entomological Society of America* 79: 84-90.
- Wang, Y., Kleespies, R.G., Huger, A.M. and Jehle, J.A., 2007. The genome of *Gryllus bimaculatus* nudivirus indicates an ancient diversification of baculovirus-related nonoccluded nudiviruses of insects. *Journal of Virology* 81: 5395-5406.
- Watanabe, T., Ochiai, H., Sakuma, T., Horch, H.W., Hamaguchi, N., Nakamura, T., Bando, T., Ohuchi, H., Yamamoto, T., Noji, S. and Mito, T., 2012. Non-transgenic genome modifications in a hemimetabolous insect using zinc-finger and TAL effector nucleases. *Nature Communications* 3: 1017.
- Webster, J., 1994. *Animal welfare: a cool eye towards Eden*. Blackwell Science, Oxford, UK.
- Weissman, D., Gray, D., Pham, H. and Tijssen, P., 2012. Billions and billions sold: pet-feeder crickets (Orthoptera: Gryllidae), commercial cricket farms, an epizootic densovirus, and government regulations make for a potential disaster. *Zootaxa* 3504: 67-88.
- Weissman, D.B. and Gray, D.A., 2019. Crickets of the genus *Gryllus* in the United States (Orthoptera: Gryllidae: Gryllinae). *Zootaxa* 4705: 1-277.
- Weissman, D.B. and Rentz, D.C.F., 1977. Feral house crickets *Acheta domesticus* (L.) (Orthoptera: Gryllidae) in southern California. *Entomological News* 88: 246-248.
- Weissman, D.B., Rentz, D.C.F., Alexander, R.D. and Loher, W., 1980. Field crickets (*Gryllus* and *Acheta*) of California and Baja California, Mexico (Orthoptera: Gryllidae: Gryllinae). *Transactions of the American Entomological Society* 106: 327-356.
- Weissman, D.B., Walker, T.J. and Gray, D.A., 2009. The field cricket *Gryllus assimilis* and two new sister species (Orthoptera: Gryllidae). *Annals of the Entomological Society of America* 102: 367-380.
- Whattam, E.M. and Bertram, S.M., 2011. Effects of juvenile and adult condition on long-distance call components in the Jamaican field cricket, *Gryllus assimilis*. *Animal Behaviour* 81: 135-144.
- Wheeler, D.E. and Buck, N.A., 1996. Depletion of reserves in ant queens during claustral colony founding. *Insectes Sociaux* 43: 297-302.
- Wheeler, D.E. and Martinez, T., 1995. Storage proteins in ants (Hymenoptera: Formicidae). *Comparative Biochemistry and Physiology Part B: Biochemistry and Molecular Biology* 112: 15-19.
- Wiedemann, E., 2014. How American cricket farmers raise bugs for us to eat. *Vice*. Available at: <https://www.vice.com/en/article/nz9x8d/how-american-cricket-farmers-raise-bugs-for-us-to-eat>
- Williams, C.M., 1946. Continuous anesthesia for insects. *Science* 103: 57.
- Wilson, A.D.M., Whattam, E.M., Bennett, R., Visanuvmol, L., Lauzon, C. and Bertram, S.M., 2010. Behavioral correlations across activity, mating, exploration, aggression, and antipredator contexts in the European house cricket, *Acheta domesticus*. *Behavioral Ecology and Sociobiology* 64: 703-715.
- Wilson, J.E., Powell, M.J., Hoover, S.E. and Sarnow, P., 2000. Naturally occurring dicistronic cricket paralysis virus RNA is regulated by two internal ribosome entry sites. *Molecular and Cellular Biology* 20: 4990-4999.
- Wilson, K.M. and Walker, S.E., 2019. Age at mating and male quality influence female patterns of reproductive investment and survival. *Ecology and Evolution* 9: 5440-5449.
- Winship, K.A. and Jones, B.L., 2023. Acoustic monitoring of professionally managed marine mammals for health and welfare insights. *Animals*: 2124.

- Woodring, J.P., McBride, L.A. and Fields, P., 1989. The role of octopamine in handling and exercise-induced hyperglycaemia and hyperlipaemia in *Acheta domesticus*. *Journal of Insect Physiology* 35: 613-617.
- Woodring, J.P., Meier, O.W. and Rose, R., 1988. Effect of development, photoperiod, and stress on octopamine levels in the house cricket, *Acheta domesticus*. *Journal of Insect Physiology* 34: 759-765.
- Wu, H.J., Zhu, D.H., Zeng, Y., Zhao, L.Q. and Sun, G.X., 2014. Brachypterizing effect of high density and its relationship with body injury in cricket species *Velarifictorus micado* (Orthoptera: Gryllidae). *Annals of the Entomological Society of America* 107: 113-118.
- Ylla, G., Nakamura, T., Itoh, T., Kajitani, R., Toyoda, A., Tomonari, S., Bando, T., Ishimaru, Y., Watanabe, T., Fuketa, M., Matsuoka, Y., Barnett, A.A., Noji, S., Mito, T. and Extavour, C.G., 2021. Insights into the genomic evolution of insects from cricket genomes. *Communications Biology* 4: 733.
- Zandbergen, R., 2022. Massive cricket-processing facility comes online in London, Ont. CBC News. Available at: <https://www.cbc.ca/news/canada/london/cricket-farm-london-ontario-1.6506606#:~:text=Crickets%20can%20produce%20protein%20much%20more%20efficiently%20and,pound%2C%20more%20meat%20than%20their%20owners%2C%22%20he%20said>.
- Zefa, E., De Pinho Martins, L., Demari, C.P., Acosta, R.C., Centeno, E., Castro-Souza, R.A., De Oliveira, G.L., Miyoshi, A.R., Fianco, M., Redü, D.R., Timm, V.F., Costa, M.K.M.D. and Szinwelski, N., 2022. Singing crickets from Brazil (Orthoptera: Gryllidea), an illustrated checklist with access to the sounds produced. *Zootaxa* 5209: 211-237.
- Zera, A.J. and Harshman, L.G., 2001. The physiology of life history trade-offs in animals. *Annual Review of Ecology and Systematics* 32: 95-126.
- Zera, A.J., Potts, J. and Kobus, K., 1998. The physiology of life-history trade-offs: experimental analysis of a hormonally induced life-history trade-off in *Gryllus assimilis*. *The American Naturalist* 152: 7-23.
- Zhang, Z., Gao, L. and Zhang, X., 2022. Environmental enrichment increases aquatic animal welfare: a systematic review and meta-analysis. *Reviews in Aquaculture* 14: 1120-1135.
- Zhukovskaya, M., Yanagawa, A. and Forschler, B., 2013. Grooming behavior as a mechanism of insect disease defense. *Insects* 4: 609-630.
- Zielińska, E., Baraniak, B., Karaś, M., Rybczyńska, K. and Jakubczyk, A., 2015. Selected species of edible insects as a source of nutrient composition. *Food Research International* 77: 460-466.
- Zimmermann, G., 2007. Review on safety of the entomopathogenic fungus *Metarhizium anisopliae*. *Bio-control Science and Technology* 17: 879-920.